

BILKENT CALCULUS I EXAMS

1989 – 2023

Fall 2024 Midterm 1 Version

Fall 2024 Midterm 1	1	Fall 2009 Final	66
Fall 2023 Midterm 1	2	Fall 2008 Midterm I	67
Fall 2023 Midterm 2	3	Fall 2008 Midterm II	68
Fall 2023 Final	4	Fall 2008 Final	69
Fall 2022 Midterm 1	6	Fall 2007 Midterm I	70
Fall 2022 Midterm 2	7	Fall 2007 Midterm II	71
Fall 2022 Final	8	Fall 2007 Final	72
Fall 2021 Midterm	9	Fall 2006 Midterm I	73
Fall 2021 Final	11	Fall 2006 Midterm II	74
Fall 2020 Midterm	12	Fall 2006 Final	74
Fall 2020 Final	14	Fall 2005 Midterm I	75
Fall 2019 Midterm I	14	Fall 2005 Midterm II	76
Fall 2019 Midterm II	16	Fall 2005 Final	77
Fall 2019 Final	17	Fall 2004 Midterm I	78
Fall 2018 Midterm I	19	Fall 2004 Midterm II	79
Fall 2018 Midterm II	20	Fall 2004 Final	79
Fall 2018 Final	21	Fall 2003 Midterm I	80
Fall 2017 Midterm I	23	Fall 2003 Midterm II	81
Fall 2017 Midterm II	24	Fall 2003 Final	81
Fall 2017 Final	25	Fall 2002 Midterm I	82
Fall 2016 Midterm I	27	Fall 2002 Midterm II	83
Fall 2016 Midterm II	29	Fall 2002 Final	83
Fall 2016 Final	30	Fall 2001 Midterm I	84
Fall 2015 Midterm I	31	Fall 2001 Midterm I	84
Fall 2015 Midterm II	33	Fall 2001 Final	85
Fall 2015 Final	36	Fall 2000 Midterm I	86
Fall 2014 Midterm I	37	Fall 2000 Midterm II	87
Fall 2014 Midterm II	38	Fall 2000 Final	88
Fall 2014 Final	40	Fall 1999 Midterm I	88
Fall 2013 Midterm I	44	Fall 1999 Midterm II	89
Fall 2013 Midterm II	46	Fall 1999 Final	90
Fall 2013 Final	48	Fall 1997 Midterm I	90
Fall 2012 Midterm I	51	Fall 1997 Midterm II	91
Fall 2012 Midterm II	53	Fall 1997 Final	92
Fall 2012 Final	55	Fall 1996 Midterm I	92
Fall 2011 Midterm I	56	Fall 1996 Midterm II	93
Fall 2011 Midterm II	58	Fall 1996 Final	93
Fall 2011 Final	59	Fall 1995 Midterm I	94
Fall 2010 Midterm I	60	Fall 1995 Midterm II	94
Fall 2010 Midterm II	61	Fall 1995 Final	95
Fall 2010 Final	63	Fall 1994 Midterm I	95
Fall 2009 Midterm I	65	Fall 1994 Midterm II	96
Fall 2009 Midterm II	65	Fall 1994 Final	96

Fall 1993 Midterm I	97
Fall 1993 Midterm II	98
Fall 1993 Final	99
Fall 1992 Midterm I	99
Fall 1992 Midterm II	100
Fall 1992 Final	101
Fall 1991 Midterm I	102

Fall 1991 Midterm II	103
Fall 1991 Final	103
Fall 1990 Midterm I	104
Fall 1990 Midterm II	105
Fall 1990 Final	106
Fall 1989 Midterm	106

Fall 2024 Midterm 1

1. Evaluate the following limits.

[Do not use L'Hôpital's Rule!]

a. $\lim_{x \rightarrow 0^+} \frac{(x^{1/2} - x)^{1/2} - x^{1/4}}{x^{3/4}}$

b. $\lim_{x \rightarrow 0} \frac{\sin(x^2) \cos(\cot x)}{x}$

2. Find $\left. \frac{d^2 y}{dx^2} \right|_{(x,y)=(0,1)}$ if y is a differentiable function of x satisfying the equation $y^3 = y^2 \cos x + \sin x$.

3. The lengths of the sides of a right triangle are changing as differentiable functions of time. At a certain moment,

- the length of one of the legs is increasing at a rate of 1 cm/s,

- the length of the other leg is decreasing at a rate of 2 cm/s,
- the length of the hypotenuse is decreasing at a rate of 1 cm/s, and
- the area of the triangle is decreasing at a rate of $1/2$ cm²/s.

Find the length of the shortest side at this moment.

4. Let $f(x) = 2x^4 - 7x^2 + 6x$.

a. Compute $f'(x)$.

b. Write an equation for the tangent line to the graph of $y = f(x)$ at the point with $x = -1$.

 Write only the equation (and nothing else) in the box!

c. Find the absolute maximum and minimum values of f on the interval $[-2, 2]$.

Fall 2023 Midterm 1

1. Evaluate the following limits.

[Do not use L'Hôpital's Rule!]

a. $\lim_{x \rightarrow 1} \frac{2x^3 - 3x^2 + 1}{x^3 - 3x + 2}$

b. $\lim_{x \rightarrow 0^-} \frac{\sqrt{\cos(x) - \cos(2x)}}{x}$

2. Consider the function $f(x) = \frac{\sin(\pi x^3)}{x^2 + 1}$.

a. Find an equation for the tangent line to the graph of $y = f(x)$ at the point with $x = 1$.

b. Show that the equation $f(x) = \frac{3}{5}$ has at least two solutions.

3. Find $\left. \frac{d^2 y}{dx^2} \right|_{(x,y)=(0,1)}$ if y is a differentiable function of x satisfying the equation:

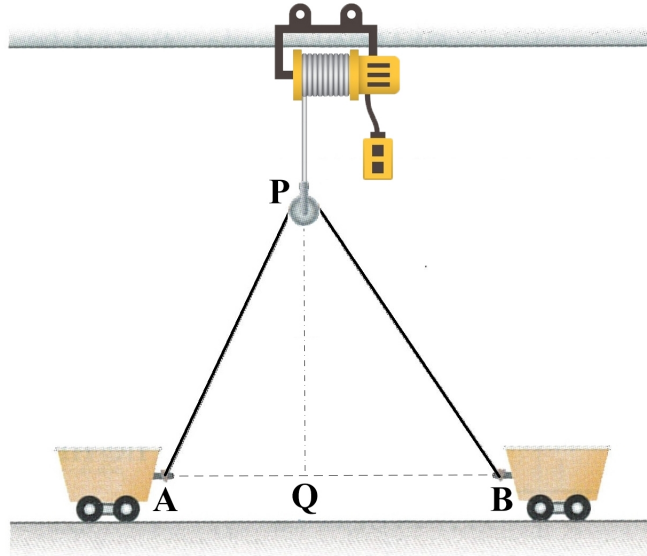
$$xy^2 + \sin(\pi y) = x^3$$

4. A 13 m long rope is tied to two carts at the points A and B , and passes over a pulley at the point P as shown in the figure. A winch moves the pulley along a fixed vertical line. The point Q lies directly below the point P and on the line connecting the points A and B .

At a certain moment when the winch is raising the pulley at a rate of 1 m/s; P is 4 m away from Q , A is 3 m away from Q , and the distance between the points A and Q is decreasing at a rate of 2 m/s.

Determine how fast the distance between the points B and Q is changing at this moment.

[Do not assume anything about how the quantities in the question depend on time beyond what is already given in the question.]



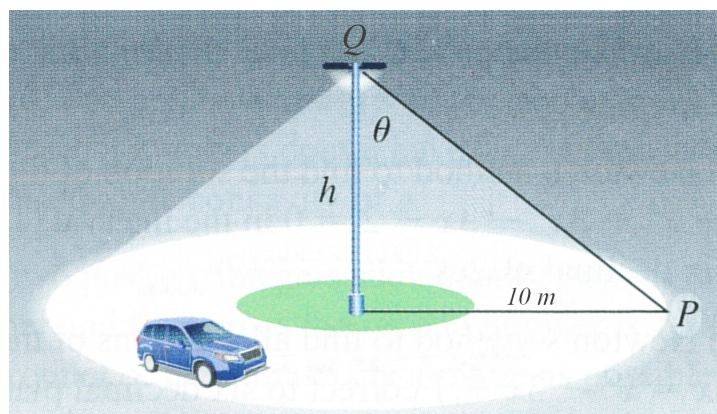
Fall 2023 Midterm 2

1. Sketch the graph of $y = x^{1/3} + x^{-2/3}$ by computing y' and y'' , and determining their signs; finding the critical points, the inflections points, the intercepts, and the asymptotes; and clearly labeling them in the picture.

2. A lamp post will be erected in the center of a traffic circle with radius 10 m. The municipal regulations require the height h of the lamp post to be between 5 m and 10 m. The intensity of illumination I at a point P at the edge of the traffic circle is given by

$$I = c \frac{\cos \theta}{(\text{the distance from } Q \text{ to } P)^2}$$

where c is a positive constant, and the point Q and the angle θ are as shown in the figure.



Determine how tall the lamp post should be to provide **(a)** the maximum and **(b)** the minimum illumination at P .

3. Evaluate the following integrals.

a. $\int \frac{dx}{\tan x + \cot x}$

b. $\int_{1/3}^1 x^{-4}(x^{-7} + 7x^{-3})^{1/3} dx$

4a. Find $y(4)$ if $\frac{dy}{dx} = \sqrt{xy}$ and $y(0) = \frac{1}{9}$.

4b. Suppose f is a continuous function such that

- $f(0) = 2017$, $f(2) = 2020$, and $f(4) = 2023$, and
- the average value of f on the interval $[0, 4]$ is 1.

Find $g'(2)$ where $g(x) = \int_0^x f(xt) dt$.

Fall 2023 Final

1. Evaluate the following limit $\lim_{x \rightarrow 0} \left(\frac{1}{2x^2} + \frac{1}{x^3} + \frac{1}{x^4} - \frac{e^{\sin x}}{x^4} \right)$.

2a. Evaluate the integral $\int \frac{dx}{(\tan x + \cot x)^2}$.

2b. A function f with a continuous derivative satisfies:

$$f(0) = 1, \quad f(\pi/4) = 2, \quad \int_0^{\pi/4} f(x) dx = 3, \quad \int_0^{\pi/4} f(x) \tan^2 x dx = 4$$

Evaluate $\int_0^{\pi/4} f'(x) \tan x dx$.

3a. The area of the region lying above the parabola $y = x^2$ and below a line passing through the origin is 1 square unit. Find all possible values of the slope of this line.

3b. Find the volume of the solid generated by revolving the region between the graph of $y = 1/(1 + e^x)$ and the x -axis for $x \geq 0$ about the x -axis.

4. In each of the following, if the given statement is true for all f , then mark the ☐ to the left of TRUE with a **X**; otherwise, mark the ☐ to the left of FALSE with a **X** and give a counterexample.

a. If f has a derivative on $(-\infty, \infty)$, then f has an antiderivative on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. If f has an antiderivative on $(-\infty, \infty)$, then f has a derivative on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. If $f'(x + 2\pi) = f'(x)$ for all x , then $f(x + 2\pi) = f(x)$ for all x .

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. If $f(n) \geq n$ for all positive integers n , then $\lim_{x \rightarrow \infty} f(x) = \infty$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. If f is continuous on $(-\infty, \infty)$, then $\frac{d}{dx} \int_0^x f(xt) dt = f(x^2)$ for all x .

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

Fall 2022 Midterm 1

1a. Evaluate the limit $\lim_{x \rightarrow 1} \frac{x^2 + 2 - \frac{12}{x+3}}{x^2 + 3 - \frac{12}{x+2}}$. (Do not use L'Hôpital's Rule!)

1b. Suppose that a differentiable function f satisfies

$$f'(x) + 3f(x^2) = f(x)^2$$

for all $x > 0$ and $f(1) = 5$. Find $f''(1)$.

2a. Find an equation for the tangent line to the graph of $y = \tan^3\left(\frac{\pi}{x}\right)$ at the point with $x = 3$.

2b. Suppose that a differentiable function f satisfies

$$-|x| - 2 < f(x) < |x|$$

for all x . Show that there is c such that $f'(c) = 0$.

3. For a point P on the curve $y = x^2 + 5$ different from $(0, 5)$, let T and Q be points on the x -axis such that the line PT is tangent to the curve at P and PQ is perpendicular to the x -axis. Let A be the area of the triangle PQT . Assume that the coordinates are measured in centimeters and the time is measured in seconds.

Find all possible values of the x -coordinate of the point P at a moment when A is decreasing at a rate of $6 \text{ cm}^2/\text{s}$ and the x -coordinate of the point P is increasing at a rate of 2 cm/s .

4. A function f , which is continuous on $[0, \infty)$ and twice-differentiable on $(0, \infty)$, satisfies the following conditions:

- ① $f(0) = 2$, $f(3 - \sqrt{5}) = 1 + \sqrt{5}$, $f(2) = 3$
- ② $\lim_{x \rightarrow \infty} f(x) = 0$
- ③ $f'(x) > 0$ for $0 < x < 3 - \sqrt{5}$, and $f'(x) < 0$ for $x > 3 - \sqrt{5}$
- ④ $\lim_{x \rightarrow 0^+} f'(x) = \infty$
- ⑤ $f''(x) < 0$ for $0 < x < 2$, and $f''(x) > 0$ for $x > 2$

a. Sketch the graph of $y = f(x)$ making sure that all important features are clearly shown.

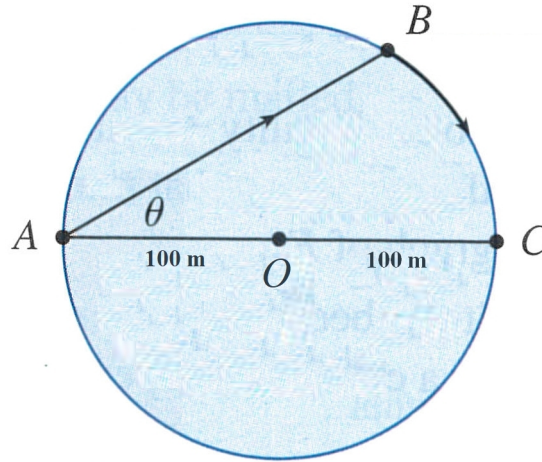
b. Fill in the boxes to make the following a true statement. No explanation is required.

The function $f(x) = \frac{a\sqrt{x} + b}{x + c}$ satisfies the conditions ①-⑤ at all points in its domain if a , b and c are chosen as

$$a = \boxed{}, \quad b = \boxed{} \quad \text{and} \quad c = \boxed{}.$$

Fall 2022 Midterm 2

1. You are standing at a point A on the shore of a circular lake with radius 100 m, and you are planning to go to the point C , where $[AC]$ is a diameter of the circle. You will swim from A to a point B on the shore along a straight line and then either walk or run from B to C along the shore.



- You can swim with a speed of 1 m/s.
- You can walk with a speed of $\sqrt{2}$ m/s.
- You can run with a speed of 2 m/s.

Determine the angle $\theta = \widehat{CAB}$ that will take you from A to C in the shortest possible time

- if you walk from B to C , and
- if you run from B to C .

2a. Suppose that f is a continuous function satisfying

$$f(x) = x - x^2 - x \int_0^x f(t) dt$$

for all x , and c is a real number such that $f'(c) = 0$. Express $f(c)$ in terms of c only.

2b. Evaluate the limit $\lim_{n \rightarrow \infty} \frac{1}{n^3} \left(\sum_{i=1}^{3n} \sqrt{i} \right)^2$.

3. Evaluate the following integrals.

a. $\int (3x - 1)(x + 2)(x^2 - 2)^{2022}(2x + 5)^{2022} dx$

b. $\int_{\pi/4}^{5\pi/4} \frac{\cos^2 x}{(x + \sin x \cos x)^2} dx$

- Compute the volume $V(a)$ of the solid generated by revolving $R(a)$ about the x -axis.
- Compute the volume $W(a)$ of the solid generated by revolving $R(a)$ about the y -axis.
- Find all values of a for which $V(a) = W(a)$.

Fall 2022 Final

$$\lim_{x \rightarrow 0} \frac{xe^{ax^2} - \sin x}{x^5}$$

2a. Evaluate the improper integral $\int_0^{\infty} \frac{dx}{(x+1)(2x+3)}$.

2b. Let $A = \int_1^{\sqrt{3}} \frac{\ln x}{x^2 + 1} dx$ and $B = \int_2^{2\sqrt{3}} \frac{\ln x}{x^2 + 4} dx$. Express B in terms of A .

$$\begin{array}{lll} \int_0^{\pi/2} \sin(x)f(x) \, dx = 1 & \int_0^{\pi/2} \sin(x)f'(x) \, dx = 3 & \int_0^{\pi/2} \sin(2x)f(x) \, dx = 5 \\ \int_0^{\pi/2} \cos(x)f(x) \, dx = 2 & \int_0^{\pi/2} \cos(x)f'(x) \, dx = 4 & \int_0^{\pi/2} \cos(2x)f(x) \, dx = 6 \end{array}$$

Evaluate $\int_0^{\pi/2} \sin(2x)f''(x) \, dx$.

a. f is continuous on $(-\infty, \infty)$ and f does not have an antiderivative on $(-\infty, \infty)$.

$$f(x) =$$

b. f is positive and differentiable on $(-\infty, \infty)$ and $\int \frac{dx}{f(x)} \neq \ln(f(x)) + C$.

$$f(x) =$$

c. f is continuous on $[0, \pi]$ and $\int_0^\pi |f(x)| dx \neq \left| \int_0^\pi f(x) dx \right|$.

$$f(x) =$$

d. f is differentiable on $(0, \infty)$, $\lim_{x \rightarrow \infty} f'(x) = 0$, and $\lim_{x \rightarrow \infty} f(x)$ does not exist.

$$f(x) =$$

e. f is differentiable on $(0, \infty)$, $\lim_{x \rightarrow \infty} f(x) = 0$, and $\lim_{x \rightarrow \infty} f'(x)$ does not exist.

$$f(x) =$$

Fall 2021 Midterm

1. Evaluate the following limits.

[Do not use L'Hôpital's Rule!]

a. $\lim_{x \rightarrow -\infty} \frac{\sin(1/x)}{\sqrt{x^2 + 1} + x}$

b. $\lim_{x \rightarrow \pi/2} \frac{1 + \cos 2x}{1 - \sqrt[3]{\sin x}}$

2. Find $\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(\pi/6, \pi/3)}$ if y is a differentiable function of x satisfying the equation:

$$y \sin(2y - x) = 2x$$

3. The points P and Q are moving along the graph of a twice-differentiable function $y = f(x)$ in the xy -plane in such a way that their coordinates are differentiable functions of time t , and the tangent line to the graph at the point P intersects the graph also at the point Q at all times. (Assume that the coordinates are measured in meters and the time is measured in seconds.)

Find $f''(2)$ if

- ① the x -coordinate of Q is -1 and decreasing at a rate of 3 m/s when the x -coordinate of P is 2 and increasing at a rate of 4 m/s,
- ② $y = 9x - 8$ is an equation for the tangent line to the graph of f at the point with $x = 2$, and
- ③ $y = -6x - 23$ is an equation for the tangent line to the graph of f at the point with $x = -1$.

4. In each of the following, if the given statement is true, then mark the ☐ to the left of TRUE with a **X** and prove the statement; otherwise, mark the ☐ to the left of FALSE with a **X** and give a counterexample.

a. If f is differentiable on $(0, \infty)$ and $f(1/x) = f(x)$ for all $x > 0$, then there is a c in $(0, \infty)$ such that $f'(c) = 0$. ☐ TRUE ☐ FALSE

b. If f is differentiable on $(0, \infty)$ and $f(x^2) = (f(x))^3$ for all $x > 0$, then there is a c in $(0, \infty)$ such that $f'(c) = 0$. ☐ TRUE ☐ FALSE

c. If f is differentiable on $(0, \infty)$ and $f(x)f(2x) \geq 0$ for all $x > 0$, then there is a c in $(0, \infty)$ such that $f'(c) = 0$. ☐ TRUE ☐ FALSE

d. If f is differentiable on $(0, \infty)$ and $f(x)f(2x) \leq 0$ for all $x > 0$, then there is a c in $(0, \infty)$ such that $f'(c) = 0$. ☐ TRUE ☐ FALSE

5. A twice-differentiable function f on $(-\infty, \infty)$ satisfies the following conditions:

- ① $f(-5) = 0$, $f(-3) = A$, $f(0) = B$, $f(3) = C$, where A, B, C are real numbers such that $2 < A < C$
- ② $\lim_{x \rightarrow -\infty} f(x) = -2$, $\lim_{x \rightarrow \infty} f(x) = 2$
- ③ $f'(x) > 0$ for $x < 0$, $f'(x) < 0$ for $x > 0$
- ④ $f''(0) = 0$, $f''(x) > 0$ for $x < -3$ and for $x > 3$, $f''(x) < 0$ for $-3 < x < 0$ and for $0 < x < 3$

- a.** Sketch the graph of $y = f(x)$ making sure that all important features are clearly shown.
- b.** Fill in the boxes to make the following a true statement. No explanation is required.

The function $f(x) = \frac{ax^3 + b}{|x|^3 + c}$ satisfies the conditions ①-④ if a , b and c are chosen as

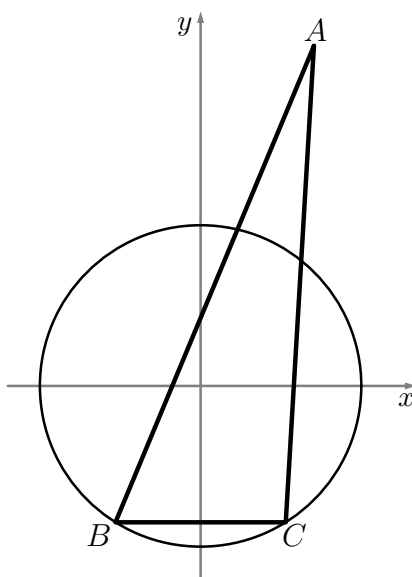
$$a = \boxed{}, \quad b = \boxed{} \text{ and } c = \boxed{}.$$

Fall 2021 Final

1. A triangle ABC in the xy -plane satisfies the following conditions:

- ❶ The coordinates of the vertex A are $(1, 3)$.
- ❷ The vertices B and C lie on the circle $x^2 + y^2 = 2$.
- ❸ The side $[BC]$ is parallel to the x -axis.

Find the largest and smallest possible values of the area of the triangle ABC .



2. Evaluate the following integrals.

a. $\int \frac{\sin^3 x}{1 + \cos x} dx$

b. $\int_0^{\pi/2} \frac{\sin^3 x}{1 + \cos^2 x} dx$

3. Consider the function

$$f(x) = \exp(-(\ln x)^k) = e^{-(\ln x)^k}$$

where k is a constant.

a. In this part, let $k = 3$ and compute $f'(e^2)$.

b. In this part, let $k = 2$ and compute $\int_1^\infty f(x) \frac{\ln x}{x} dx$.
 [Here you may use the fact that $\int_a^\infty e^{-x} dx = e^{-a}$.]

c. In this part, assume that $f(x^2) = (f(x))^3$ for all $x > 1$ and find k .

4a. Find $f(8)$ if f is a continuous function satisfying

$$\int_0^{x^3} f(t) dt + \int_0^x f(t^3) dt = x$$

for all x .

4b. Evaluate the limit $\lim_{x \rightarrow 0} \frac{e^{x - \frac{1}{2}x^2} - x - 1}{x^3}$.

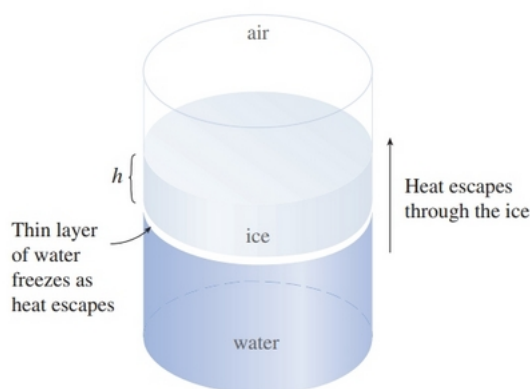
5. The thickness h of a sheet of sea ice as a function of time t satisfies the equation

$$\frac{dh}{dt} = \frac{k}{LDh} (T_w - T_a)$$

for $h > 0$ where

- T_a is the air temperature,
- T_w is the water temperature,
- k is the thermal conductivity of the ice,
- L is the latent heat of the water, and
- D is the mass density of the water.

In this question we will assume that T_a , T_w , k , L , and D are constants.



We measure the thickness of a sheet of sea ice daily, and observe that it is 3 cm on day 0, and 5 cm on day 10.

Determine when the thickness of the sheet will be 13 cm.

Fall 2020 Midterm

1. A differentiable function f and the polynomials p , q , r , s satisfy

$$-x^2 + \frac{9}{2}x - \frac{1}{2} = p(x) \leq f(x) \leq q(x) = x^2 + 2 \quad \text{for all } x < 1,$$

and

$$-\frac{3}{2}x^3 + 6x^2 - 6x + \frac{9}{2} = r(x) \leq f(x) \leq s(x) = -x^3 + 5x^2 - 5x + 4 \quad \text{for all } x > 1.$$

a. Find $f(1)$.

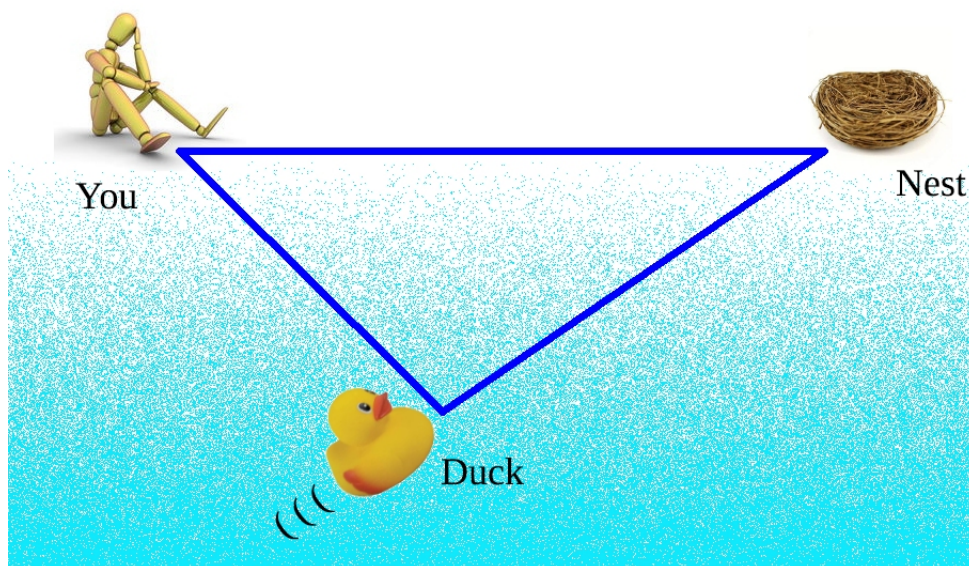
b. Find $f'(1)$.

c. Show that there is a point c such that $f'(c) = 0$.

2. Find $\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1/2,1)}$ if y is a differentiable function of x satisfying the equation:

$$\sin(\pi xy) = \frac{1}{x} - \frac{1}{y}$$

3. As you sit on a shore, you notice a duck swimming towards its nest. At that moment, the duck is 7 m away from its nest and swimming with a speed of 2 m/s, and the distance between you and the duck is 8 m and increasing at a rate of 1 m/s. Determine how far you are from the duck's nest.



4. Find the absolute maximum and minimum values of $f(x) = -x^4 + \frac{21}{2}x^2 - 10x$ on the interval $[-3, 3]$.

5. A function f , which is defined and continuous for all $x \neq 2$, satisfies the following conditions:

① $f(0) = 2$, $f(\sqrt{6} - 2) = (22 + 8\sqrt{6})/25$

② $\lim_{x \rightarrow 2^-} f(x) = \infty$, $\lim_{x \rightarrow 2^+} f(x) = -\infty$, $\lim_{x \rightarrow -\infty} f(x) = 0$, $\lim_{x \rightarrow \infty} f(x) = 0$

- ③ $f'(x) > 0$ for $x < 0$, and for $x > \sqrt{6} - 2$ and $x \neq 2$; and $f'(x) < 0$ for $0 < x < \sqrt{6} - 2$
- ④ $\lim_{x \rightarrow 0^-} f'(x) = 4$, $\lim_{x \rightarrow 0^+} f'(x) = -2$
- ⑤ $f''(x) > 0$ for $x < 2$ and $x \neq 0$, $f''(x) < 0$ for $x > 2$

a. Sketch the graph of $y = f(x)$ making sure that all important features are clearly shown.

b. Fill in the boxes to make the following a true statement. No explanation is required.

The function $f(x) = \frac{ax + b}{x^2 + c|x| + d}$ satisfies the conditions ①-⑤ at all points in its domain if a , b , c and d are chosen as

$$a = \boxed{}, \quad b = \boxed{}, \quad c = \boxed{} \quad \text{and} \quad d = \boxed{}.$$

Fall 2020 Final

1. Evaluate the following integrals:

a. $\int_{-1}^1 (x^3 - 3x^2 + 2)(x^2 - 2x - 2)^{2021} dx$

b. $\int \cos 2x \tan^3 x dx$

2. Evaluate the limit $\lim_{x \rightarrow 0} \frac{x^2 \cos x - x \sin x}{\ln(1 + x^4)}$.

3. Let $R(a)$ be the region between the curve $y = x^{1/2}(9 - x)^{1/3}$ and the x -axis for $0 \leq x \leq a$. Let $V(a)$ be the volume of the solid generated by revolving $R(a)$ about the x -axis. Let $W(a)$ be the volume of the solid generated by revolving $R(a)$ about the y -axis.

a. Compute $V(9)$.

b. Find all $a \geq 0$ for which $\frac{dV}{da} = \frac{dW}{da}$.

4. For $t \geq 0$, let $A(t)$ denote the area of the triangle formed by the coordinate axes and the tangent line to the curve $y = e^{-x}$ at the point (t, e^{-t}) . Find the absolute maximum and minimum values of A .

Fall 2019 Midterm I

1. Suppose that $\lim_{x \rightarrow 0^+} f(x) = A$, $\lim_{x \rightarrow 0^-} f(x) = B$, $f(0) = C$, where A , B , C are distinct real numbers.

In each of the following, fill in the corresponding box by:

- Expressing the limit in terms of A , B , C if it is possible to do so using the given information;
- Writing DNE if it is possible to conclude that the limit does not exist using the given information;

or

- Putting a **X**, otherwise.

No explanation is required. No partial points will be given.

a. $\lim_{x \rightarrow 0^+} f(x - \sqrt{x}) =$

b. $\lim_{x \rightarrow 0^+} f(x \sin^2(1/x)) =$

c. $\lim_{x \rightarrow 0^+} f(x - x^2 \sin(1/x)) =$

d. $\lim_{x \rightarrow 0^+} f(x - \sin(x)) =$

e. $\lim_{x \rightarrow 0^+} f(x - \tan(x)) =$

2. Consider the function $f(x) = \frac{\sin(\pi x)}{\sqrt{x^2 + x^3 - x}}$.

a. Find the domain of f .

b. Compute $f(5/4)$.

c. Show that there is a positive real number x such that $f(x) = -1$.

d. Evaluate $\lim_{x \rightarrow 0^-} f(x)$.

[Do not use L'Hôpital's Rule!]

e. Evaluate $\lim_{x \rightarrow 0^+} xf(x)$.

[Do not use L'Hôpital's Rule!]

3. Suppose that f is a differentiable function and $g(x) = f(xf(x^2) + f(x))$.

Find $f(5)$ and $f'(5)$ if

❶ $f(2) = 3$, $f(3) = 5$, $f(4) = 1$, $f'(2) = -2$, $f'(3) = -4$, $f'(4) = -3$; and

❷ $y = 20x + 19$ is an equation for the tangent line to the graph of $y = g(x)$ at the point with $x = 2$.

4. The points P and Q are moving along the parabola $y = x^2$ in the xy -plane in such a way that their coordinates are differentiable functions of time and the distance between them is constant. (Assume that the coordinates are measured in meters and the time is measured in seconds.)

Determine the rate of change of the distance between the point Q and the origin at the moment when P is at $(2, 4)$, Q is at $(-1, 1)$, and the distance between the point P and the origin is increasing at a rate of 3 m/s.

Fall 2019 Midterm II

1. A continuous function f on $(-\infty, \infty)$ satisfies the following conditions:

❶ $f(0) = 6$, $f(3) = 0$

❷ $f'(x) > 0$ for $x < 0$ and for $3 < x$; $f'(x) < 0$ for $0 < x < 3$

❸ $f''(x) > 0$ for $x < 0$ and for $0 < x < 3$; $f''(x) < 0$ for $3 < x$

❹ $\lim_{x \rightarrow -\infty} f(x) = 4$, $\lim_{x \rightarrow \infty} f(x) = 4$

❺ $\lim_{x \rightarrow 0^-} f'(x) = 1$, $\lim_{x \rightarrow 0^+} f'(x) = -5$; $\lim_{x \rightarrow 3^-} f'(x) = -4/5$, $\lim_{x \rightarrow 3^+} f'(x) = 4/5$

a. Sketch the graph of $y = f(x)$ making sure that all important features are clearly shown.

b. Fill in the boxes to make the following a true statement. No explanation is required.

The function $f(x) = \frac{|x - a|}{b|x| + c}$ satisfies the conditions ❶–❺ if a , b and c are chosen as

$$a = \boxed{}, \quad b = \boxed{} \quad \text{and} \quad c = \boxed{}.$$

2. Suppose f is a twice-differentiable function on $(-\infty, \infty)$ satisfying the following conditions:

❶ $x = 4$ and $x = 11$ are the only critical points of f in the interval $(0, 15)$.

❷ $f(0) = 1$, $f(4) = -2$, $f(11) = 4$, $f(15) = -1$.

❸ $f'(0) = -1$, $f'(15) = -2$.

❹ $|f''(x)| \leq 1$ for all x in the interval $[0, 15]$.

a. Let $g(x) = 3f(x) - (f'(x))^2$. Show that the information given above is sufficient to determine the absolute maximum and minimum values of the function g on the interval $[0, 15]$, and find them.

b. Show that there is a point c in the interval $(0, 15)$ such that $f''(c) = -1/2$.

3a. The slope of the tangent line at each point (x, y) on the graph of a differentiable function $y = f(x)$ is proportional to $x^2 - 5$. If $f(1) = 1$ and $f(3) = 3$, find $f(2)$.

3b. Suppose that a continuous function g satisfies:

$$\int_0^3 g(x) dx = 7 \quad \text{and} \quad \int_0^6 g(2x) dx = 5$$

Find $\int_1^2 x g(3x^2) dx$.

4. Evaluate the following integrals.

a. $\int_0^{\pi/4} \frac{\sec^2 x}{(2 \tan x + 1)^2} dx$

b. $\int_0^{\pi/4} \frac{\sin x}{(2 \sin x + \cos x)^3} dx$

Fall 2019 Final

1. Find $y'''|_{x=1}$ if y is a differentiable function of x satisfying the identity $y^3 + xy + x^2 = 1$.

2. Find all pairs (a, b) of constants for which the limit

$$\lim_{x \rightarrow 0} \frac{e^x - xe^{ax} - \cos bx}{x^4}$$

exists, and find the value of the limit for each of these pairs of constants.

3. In each of the following, if the given statement is true for all continuous functions f that are defined on $(-\infty, \infty)$, then mark the $\square$ to the left of TRUE with a **X**; otherwise, mark the $\square$ to the left of FALSE with a **X** and give a counterexample.

a. f has a derivative on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. f has an antiderivative on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. $\frac{d}{dx} \int_0^1 f(x) dx = f(x)$ for all $0 \leq x \leq 1$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. If f is increasing on $(-\infty, \infty)$, then f^2 is increasing on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. If f is decreasing on $(-\infty, \infty)$, then e^f is decreasing on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

4a. Evaluate the integral $\int \frac{\tan x}{2 \tan^2 x - 1} dx$

4b. Suppose that a function f with a continuous derivative satisfies the conditions:

❶ $\int_1^3 f'(x) dx = 1$ ❷ $\int_1^3 f(x) f'(x) dx = 1$ ❸ $\int_1^3 x^2 f'(x) dx = 1$ ❹ $\int_1^3 (f(x))^2 dx = 2020$

Find $\int_1^3 x f(x) dx$.

5. Suppose that a function f satisfies the following conditions:

- ❶ f is positive and continuous on $[0, \infty)$.
- ❷ $f'(x) < 0$ for all $x > 0$.
- ❸ The improper integral $\int_0^\infty x f(x) dx$ converges.

Let R be the region between the graph of $y = f(x)$ and the x -axis for $x \geq 0$, and let $V(a)$ be the volume of the solid generated by revolving the region R about the line $x = a$ where a is a nonnegative real number.

a. Show that the improper integral $\int_0^\infty f(x) dx$ converges.

b. Express $V(a)$ using the cylindrical shells method.

c. Show that $V''(a) > 0$ for all $a > 0$.

Fall 2018 Midterm I

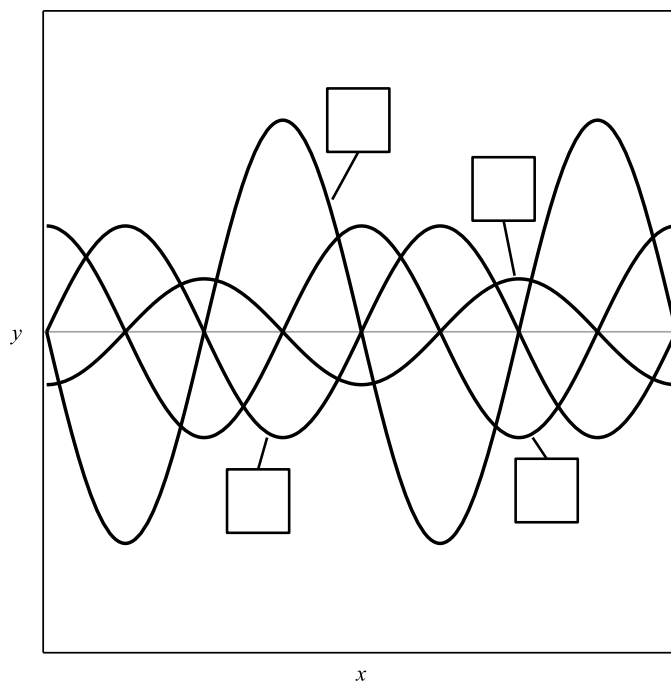
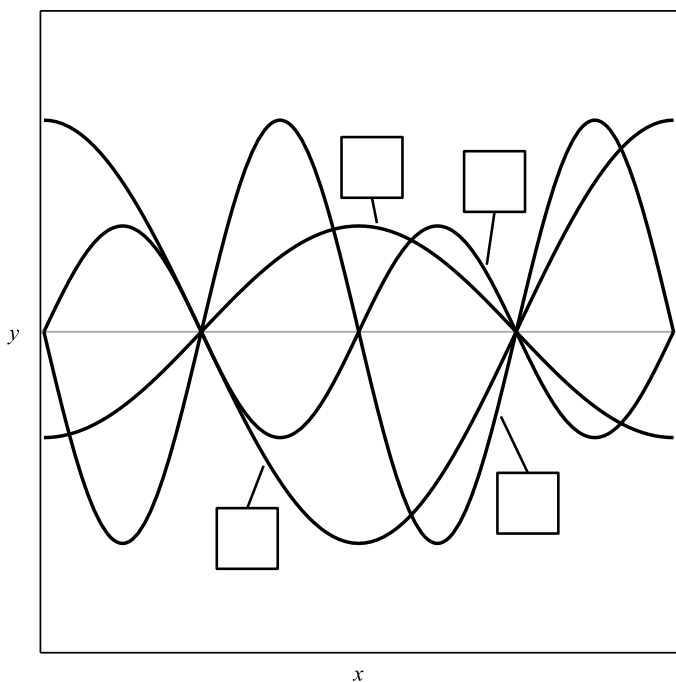
1. Evaluate the following limits by expressing the answers in terms of $A = \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$. [Do not use L'Hôpital's Rule!]

a. $\lim_{x \rightarrow 0} \frac{1 - x^2/2 - \cos x}{x^4}$

b. $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3}$

2a. Let P be the point on the graph of $y = x^5 - x^2$ with $x = 1$. Show that there is a point Q on the graph such that the tangent lines to the graph at the points P and Q are perpendicular to each other.

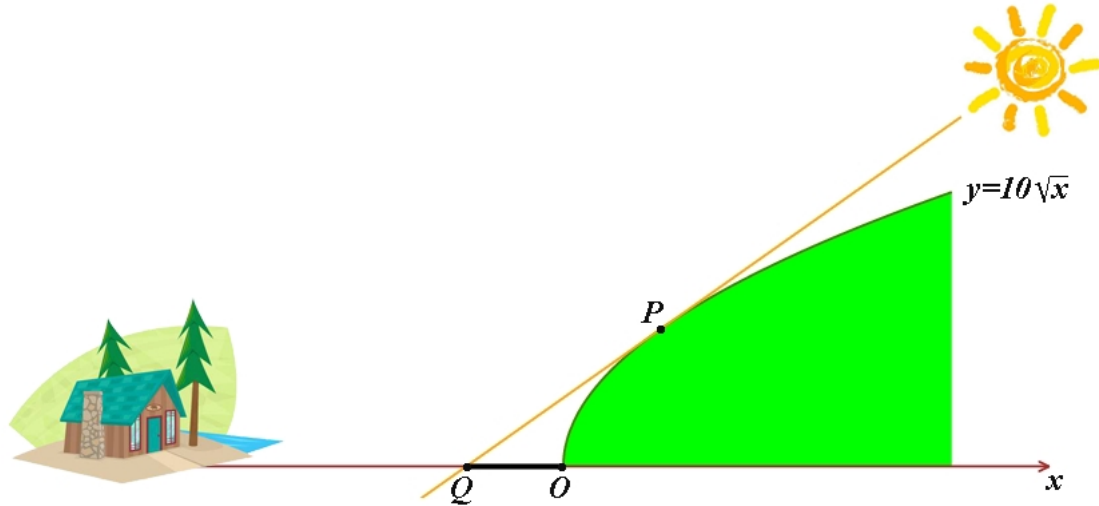
2b. In one of the following figures, the graphs of two functions f and g together with their derivatives f' and g' are shown; while in the other, the graphs of two functions h and k together with their second derivatives h'' and k'' are shown. Identify each by filling in the boxes with f , g , f' , g' , h , k , h'' , and k'' .



3. Find $\left. \frac{d^2 y}{dx^2} \right|_{(x,y)=(\pi/2, \pi/6)}$ if y is a differentiable function of x satisfying the equation:

$$2 \sin^2(x + y) = \sin x + \sin y$$

4. You have a cabin on the negative x -axis. A hill whose height is given by $y = 10\sqrt{x}$ for $x \geq 0$ lies to the west along the positive x -axis. (All coordinates are measured in meters.) As the sun starts to set, the hill casts a shadow as shown in the figure. Determine how fast the shadow is approaching your cabin at the moment when the sunrays are making a 30° angle with the horizontal and this angle is decreasing at a rate of $1/4^\circ/\text{min}$. Express your answer in units of meters per minute.



Fall 2018 Midterm II

1. A function f that is twice-differentiable on the entire real line satisfies the following conditions:

- ❶ $f(0) = 0$, $f(\sqrt[3]{2}) = 3\sqrt[3]{4}$, $f(2) = 8$, $f(\sqrt[3]{20}) = 0$
- ❷ $f'(x) < 0$ for $x < 0$ and for $x > 2$; $f'(x) > 0$ for $0 < x < 2$
- ❸ $f''(x) > 0$ for $x < \sqrt[3]{2}$; $f''(x) < 0$ for $x > \sqrt[3]{2}$

a. Sketch the graph of $y = f(x)$ making sure that all important features are clearly shown.

b. Fill in the boxes to make the following a true statement. No explanation is required.

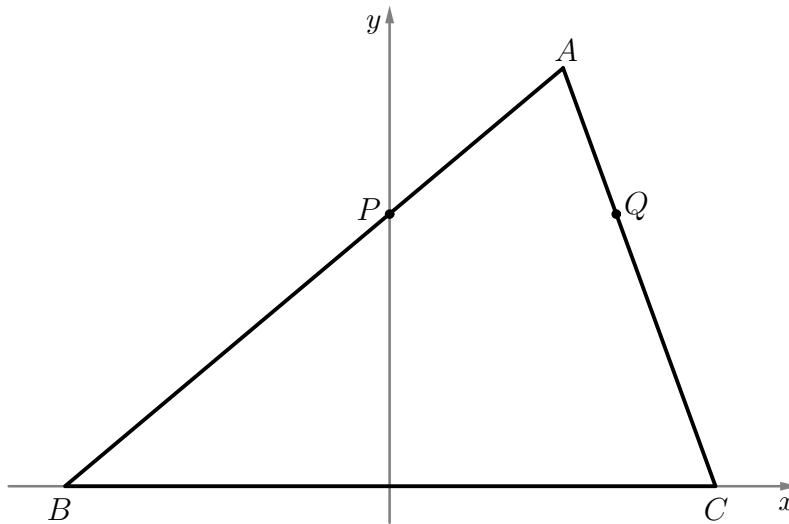
The function $f(x) = ax^b + cx^d$ satisfies the conditions ❶-❸ if a , b , c and d are chosen as

$$a = \boxed{}, \quad b = \boxed{}, \quad c = \boxed{} \quad \text{and} \quad d = \boxed{}.$$

2. An isosceles triangle ABC with $|AB| = |BC|$ in the xy -plane satisfies the following conditions:

- ❶ The side $[BC]$ lies along the x -axis
- ❷ The side $[AB]$ passes through the point $P(0, 6)$
- ❸ The side $[AC]$ passes through the point $Q(5, 6)$

Determine the largest and smallest possible values of the area of the triangle ABC .



3a. Suppose that a continuous function f satisfies the equation

$$f'(x) = f(x) \int_0^x f(t) dt$$

for all x .

❶ In this part assume that $f(0) = -1$. Show that f has a critical point at $x = 0$, and determine whether it is a local maximum, a local minimum, or neither.

❷ In this part assume that $f''(2) = 0$. Express $f(2)$ in terms of $A = f'(2)$ only.

3b. Suppose that a function g with continuous second derivative on $[0, 1]$ satisfies:

$$g(0) = 2, \quad g'(0) = 3, \quad g''(0) = 4, \quad g(1) = -4, \quad g'(1) = -3, \quad g''(1) = -2$$

Evaluate $\int_0^1 g(x) g'(x) (1 + g'(x)^2 + g(x)g''(x)) dx$.

4. Evaluate the following integrals:

a. $\int_{-1}^2 (|2x - 3| - 2|x| + 3) dx$

b. $\int \frac{\tan \theta \sec^2 \theta}{(1 - \tan^2 \theta)^2} d\theta$

Fall 2018 Final

1. In each of the following, if the given statement is true for all functions f that are defined on $(-\infty, \infty)$, then mark the $\square$ to the left of TRUE with a **X**; otherwise, mark the $\square$ to the left of FALSE with a **X** and give a counterexample.

a. If $f(2x) = f(x)$ for all x , then f is constant on $(-\infty, \infty)$.

$\square$ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. If f is continuous on $(-\infty, \infty)$, then f has a derivative on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. If f is continuous on $(-\infty, \infty)$, then f has an antiderivative on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. If f is continuous on $(-\infty, \infty)$, then $\int f(x) dx = \frac{1}{2} f(x)^2 + C$ on $(-\infty, \infty)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. If $\int_{-1}^1 f(x) dx = 0$, then $\int_{-1}^1 f(x)^3 dx = 0$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

2. Evaluate the following limits:

a. $\lim_{x \rightarrow 0} \frac{1 - x^2/2 - \cos x}{x^4}$

b. $\lim_{x \rightarrow 0} \frac{x - \tan x}{x^3}$

3. Evaluate the following integrals:

a. $\int_0^\pi \sin x \cos x \sin(\pi \cos x) dx$

b. $\int \frac{\tan \theta}{(1 - \tan^2 \theta)^2} d\theta$

4. The coordinates (x_c, y_c) of the *center* of a region R contained in the first quadrant of the xy -plane are defined by

$$x_c = \frac{W}{2\pi A} \quad \text{and} \quad y_c = \frac{V}{2\pi A}$$

where

- A is the area of R ,
- V is the volume of the solid generated by revolving R about the x -axis, and
- W is the volume of the solid generated by revolving R about the y -axis.

Let R be the region lying between the graph of $y = (x^2 + 1)^{-3/2}$ and the x -axis for $x \geq 0$. Compute one of the coordinates $\square x_c$ or $\square y_c$ of the center of R .

[Indicate the one you are computing by putting a $\checkmark$ in the $\square$ to the left of it.]

5. Find all values of the constant k for which the function $f(x) = \frac{k}{(e^x + e^{-x})^2}$ satisfies the equation

$$f'(x) = f(x) \int_0^x f(t) dt$$

for all x .

Fall 2017 Midterm I

1. Evaluate the following limits without using L'Hôpital's Rule.

a. $\lim_{x \rightarrow 2} \frac{x^3 - 3x^2 + 4}{\sqrt{x^3 + 8x} - \sqrt{5x^2 + 4}}$

b. $\lim_{x \rightarrow 0} \frac{\cos 8x - \cos 5x \cos 3x}{x^2}$

2. Suppose that f is a differentiable function with $f(1) = 1/2$, $f'(1) = 3$, $f(2) = 7$, $f'(2) = -9$, $f(5) = 1/2$, $f'(5) = -4$.

a. Find an equation for the tangent line to the graph of $y = f(x)$ at the point with $x = 2$.

b. Compute $f(1.1)$ approximately.

c. Find $\left. \frac{du}{dt} \right|_{(t,u)=(2,1)}$ if u is a differentiable function of t satisfying the relation $f(tf(u)) = f(1 + t^2 u^2)$.

3. A function f satisfies the following conditions:

❶ f is differentiable on $(-\infty, \infty)$.

❷ $f(x) = 5x + 4$ for $x \leq -1$.

❸ $f(x) = x$ for $x \geq 1$.

a. Show that there is a real number A such that $f(A) = 0$.

b. Show that there is a real number B such that $f'(B) = 2$.

c. Give an example of a function satisfying the conditions ❶, ❷ and ❸ by explicitly defining $f(x)$ for $-1 < x < 1$.

4. At a certain moment the sides of a triangle are changing as follows:

❶ The length of the first side is 3 m and increasing at a rate of 5 m/s.

❷ The length of the second side is 8 m and increasing at a rate of 2 m/s.

❸ The length of the third side is 7 m and increasing at a rate of 1 m/s.

Determine how fast the angle between the first two sides is changing at this moment. Express your answer in units of $^\circ/\text{s}$.

Fall 2017 Midterm II

1. A function f that is defined and continuous for $x \neq 0$ satisfies the following conditions:

❶ $f(2 - \sqrt{2}) = \sqrt[3]{1 - 1/\sqrt{2}}$, $f(2/3) = 0$, $f(2) = \sqrt[3]{2}$, $f(2 + \sqrt{2}) = \sqrt[3]{1 + 1/\sqrt{2}}$

❷ $\lim_{x \rightarrow 0^-} f(x) = -\infty$, $\lim_{x \rightarrow 0^+} f(x) = \infty$, $\lim_{x \rightarrow -\infty} f(x) = 0$, $\lim_{x \rightarrow \infty} f(x) = 0$

❸ $f'(x) < 0$ for $x < 2/3$ and $x \neq 0$, and for $x > 2$; $f'(x) > 0$ for $2/3 < x < 2$

❹ $\lim_{x \rightarrow (2/3)^-} f'(x) = -\infty$, $\lim_{x \rightarrow (2/3)^+} f'(x) = \infty$

❺ $f''(x) < 0$ for $x < 0$, and for $2 - \sqrt{2} < x < 2 + \sqrt{2}$ and $x \neq 2/3$; $f''(x) > 0$ for $0 < x < 2 - \sqrt{2}$ and for $x > 2 + \sqrt{2}$

a. Sketch the graph of $y = f(x)$ making sure that all important features are clearly shown.

b. Fill in the boxes to make the following a true statement. No explanation is required.

The function $f(x) = (ax + b)^c x^d$ satisfies the conditions ❶-❺ if a , b , c and d are chosen as

$$a = \boxed{}, \quad b = \boxed{}, \quad c = \boxed{} \quad \text{and} \quad d = \boxed{}.$$

2. Find the largest and smallest possible values of the area of the triangle cut off from the first quadrant by a line L which is tangent to the parabola $y = 15 - 2x - x^2$ at a point in the first quadrant.

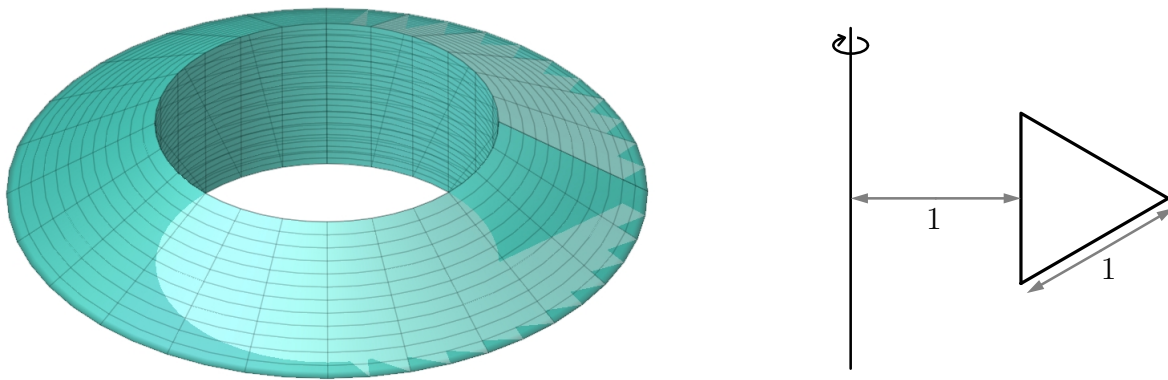
3. Evaluate the following integrals.

a. $\int_0^1 x^{4035} (x^4 + 1)^{2017} (3x^4 + 1) dx$

b. $\int \frac{\sin x - \cos x}{1 + \sin 2x} dx$

4a. A solid is generated by revolving an equilateral triangle with unit side length about a line at a unit distance from one of its sides as shown in the figure. Express the volume V of the solid as an integral using either ☐ the **washer** method or ☐ the **cylindrical shells** method by carefully defining your variable of integration, drawing a typical rectangle that generates a washer or a cylindrical shell and showing the relevant lengths and distances on the figure.

[Indicate your method by **X**ing the corresponding ☐. Do not evaluate the integral!]



4b. We start a rabbit farm with a pair of rabbits. Assume that at any moment the rabbit population is increasing at a rate proportional to the square of the rabbit population at that moment. Show that we will have infinitely many rabbits after a finite period of time.

Fall 2017 Final

1. Find $\left. \frac{d^2 y}{dx^2} \right|_{(x,y)=(2,1/2)}$ if y is a differentiable function of x satisfying the equation:

$$\sin(\pi xy) = \sin(\pi x) + \cos(\pi y)$$

2. Evaluate the following.

a. $\int \frac{\tan x}{1 + \sec 2x} dx$

b. $\int_1^2 \ln(x^2 + x) dx$

c. $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{in^2}{n^4 + i^4}$

3. In each of the following, if the given statement is true for all functions f that are defined on $(-\infty, \infty)$, then mark the $\square$ to the left of TRUE with a **X**; otherwise, mark the $\square$ to the left of FALSE with a **X** and give a counterexample.

a. If $\lim_{x \rightarrow 0} f(x) = f(0)$, then $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x}$ exists.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. If the graph of $y = f(x)$ has an inflection point at $(0, f(0))$ and $f(0) \neq 0$, then the graph of $y = 1/f(x)$ has an inflection point at $(0, 1/f(0))$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. If f is continuous on $[-1, 1]$, then $\int_{-1}^1 (f(x))^2 dx = \left(\int_{-1}^1 f(x) dx \right)^2$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. If f is continuous on $(-\infty, \infty)$ and $\int_{-x}^x f(t) dt = 0$ for all $x > 0$, then f is an odd function.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. If f is integrable on $[-a, a]$ and $\int_{-a}^a f(t) dt = 0$ for all $a > 0$, then $f(0) = 0$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

4. For a positive continuous function f on $(-\infty, \infty)$, let

- $R(a)$ be the region between the graph of $y = f(x)$ and the x -axis for $x \leq a$,
- $A(a)$ be the area of $R(a)$, and
- $V(a)$ be the volume of the solid generated by revolving $R(a)$ about the x -axis.

a. Show that if $f(x) = e^{2x/\pi}$, then $V(a) = (A(a))^2$ for all a .

b. Show that $f(x) = e^{2x/\pi}$ is the only positive continuous function on $(-\infty, \infty)$ for which $A(a)$ and $V(a)$ are finite and satisfy $V(a) = (A(a))^2$ for all a , and $f(0) = 1$.

5. The towns A and B lie inland and the town C lies on the coast as shown in *Figures 1* and *2*. A port P will be built on the coast and connected to the towns A , B , C with straight roads. The cost of constructing the roads connecting P to A and P to B is 10^5 ₺/km, and the cost of constructing the road connecting P to C is $k \times 10^5$ ₺/km. The port will be built at the location for which the total cost of constructing these roads is the smallest.

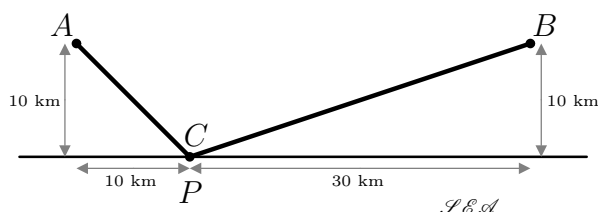


Figure 1

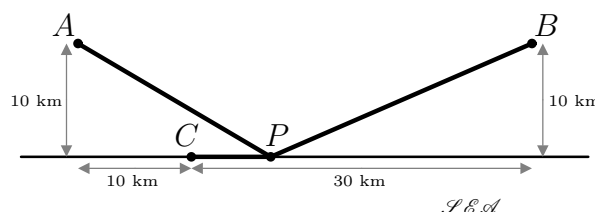


Figure 2

For some values of k the smallest total cost will be achieved when P is built at C , as in *Figure 1*; whereas for others it will be achieved, if at all, when P is built somewhere else, as in *Figure 2*. Determine the smallest value of k for which the lowest total cost is achieved by building P at C .

Fall 2016 Midterm I

1. Consider the functions

$$f(x) = \sqrt{x^2 + a} + bx \quad \text{and} \quad g(x) = x^3 + cx^2 + d$$

where a, b, c, d are nonzero real constants.

a. Give values to a, b, c, d by filling in the boxes below so that the resulting functions satisfy $f(1) = 0$ and $g(1) = 0$. No explanation is required in this part.

$$a = \boxed{} \quad b = \boxed{} \quad c = \boxed{} \quad d = \boxed{}$$

b. Compute the limit $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$ where a, b, c, d have the values given in **Part a**.

2. In each of the following, if the given statement is true for all functions f that are defined on the entire real line, then mark the $\square$ to the left of TRUE with a $\mathbf{X}$; otherwise, mark the $\square$ to the left of FALSE with a $\mathbf{X}$ and give a counterexample. No explanation is required.

a. If f is continuous on $(-\infty, \infty)$, then $f'(\pi)$ exists.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. If $f(-1) = -1$ and $f(1) = 1$, then $f(c) = 0$ for some c in $(-1, 1)$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. If $f''(x) = -f(x)$ for all x , then $f(x) = \sin x$ or $f(x) = \cos x$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. If f is differentiable on $(-\infty, \infty)$, then $f'(\pi)$ exists.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. If $\lim_{x \rightarrow 0} |f(x)| = 1$, then $\lim_{x \rightarrow 0} f(x) = 1$ or $\lim_{x \rightarrow 0} f(x) = -1$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

3. Suppose that f is a twice-differentiable function satisfying the following conditions:

- $y = x/2 + 1$ is an equation for the tangent line to the graph of $y = f(x)$ at the point with $x = 4$.
- $y = x/4 - 3$ is an equation for the tangent line to the graph of $y = f(x)$ at the point with $x = 8$.
- $y = -x/3 + 2$ is an equation for the tangent line to the graph of $y = f(x)$ at the point with $x = 12$.

Consider the function $g(x) = (f(x^3))^2$.

- a. Find an equation for the tangent line to the graph of $y = g(x)$ at the point with $x = 2$.
- b. Suppose that $g''(2) = 0$. Find $f''(8)$.

4. Air is escaping from a balloon that is hanging at the end of a 75 cm long string attached to a nail on a wall. Assume that at all times the balloon has the shape of a sphere which is tangent to the wall, and the string extends along a line which passes through the center of the sphere and lies in a vertical plane perpendicular to the wall.

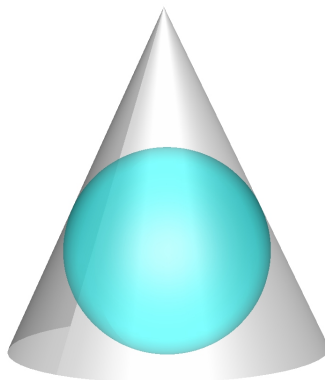
Suppose that at a certain moment the volume of the balloon is decreasing at a rate of 1800π cm³/s and its surface area is decreasing at a rate of 240π cm²/s. Determine how fast the angle between the string and the wall is changing at this moment. Express your answer in units of degrees per second.

Fall 2016 Midterm II

1a. Sketch the graph of $y = 1/(x^2 + 1)$ by computing y' and y'' , and determining their signs; finding the critical points, the inflections points, the intercepts, and the asymptotes; and clearly labeling them in the picture.

1b. Show that $\left| \frac{1}{a^2 + 1} - \frac{1}{b^2 + 1} \right| \leq \frac{3\sqrt{3}}{8} |a - b|$ for all real numbers a, b .

2a. Find the dimensions of the cone with the smallest possible lateral surface area that surrounds a sphere of radius 1.



2b. Having solved **Part 2a** correctly, after the exam you run into a friend who mistakenly minimized the total surface area $A = \pi r\sqrt{r^2 + h^2} + \pi r^2$ instead of the lateral surface area $A = \pi r\sqrt{r^2 + h^2}$.

Let h_y and r_y be respectively the height and the radius you found, and let h_f and r_f be the ones your friend did. Assuming that your friend also solved the corresponding problem correctly, which one of the following is true? No explanation is required.

- | | | |
|--|--|--|
| ☐ $h_y > h_f$ and $r_y > r_f$ | ☐ $h_y > h_f$ and $r_y = r_f$ | ☐ $h_y > h_f$ and $r_y < r_f$ |
| ☐ $h_y = h_f$ and $r_y > r_f$ | ☐ $h_y = h_f$ and $r_y = r_f$ | ☐ $h_y = h_f$ and $r_y < r_f$ |
| ☐ $h_y < h_f$ and $r_y > r_f$ | ☐ $h_y < h_f$ and $r_y = r_f$ | ☐ $h_y < h_f$ and $r_y < r_f$ |

3a. Determine $f(5)$ if f is a continuous function that satisfies $\int_0^{4x+\sin(\pi x)} f(t) dt = x^2$ for all x .

3b. Determine $f(5)$ if f is a function that satisfies $\int_0^{f(x)} t^2 dt = 4x + \sin(\pi x)$ for all x .

4. Consider the region R between the graph of $y = x \sin(x^3)$ and the x -axis for $0 \leq x \leq \pi^{1/3}$.

a. Find the volume V of the solid generated by revolving R about the x -axis.

b. Find the volume W of the solid generated by revolving R about the y -axis.

Fall 2016 Final

1. Find $\left. \frac{d^2 y}{dx^2} \right|_{(x,y)=(2,1)}$ if y is a differentiable function of x satisfying the equation $1 + \ln(2x - 3y) = x^2 - 3y^2$.

2. In each of the following, indicate whether the given statement is TRUE or FALSE by marking the corresponding ☐ with a **X**, and then **explain** why it is true or false.

a. $\int \sin^3 x dx = \frac{\sin^4 x}{4} + C$ ☐ TRUE ☐ FALSE

b. $\int \sin^3 x dx = \frac{\sin^4 x}{4 \cos x} + C$ ☐ TRUE ☐ FALSE

c. $\int \sin^3 x dx = \frac{1}{6} \cos x \cos 2x - \frac{5}{6} \cos x + C$ ☐ TRUE ☐ FALSE

d. $\int \frac{dx}{x^2 + 1} = \frac{\ln(x^2 + 1)}{2x} + C$ ☐ TRUE ☐ FALSE

e. $\int \frac{dx}{x^2 + 1} = \arcsin\left(\frac{x}{\sqrt{x^2 + 1}}\right) + C$ ☐ TRUE ☐ FALSE

3. Some students believe that Bilkent Math 101 exams get more difficult as time passes. This is in fact true. The difficulty level $H(t)$ of these exams satisfies the differential equation

$$\frac{dH}{dt} = H^a$$

with the initial condition $H(0) = 1$, where t is time measured from Fall 1986 in academic years and a is a constant whose value is a secret.

There is a quatrain in Nostradamus's *Les Propheties* which can be interpreted to be about Bilkent Math 101 exams.

a. According to one interpretation, the exams will become infinitely difficult in Fall 2021. Accepting this interpretation, find the difficulty level of the exams in Fall 2016.

b. According to another interpretation, the exams will be twice as difficult in Fall 2021 as they were in Fall 1986. Show that a must satisfy $-8 < a < -7$ if this is the case.

4. Evaluate the limit $\lim_{x \rightarrow 0} \frac{e^{x - \frac{1}{2}x^2} - \arctan x - 1}{x^4}$.

5. Evaluate the following integrals.

a. $\int_0^1 x^5 \sqrt[4]{1-x^3} dx$

b. $\int \sin 2x \tan^2 x dx$

Fall 2015 Midterm I

1. Suppose you are told that a function f satisfies the condition:

$$x^3 - 3x - 3 \leq f(x) \leq 9x^2 - 27x + 17 \quad \text{for } |x - 2| < 1$$

a. Is it possible to determine the value of $\lim_{x \rightarrow 2} f(x)$ using only this information?

b. At this point you peek at your neighbor's paper, and see that Yes is marked in **Part 1c** and then the solution starts with the sentence:

Taking derivatives of all sides, we obtain

$$(x^3 - 3x - 3)' \leq f'(x) \leq (9x^2 - 27x + 17)'$$

for $|x - 2| < 1$.

You immediately realize that this cannot be true, because

$$\frac{d}{dx}(x^3 - 3x - 3) = \boxed{} \quad \text{and} \quad \frac{d}{dx}(9x^2 - 27x + 17) = \boxed{},$$

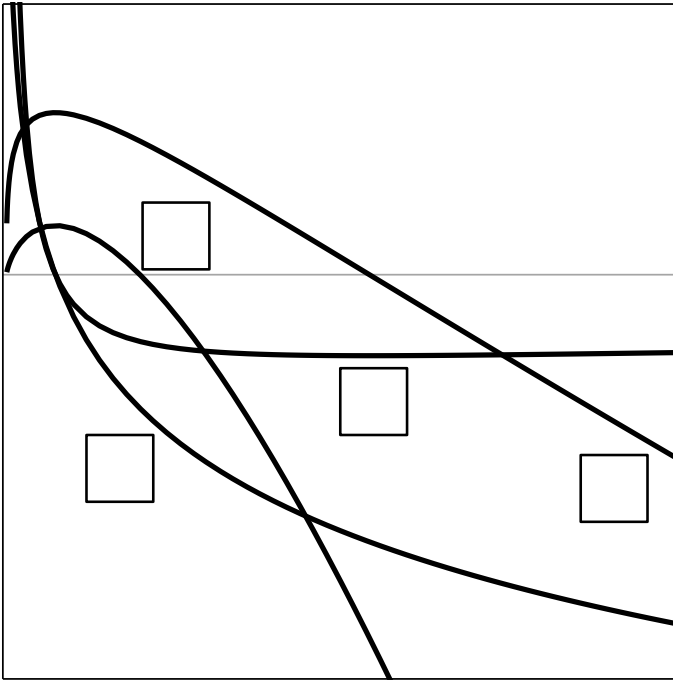
and if we take $a = \boxed{}$, then $|a - 2| < 1$ but

$$\left. \frac{d}{dx}(x^3 - 3x - 3) \right|_{x=a} = \boxed{} > \boxed{} = \left. \frac{d}{dx}(9x^2 - 27x + 17) \right|_{x=a}.$$

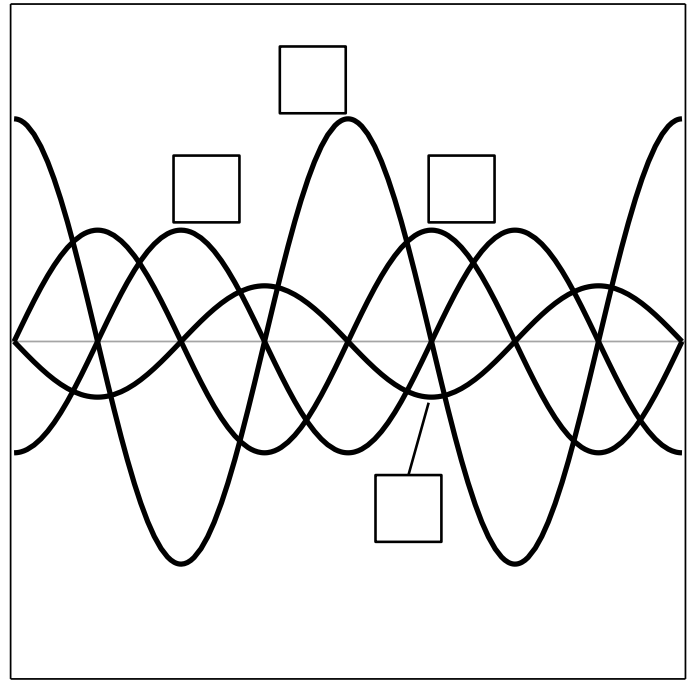
c. Is it possible to determine the value of $f'(2)$ using only this information?

2. In each of the following figures, the graphs of two functions f and g together with their derivatives f' and g' are shown. Identify each by filling in the boxes with f , g , f' and g' .

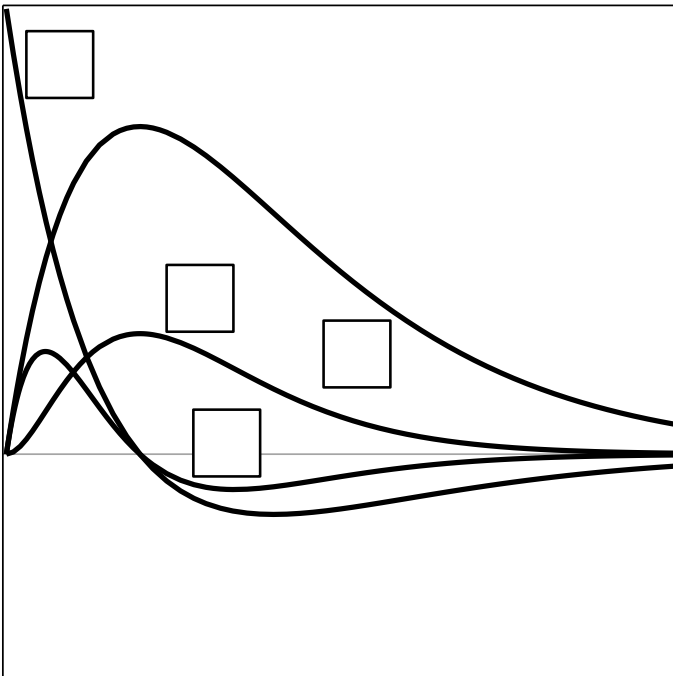
a.



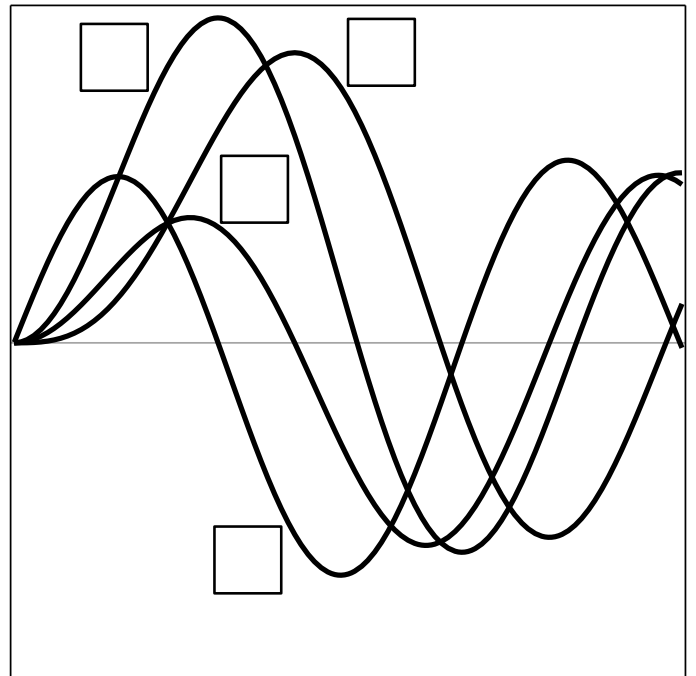
b.



c.



d.



3a. Compute $\frac{d}{dx} \sin(x + \sin(x + \sin x))$.

3b. Find an equation for the tangent line to the graph of $y = \tan\left(\frac{\pi}{\sqrt{25-x^2}}\right)$ at the point with $x = 3$.

4. At a certain moment the dimensions of a cylinder are changing as follows:

- The radius is decreasing at a rate of 3 cm/s.
- The height is increasing at a rate of 15 cm/s.
- The volume is increasing at a rate of 2π cm³/s.
- The surface area is increasing at a rate of 7π cm²/s.

Find the radius at this moment.

Fall 2015 Midterm II

1. A function f that is continuous on $(-\infty, \infty)$ and twice-differentiable for $x \neq A$ satisfies the following conditions:

- ① $f(-\sqrt{5}-1) = B$, $f(0) = -2$, $f(\sqrt{5}-1) = C$, $f(4) = D$, $f(A) = 0$ where $B < D$
- ② The line $y = x - 2$ is a slant asymptote of the graph of $y = f(x)$ both as $x \rightarrow -\infty$ and as $x \rightarrow \infty$
- ③ $\lim_{x \rightarrow A} f'(x) = \infty$
- ④ $f'(x) > 0$ for $x < 0$, and for $x > 4$ and $x \neq A$; $f'(x) < 0$ for $0 < x < 4$
- ⑤ $f''(x) > 0$ for $x < -\sqrt{5}-1$ and for $\sqrt{5}-1 < x < A$; $f''(x) < 0$ for $-\sqrt{5}-1 < x < \sqrt{5}-1$ and for $x > A$

a. Sketch the graph of $y = f(x)$. Make sure to clearly show all important features.

b. Fill in the boxes to make the following a true statement. No explanation is required.

The function $f(x) = \sqrt[3]{ax^3 + bx^2 + cx + d}$ satisfies the conditions ①–⑤ for suitable real numbers

A, B, C, D if the constants a, b, c and d are chosen as

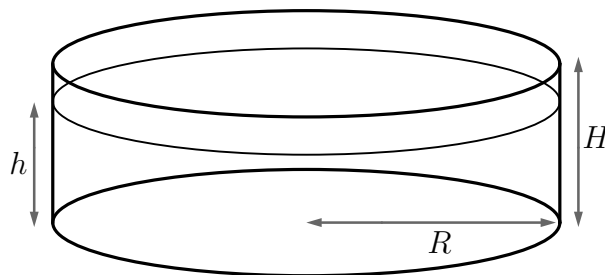
$$a = \boxed{}, \quad b = \boxed{}, \quad c = \boxed{} \quad \text{and} \quad d = \boxed{}.$$

2. A pool, like the one in front of the Faculty of Science Building A, loses water from its sides and its bottom due to seepage, and from its top due to evaporation. For a pool with radius R and depth H in meters, the rate of this loss in m^3/hour is given by an expression of the form

$$aR^2 + bR^2h + cRh^2$$

where h is the depth of the water in meters, and a , b , c are constants independent of R , H and h . Due to this loss, water must be pumped into the pool to keep it at the same level even when the drains are closed.

Suppose that $a = 1/300$ m/hour and $b = c = 1/150$ $1/\text{hour}$. Find the dimensions of the pool with a volume of $45\pi \text{ m}^3$ which will require the water to be pumped at the slowest rate to keep it completely full.



3a. Find the average value of $|x|$ on the interval $[-1, 2]$.

3b. Suppose that a continuous function f satisfies the equation

$$f(x) = x^2 - x + (1-x) \int_0^x t^2 f(t) dt + x \int_x^1 (t-t^2) f(t) dt$$

for all x . Express $f''(1/2)$ in terms of $A = f(1/2)$.

4. An island has the shape of a $10 \text{ hm} \times 10 \text{ hm}$ square and its landscape consists of a mountain whose height h at a horizontal distance x from the shore is given by $h = x^2$ where both h and x are measured in hectometers ($=\text{hm}$). Let V be the volume of the mountain.

a. Express V as an integral with respect to h by considering cutting the mountain into slices as shown in *Figure a*.

b. Express V as an integral with respect to x by considering cutting the mountain into shells as shown in *Figure b*.

c. Compute V .

[The figures are on the next page.]

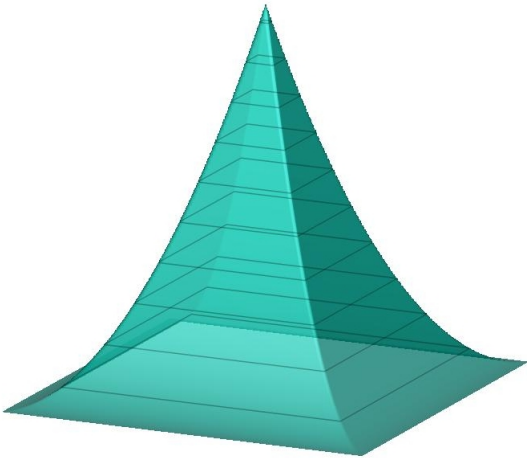


Figure a:

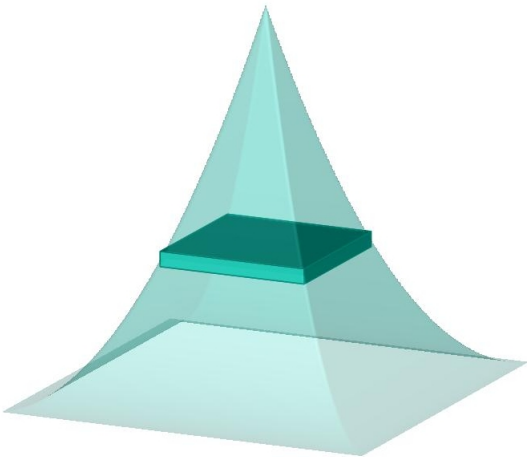
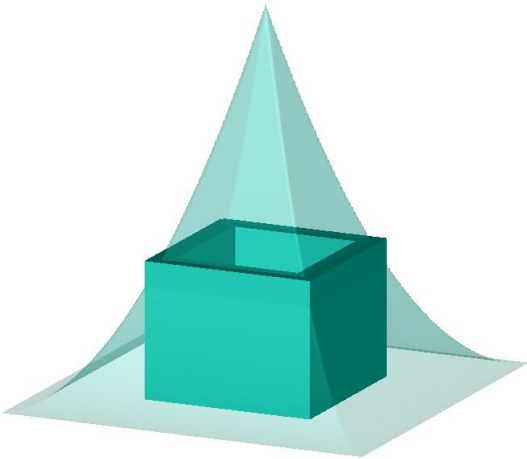


Figure b:



Fall 2015 Final

1. Let $f(x) = x^5 - 3x^3 - x + 4$.

a. Compute $f'(x)$.

b. Find the x -coordinates of all points on the graph of $y = f(x)$ where the tangent line is horizontal.

c. Show that the equation $f(x) = 0$ has at most three real solutions.

d. Show that the equation $f(x) = 0$ has at least three real solutions.

2. Find $\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(\sqrt{e}, 1/e)}$ if y is a differentiable function of x satisfying the equation:

$$\ln(xy) = (\ln x)(\ln y)$$

3. A pool, like the one in front of the Faculty of Science Building A, loses water from its sides and its bottom due to seepage. For a pool with radius R and depth H in meters, the rate of this loss in m^3/hour is given by

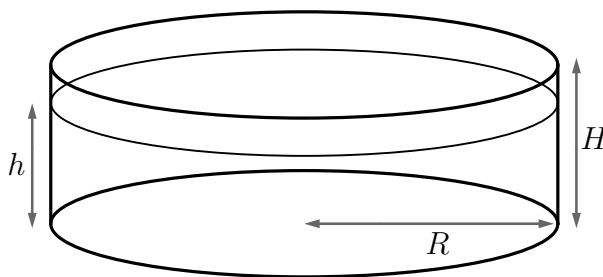
$$\frac{dV}{dt} = -aR^2h - bRh^2$$

where V is the volume of the water in cubic meters, t is the time in hours, h is the depth of the water in meters, and a and b are constants independent of R , H and h .

Consider a pool with $H = 1$ m, $R = 6$ m and $a = b = \pi/500$ 1/hour.

a. Find the depth of the water h as a function of time t if $h = 1$ m when $t = 0$ hour.

b. The term aR^2h on the right side of the equation represents the rate of loss due to seepage through the bottom of the pool. Find the total volume of the water that seeps through the bottom while the initially full pool completely empties.



4. Evaluate the limit $\lim_{x \rightarrow \infty} \left(x \left(x \left(x \left(e^{1/x} - 1 \right) - 1 \right) - \frac{1}{2} \right) \right)$.

5. Evaluate the following integrals.

a. $\int \frac{\tan^2 x}{1 + \sin x} dx$

b. $\int_1^8 \sqrt{4 - \frac{3}{x^2} + \frac{1}{x^3}} dx$

Fall 2014 Midterm I

1. In each of the following, if the given statement is true for all functions $f(x)$ that are defined for all $x \neq 0$, then mark the $\square$ to the left of TRUE with a $\mathbf{X}$; otherwise, mark the $\square$ to the left of FALSE with a $\mathbf{X}$ and give a counterexample.

a. $\lim_{x \rightarrow 0} f(1/x)$ does not exist.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. $\lim_{x \rightarrow 0} xf(1/x)$ exists.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. If $-x^2 < f(x) < x^2$ for all $x \neq 0$, then $\lim_{x \rightarrow 0} f(x) = 0$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. If $-1 \leq f(x) \leq 1$ for all $x \neq 0$, then $-1 \leq \lim_{x \rightarrow 0} f(x) \leq 1$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. If $\lim_{x \rightarrow 0} \frac{f(x)}{x}$ does not exist, then $\lim_{x \rightarrow 0} f(x) \neq 0$.

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$



2. Consider the function $f(x) = \sin\left(\frac{\pi}{\sin(\pi/x)}\right)$.

a. Find an equation of the tangent line to the graph of $y = f(x)$ at the point with $x = 6$.

b. Choose one of the options by putting a ☒ in the box to the left of it, and then explain your reasoning in one sentence.

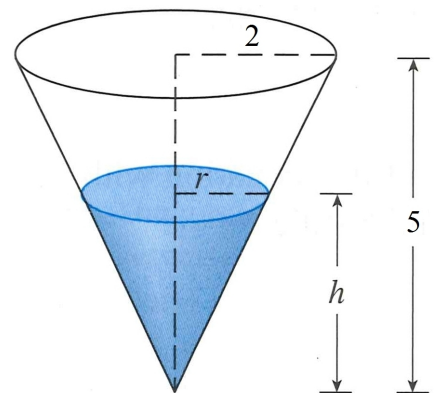
I expect $f(101\pi)$ to be ☐ positive ☐ zero ☐ negative, because:

3. Consider the curve defined by the equation $x^4 + y^4 = 4xy + 52$.

a. Find all points on the curve at which the tangent line is horizontal.

b. Find $\frac{d^2y}{dx^2}$ at all points where the tangent line is horizontal.

4. A water tank has the shape of an upside-down cone with radius 2 m and height 5 m. The water is running out of the tank through a small hole at the bottom. Assume that the speed of the water flowing through the hole is proportional to the square root of the depth of the water in the tank.



a. In this part, suppose that the water is running out at a rate of $3 \text{ m}^3/\text{min}$ when the depth of the water in the tank is 4 m. Find the rate at which the water level is changing at this moment.

b. In this part, suppose that the water level is falling at a rate of $1/3 \text{ m/min}$ when the tank is full. Find the rate at which the water level is changing when the depth of the water in the tank is 4 m.

Fall 2014 Midterm II

1. A twice-differentiable function f on $(-\infty, \infty)$ satisfies the following conditions:

① $f(-2) = A$, $f(-1) = B$, $f(0) = C$, $f(1/2) = -1$, $f(D) = 0$ where $A < C$

② $\lim_{x \rightarrow -\infty} f(x) = -1$, $\lim_{x \rightarrow \infty} f(x) = 1$

③ $f'(x) < 0$ for $x < -1$, $f'(x) > 0$ for $x > -1$,

④ $f''(x) < 0$ for $x < -2$ and for $x > 1/2$, $f''(x) > 0$ for $-2 < x < 1/2$

a. Sketch the graph of $y = f(x)$. Make sure to clearly show all important features.

b. Fill in the boxes to make the following a true statement.

The function $f(x) = \frac{ax+b}{\sqrt{x^2+c}}$ satisfies the conditions ①-④ for suitable real numbers A, B, C, D if

the constants a, b and c are chosen as

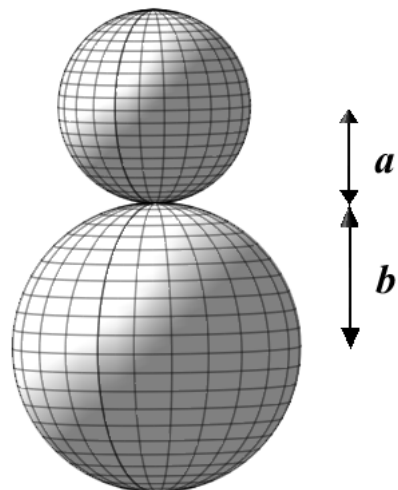
$$a = \boxed{}, \quad b = \boxed{} \quad \text{and} \quad c = \boxed{}.$$

2. A snowman is an anthropomorphic sculpture made from snow as well as some pieces of coal, a carrot, a hat and a scarf. For the purposes of this question, we consider a snowman to consist of a spherical head of radius a and a spherical body of radius b , and we also assume that the snow does not melt and its density does not change while it is being sculpted.

The research shows that the *cuteness* $\mathcal{K}$ of a snowman is given by

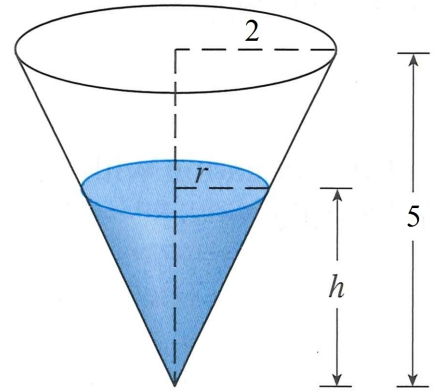
$$\mathcal{K} = \begin{cases} \left(\frac{a}{b}\right)^2 \left(1 - \frac{a}{b}\right) (a^2 + ab + b^2) & \text{if } 0 \leq a < b, \\ 0 & \text{if } 0 < b \leq a. \end{cases}$$

Find the dimensions of the cutest snowman that can be built with $4\pi/3 \text{ m}^3$ of snow.



3. A water tank has the shape of an upside-down cone with radius 2 m and height 5 m. The water is running out of the tank through a small hole at the bottom. Assume that the speed of the water flowing through the hole is proportional to the square root of the depth of the water in the tank.

If it takes 3 minutes for the depth of the water to decrease from 5 m to 4 m, find how long it takes for the full tank to completely drain.



4. Evaluate the following integrals.

a. $\int_0^1 \frac{dx}{(1 + \sqrt{x})^4}$

b. $\int x^3 \sqrt{x^2 + 1} dx$

Fall 2014 Final

1. Evaluate the following limits.

a. $\lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{x^3}$

b. $\lim_{x \rightarrow 0} \frac{x - \tan x}{e^x - e^{-x} - 2x}$

2. In each of the following, if the given statement is true for all functions f that are continuous on the entire real line, then mark the ☐ to the left of TRUE with a **X**; otherwise, mark the ☐ to the left of FALSE with a **X** and give a counterexample.

a. $\int_0^1 |f(x)| dx = \left| \int_0^1 f(x) dx \right|$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. $\frac{d}{dx} \int_0^x f(t) dt = f(x) - f(0)$ for all x

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. $\frac{d}{dx} \int_0^1 f(t) dt = f(1) - f(0)$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. $\frac{d}{dt} \int_0^x f(t) dt = f(x) - f(0)$ for all x

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. $\int_0^1 f(x)^2 dx = \frac{f(1)^3 - f(0)^3}{3}$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

f. $\int f(x)^2 dx = \frac{f(x)^3}{3f'(x)} + C$ for all x for which $f'(x) \neq 0$

☐ TRUE

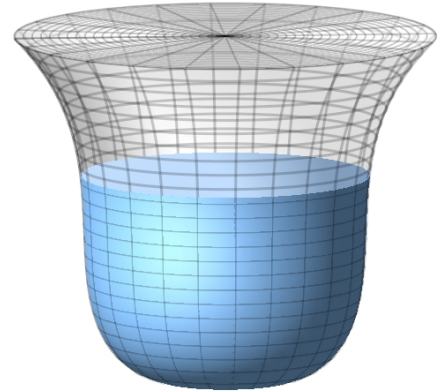
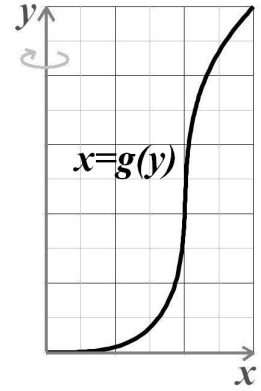
☐ FALSE, because it does not hold for $f(x) =$

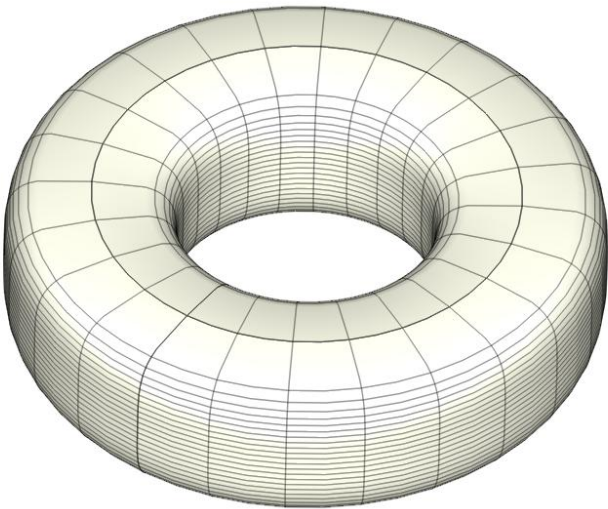
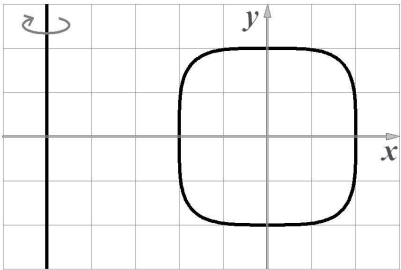
3. The side surface of a water tank has the shape generated by revolving the graph of a continuous nonnegative function $x = g(y)$ for $0 \leq y \leq 5$ with $g(0) = 0$ and $g(5) = 3$ about the y -axis where all units are in meters. Assume that:

① As water runs out of a small hole at the bottom of tank, the speed of the water flowing through the hole at any moment is proportional to the square root of the depth of the water in the tank at that moment.

② The function g is chosen in such a way that the depth of the water changes at a constant rate at all times.

Find the volume of the tank.





[These figures belong to the question on the next page.]

4. Let V be the volume of the water-dropper shown in the figure which has the shape obtained by revolving the curve $x^4 + y^4 = 1$ about the line $x = -5/2$ where all units are in centimeters.

a. Express V as an integral using the cylindrical shells method.

$$V = 2\pi \int_{\boxed{}}^{\boxed{}} \boxed{} \, dx$$

b. Express V as an integral using the washer method.

$$V = \pi \int_{\boxed{}}^{\boxed{}} \boxed{} \, dy$$

c. Show that the improper integral $\int_0^1 u^{-3/4}(1-u)^{1/4} du$ converges.

d. Express V in terms of $A = \int_0^1 u^{-3/4}(1-u)^{1/4} du$.

5. Evaluate the following integrals.

a. $\int \cos 2x \cos^3 x \, dx$

b. $\int_0^{\pi/4} \frac{x \sin x}{\cos^3 x} \, dx$

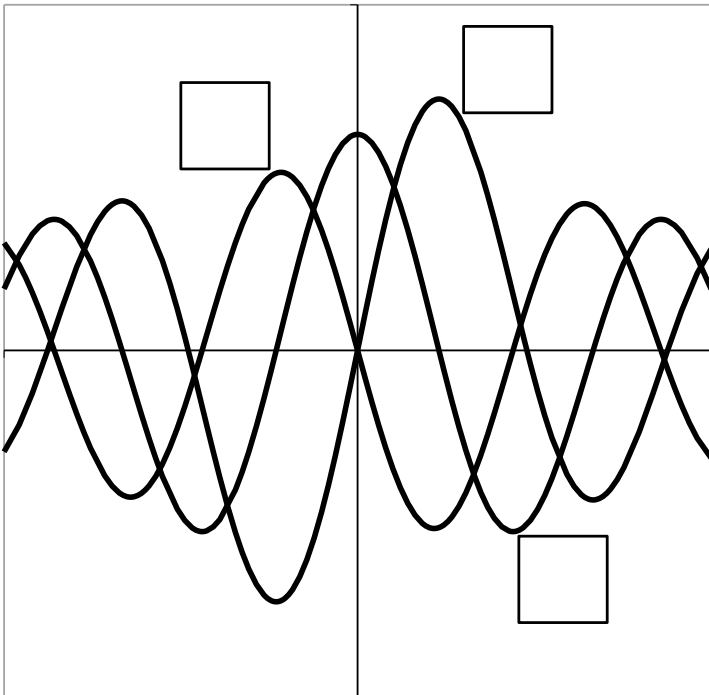
Fall 2013 Midterm I

1a. Determine the constants a, b, c if $\lim_{x \rightarrow 1} \frac{\sqrt{ax^2 + bx + c} - 3}{(x-1)^2} = 5$.

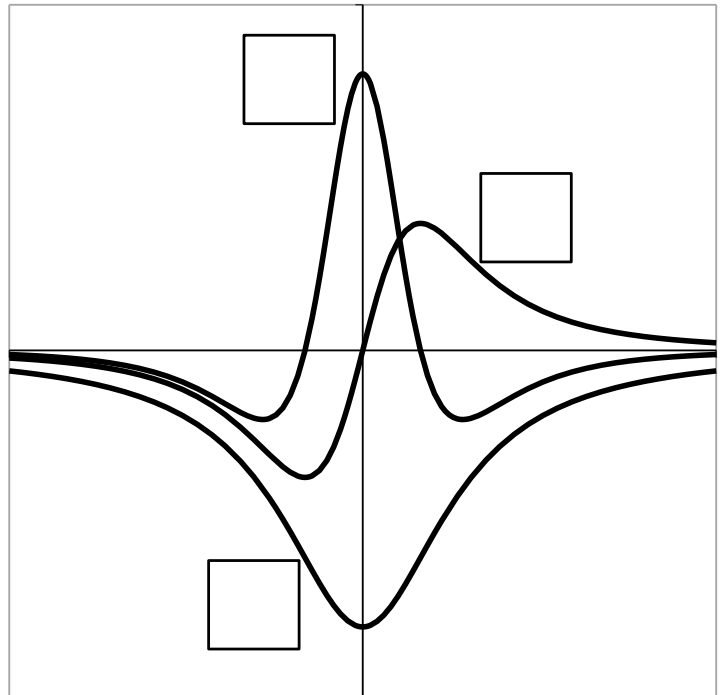
1b. Evaluate $\left. \frac{dy}{dx} \right|_{x=1/6}$ if $y = \sin(3\pi \sin(2\pi \sin(\pi x)))$.

2. In each of the following figures, the graphs of a function f together with its first and second derivatives f' and f'' are shown. Identify each by filling in the boxes with f , f' and f'' .

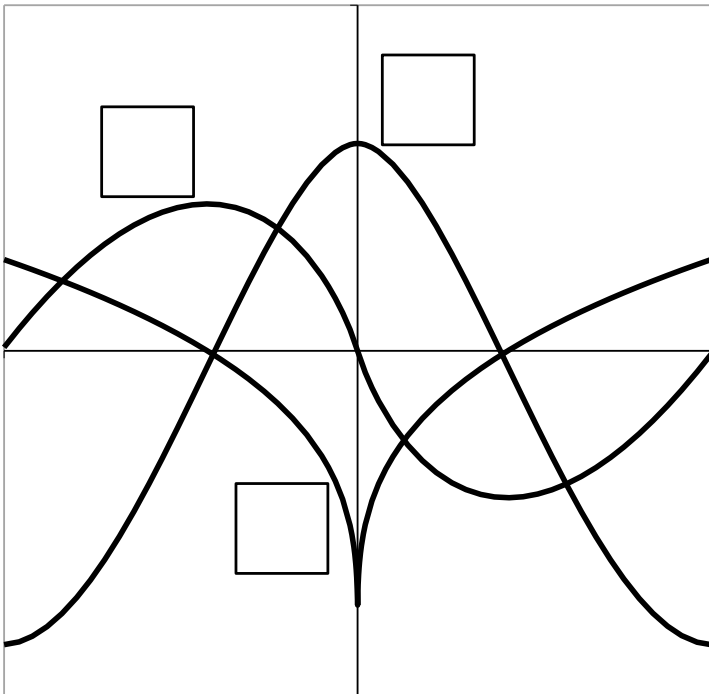
a.



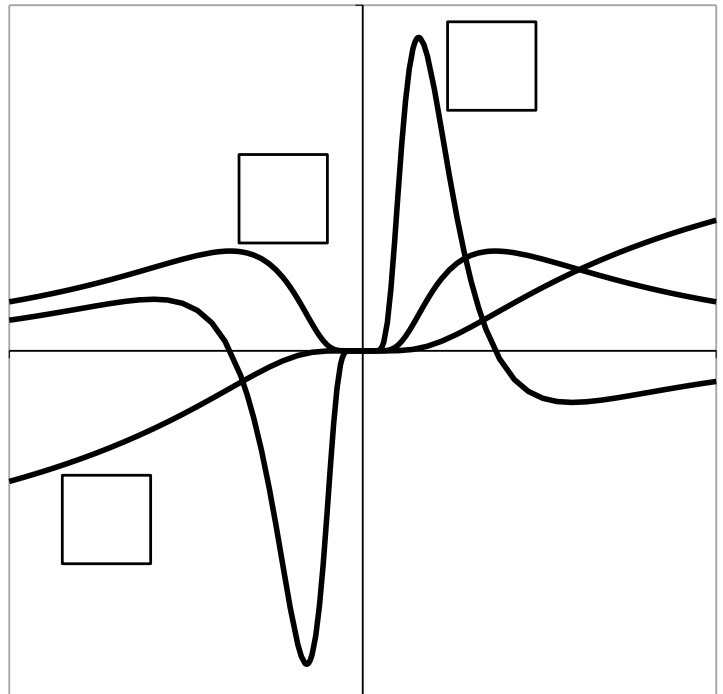
b.



c.



d.



3. A television camera is positioned 900 m from the base of a rocket launching pad. The angle of elevation of the camera has to change at the correct rate in order to keep the rocket in sight. Also, the mechanism for focusing the camera has to take into account the increasing distance from the camera to the rising rocket. Let's assume that the rocket rises vertically and its speed is 250 m/s when it has risen 1200 m.

a. How fast is the distance from the television camera to the rocket changing at that moment?

b. If the television camera is always kept aimed at the rocket, how fast is the camera's angle of elevation changing at that same moment? Express your answer in degrees per second.

4. Consider the following conditions:

- ① f is differentiable on the entire real line.
- ② $y = 3x + 7$ is an equation of the tangent line to the graph of $y = f(x)$ at the point with $x = -1$.
- ③ $y = 5x - 9$ is an equation of the tangent line to the graph of $y = f(x)$ at the point with $x = 2$.

a. Determine the following values for a function f satisfying ①–③.

$$f(-1) = \boxed{} \quad f(2) = \boxed{} \quad f'(-1) = \boxed{} \quad f'(2) = \boxed{}$$

In parts **(b)** and **(c)**, show all your work and explain your reasoning fully and in detail.

b. Estimate the value of $f(15/8)$.

c. Indicate whether each of the following statements is TRUE or FALSE by putting a **X** in the corresponding box. Then if TRUE, prove it; and if FALSE, give a counterexample.

- ❶ Every function f satisfying ①–③ takes on the value π at some point.

TRUE ☐ FALSE ☐

- ❷ Derivative f' of every function f satisfying ①–③ takes on the value -1 at some point.

TRUE ☐ FALSE ☐

Fall 2013 Midterm II

1. A continuous function f that is twice-differentiable for $x \neq 0, 3$ satisfies the following conditions:

- ① $f(0) = 0$, $f(1) = \sqrt[3]{4}$, $f(3) = 0$
- ② $f'(1) = 0$, $\lim_{x \rightarrow 0} f'(x) = \infty$, $\lim_{x \rightarrow 3^-} f'(x) = -\infty$, $\lim_{x \rightarrow 3^+} f'(x) = \infty$
- ③ $f'(x) > 0$ for $x < 1$ and $x \neq 0$; $f'(x) < 0$ for $1 < x < 3$; $f'(x) > 0$ for $3 < x$
- ④ $f''(x) > 0$ for $x < 0$; $f''(x) < 0$ for $0 < x$ and $x \neq 3$
- ⑤ The line $y = x - 2$ is a slant asymptote of the graph of $y = f(x)$ both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$

a. Sketch the graph of $y = f(x)$.

b. Fill in the boxes to make the following a true statement.

The function $f(x) = x^a(b - x)^c$ satisfies the conditions ①-⑤ if the constants a , b and c are chosen as

$$a = \boxed{}, \quad b = \boxed{} \quad \text{and} \quad c = \boxed{}.$$

2. The Rubber Duck is a sculpture designed by Florentijn Hofman and constructed from PVC. For the purposes of this question, we consider the Rubber Duck to consist of a spherical head of radius a and a spherical body of radius b . The research shows that the *cuteness* $\mathcal{K}$ of the Rubber Duck is given by

$$\mathcal{K} = \begin{cases} \frac{a}{b} \left(1 - \frac{a}{b}\right)(a + b) & \text{if } 0 \leq a < b, \\ 0 & \text{if } 0 < b \leq a. \end{cases}$$

Find the dimensions of the cutest Rubber Duck with a total surface area of $400\pi \text{ m}^2$.



3a. Suppose that f is a continuous function satisfying

$$f(x) = x \int_0^x f(t) dt + x^3$$

for all x , and c is a real number such that $f(c) = 1$. Express $f'(c)$ in terms of c only.

3b. Evaluate the integral $\int_1^9 (1 + \sqrt{x}) \sqrt{1 + \frac{2}{\sqrt{x}}} dx$.

4. Let R be the region bounded by the curve $y^2 = x^2 - x^4$ and the x -axis in the first quadrant. Let V be the volume obtained by rotating R about the x -axis. Let W be the volume obtained by rotating R about the y -axis.

a. Find an equation of the tangent line to the curve at the point $(x, y) = (-1, 0)$.

b. Find $\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(-1,0)}$.

2. Suppose that f is a continuous function that is twice-differentiable for $x \neq 0$ and $x \neq 6$ and that satisfies the following conditions:

① $f(0) = 0, f(4) = 2\sqrt[3]{4}, f(6) = 0$

(2) $f'(4) = 0$, $\lim_{x \rightarrow 0^-} f'(x) = -\infty$, $\lim_{x \rightarrow 0^+} f'(x) = \infty$, $\lim_{x \rightarrow 6} f'(x) = -\infty$

(3) $f'(x) < 0$ for $x < 0$; $f'(x) > 0$ for $0 < x < 4$; $f'(x) < 0$ for $4 < x$ and $x \neq 6$

(4) $f''(x) < 0$ for $x < 6$ and $x \neq 0$; $f''(x) > 0$ for $6 < x$

⑤ The line $y = -x + 2$ is a slant asymptote of the graph of $y = f(x)$ both as $x \rightarrow \infty$ and as $x \rightarrow -\infty$

a. Sketch the graph of $y = f(x)$.

b. Fill in the boxes to make the following a true statement.

The function $f(x) = x^a(b-x)^c$ satisfies the conditions ①-⑤ if the constants a , b and c are chosen as

$a =$, $b =$ and $c =$.

3. Consider the limit $\lim_{x \rightarrow 0} \frac{\int_0^x \sin(xt^3) dt}{x^5}$.

a. Fill in the boxes to make the following a true statement.

Both Rule and the Theorem can be used to

evaluate the limit $\lim_{x \rightarrow 0} \frac{\int_0^x \sin(xt^3) dt}{x^5}$.

b. Fill in the box to make the following a true statement.

The Fundamental Theorem of Calculus, Part 1 cannot be used directly to compute the

derivative $\frac{d}{dx} \left(\int_0^x \sin(xt^3) dt \right)$, because .

c. Evaluate the limit $\lim_{x \rightarrow 0} \frac{\int_0^x \sin(xt^3) dt}{x^5}$.

4a. Evaluate the integral $\int \sin x \cos^2 x \tan^3 x dx$.

4b. Suppose that f is a function that has a continuous second derivative and that satisfies $f(0) = 4$, $f(1) = 3$, $f'(0) = 5$, $f'(1) = 7$, $f''(0) = 8$ and $f''(1) = 11$. Show that:

$$\int_0^1 f(x)f''(x) dx \leq 1$$

5. In each of the following, if there exists a function f that satisfies the given condition, give an example of such a function; otherwise, just write DOES NOT EXIST inside the box.

a. f is continuous at 0 and f is not differentiable at 0.

$f(x) =$

b. f is continuous on $(-\infty, \infty)$, the absolute maximum value of f is 1, $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = 0$.

$f(x) =$

c. f is continuous on $[0, 1]$ and $\left(\int_0^1 f(x) dx\right)^2 < \int_0^1 (f(x))^2 dx$.

$f(x) =$

d. f is continuous on $(0, 1]$, $\int_0^1 f(x) dx = \infty$ and $\int_0^1 (f(x))^2 dx = 1$.

$f(x) =$

e. f is continuous on $(0, 1]$, $\int_0^1 f(x) dx = 1$ and $\int_0^1 (f(x))^2 dx = \infty$.

$f(x) =$

Fall 2012 Midterm I

1. Consider the unit circle $C_1 : x^2 + y^2 = 1$ and the circle C_2 with radius a and center $(1, 0)$ where $0 < a < 2$. (In parts (a), (c), (d), (e), just fill in the answer, no explanation is required; in parts (b) and (f), show all your work!)

a. An equation for the circle C_2 is

b. Find the point of intersection P of the circles C_1 and C_2 that lies above the x -axis.

c. The slope of the line ℓ that passes through the points P and $Q(1, a)$ is

$m =$

d. An equation of the line ℓ is

e. The x -intercept of the line ℓ is $b =$

f. Find the limit $\lim_{a \rightarrow 0^+} b$.

2. Consider the equation $1 - \frac{x^2}{4} = \cos x$.

a. Show that this equation has at least one real solution.

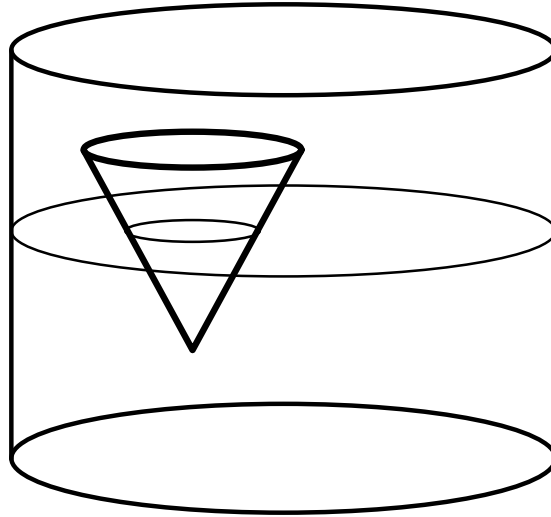
b. Show that this equation has at least two real solutions.

c. Show that this equation has at least three real solutions.

3a. Compute $\frac{d}{dx} \cos \sqrt{\sin(\tan(9x))}$. (*Do not simplify!*)

3b. Find $J''(0)$ if $xJ''(x) + J'(x) + xJ(x) = 0$ for all x , $J(0) = 1$ and $J'''(0)$ exists.

4. A cone of radius 2 cm and height 5 cm is lowered point first into a tall cylinder of radius 7 cm that is partially filled with water. Determine how fast the depth of the water is changing at the moment when the cone is completely submerged if the cone is moving with a speed of 3 cm/s at that moment.



5. In each of the following, if there exists a function f whose domain is the entire real line and that satisfies the given condition, give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No further explanation is required. No partial points will be given.

a. f is continuous at 0 and f is not differentiable at 0.

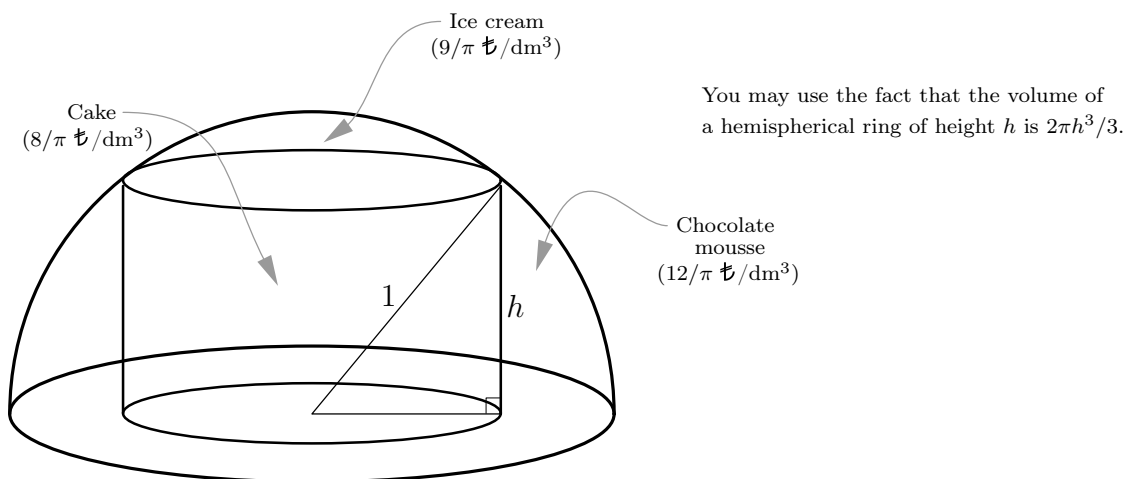
$f(x) =$

b. $f(x) < 1$ for all x and f has no absolute maximum value.

$f(x) =$

c. $f'(0) = 0$ and f does not have a local extreme value at 0.

$f(x) =$



3. Suppose that f is a twice-differentiable function satisfying $f(0) = -2$, $f'(0) = 11$, $f''(0) = -8$, $f(2) = 5$, $f'(2) = -3$, $f''(2) = 7$; and also suppose that the function

$$g(x) = \frac{1}{x} \int_0^x f(t) dt$$

has a critical point at $x = 2$.

a. Find $\lim_{x \rightarrow 0} g(x)$.

b. Find $\int_0^2 f(x) dx$.

c. Determine whether the critical point of g at $x = 2$ is a local minimum, a local maximum or neither.

4. Let R be the region between the curve $y = \sqrt{8x^2 - x^5}$ and the x -axis in the first quadrant.

a. Find the volume V generated by revolving R about the x -axis.

b. Find the volume W generated by revolving R about the y -axis.

5. In each of the following, if the given statement is true for all functions f that are differentiable on the entire real line, then mark the ☐ to the left of TRUE with a **X**; otherwise, mark the ☐ to the left of FALSE with a **X** and give a counterexample.

a. If $f(x) \geq x^2$ for all x , then $f'(x) \geq 2x$ for all x .

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

b. $\int_0^1 f(x)f'(x) dx \leq f(1)^2$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

c. $\int_0^1 f(x)^2 dx = \left(\int_0^1 f(x) dx \right)^2$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

d. $\frac{d}{dx} \int_0^x f(t) dt = f(x) - f(0)$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

e. $\frac{d}{dx} \int_0^1 f(x) dx = f(1) - f(0)$

☐ TRUE

☐ FALSE, because it does not hold for $f(x) =$

Fall 2012 Final

1. Suppose that y is a differentiable function of x satisfying the equation

$$\int_0^{x+y} e^{-t^2} dt = xy.$$

a. Find $\left. \frac{dy}{dx} \right|_{(x,y)=(0,0)}$.

b. Find $\left. \frac{d^2 y}{dx^2} \right|_{(x,y)=(0,0)}$.

2. Evaluate the limit $\lim_{x \rightarrow 0} \frac{\cos x - e^{-x^2/2}}{\sin(x^4)}$.

3. A vat with 2000 L of beer contains 4% alcohol by volume. Beer with 6% alcohol is pumped into the vat at a rate of 20 L/min and the mixture is pumped out at the same rate. How long does it take for the alcohol concentration to rise to 5%?

4. Evaluate the following integrals:

a. $\int_{\pi/4}^{\pi/3} \frac{\sqrt{\tan x}}{\sin 2x} dx$

b. $\int \frac{x-1}{x^2} e^x dx$

5. Suppose that f is a differentiable function on $(0, \infty)$ satisfying the following conditions:

① $f(1/x) = x^2 f(x)$ for all $x > 0$

② $\int_1^\infty f(x) \ln x dx$ is convergent

a. Express $f'(1)$ in terms of $A = f(1)$.

b. Express $\int_0^1 f(x) \ln x dx$ in terms of $B = \int_1^\infty f(x) \ln x dx$.

Fall 2011 Midterm I

1a. Evaluate the limit $\lim_{x \rightarrow 0} \frac{1 - \cos(x^2)}{x^2 \sin^2 x}$. (*Do not use L'Hôpital's Rule!*)

1b. Compute $\frac{d}{dx} \tan^4(5 \cos^2 7x)$. (*Do not simplify!*)

2a. Use the precise definition of the limit to show that $\lim_{x \rightarrow 2} \frac{x}{x-1} = 2$.

2b. We claim that the equation $\frac{\tan x}{x} = \frac{3}{2}$ has at least one positive root.

i. Consider the following ‘proof’ of this claim:

Let $f(x) = 2 \tan x - 3x$. Then $f(\pi/3) = 2\sqrt{3} - \pi > 0$ and $f(\pi) = -3\pi < 0$. Therefore there is c in $[\pi/3, \pi]$ such that $f(c) = 0$ by the Intermediate Value Theorem. It follows that c is a positive root of the given equation.

Explain in one sentence what is wrong with this proof.

ii. Give a correct proof of this claim.

3a. A light shines from the top of a pole 5 m high. A ball is dropped from a point 4 m away from the pole. Find how fast the shadow of the ball is moving when the ball is 2 m above the ground and moving at a speed of 9 m/sec.

3b. Compute $\sqrt[3]{63.7}$ approximately using a suitable linearization of the function $f(x) = \sqrt[3]{x}$.

4. Find the absolute maximum and minimum values of $f(x) = x^4 - 10x^2 + 8x$ on the interval $[-1, 3]$.

5. In each of the following if there exists a function f whose domain is the entire real line and that satisfies the given condition, give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No further explanation is required.

a. $f(x)$ is not differentiable at $x = 0$ and $(f(x))^2$ is differentiable at $x = 0$.

$f(x) =$

b. $f'(3) = 5$ and $\lim_{x \rightarrow 3} f(x) = \infty$.

$f(x) =$

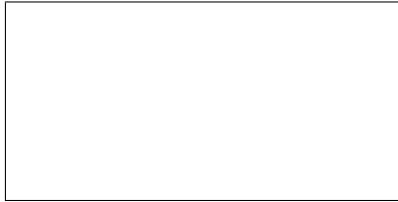
c. $f(3) = 5$ and $\lim_{x \rightarrow 3} f'(x) = \infty$.

$f(x) =$

d. f is not constant and $(f'(x))^2 - f(x)$ is constant.

$f(x) =$

e. $y = 2x + 5$ is the equation of the tangent line to the graph of $y = f(x)$ at the point with $x = 3$ and $f'(3) = 11$.

$f(x) =$


Fall 2011 Midterm II

1. Evaluate the following integrals.

a. $\int \frac{\sin x - \cos x}{\sqrt[3]{\sin x + \cos x}} dx$

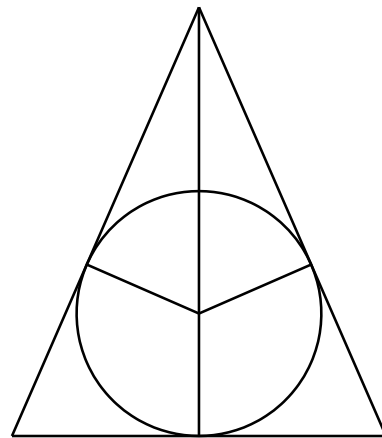
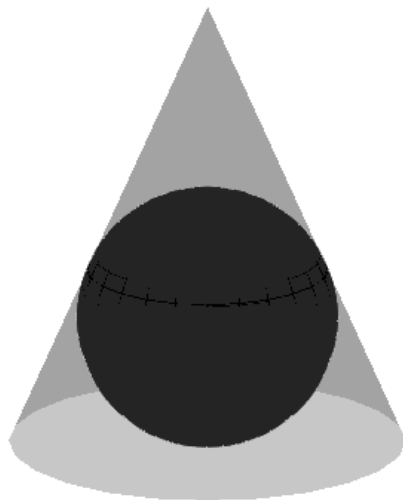
b. $\int_2^5 \frac{x}{\sqrt{x-1}} dx$

2a. Suppose that $\frac{dy}{dx} = \frac{x}{y^2 + 1}$ for all x and $y(0) = -1$. Find all x for which $y(x) = 0$.

2b. Find the area of the region that lies below the line $y = 2x$ and between the parabolas $y = x^2$ and $y = x^2/2$.

3. Suppose that f is a continuous and positive function on $[0, 5]$, and the area between the graph of $y = f(x)$ and the x -axis for $0 \leq x \leq 5$ is 8. Let $A(c)$ denote the area between the graph of $y = f(x)$ and the x -axis for $0 \leq x \leq c$, and let $B(c)$ denote the area between the graph of $y = f(x)$ and the x -axis for $c \leq x \leq 5$. Let $R(c) = A(c)/B(c)$. If $R(3) = 1$ and $\left. \frac{dR}{dc} \right|_{c=3} = 7$, find $f(3)$.

4. Find the smallest possible volume of a cone that contains a sphere of radius 1.



5. A twice-differentiable function f satisfies the following.

- $f'(x) > 0$ for $x < 0$ and $f'(x) < 0$ for $x > 0$

- $f''(x) > 0$ for $x < -2$ and $x > 1$, and $f''(x) < 0$ for $-2 < x < 1$
- $f(-2) = 2$, $f(0) = 5$, $f(1) = 4$
- $\lim_{x \rightarrow -\infty} f(x) = 0$ and $\lim_{x \rightarrow \infty} f(x) = 3$

- a. Sketch the graph of $y = f(x)$. Make sure to clearly show all important features.
- b. Explain in one sentence why f cannot be a polynomial.
- c. Explain in one sentence why f cannot be a rational function.

Fall 2011 Final

1. Evaluate the following integrals.

a. $\int \frac{\sin x + \cos x}{\tan x + \cot x} dx$

b. $\int_0^{\pi/2} \frac{\sin 2x}{1 + \cos^4 x} dx$

2a. Find a formula expressing $\int \frac{dx}{(x^2 + 1)^{n+1}}$ in terms of $\int \frac{dx}{(x^2 + 1)^n}$ for $n \geq 1$.

2b. Find y'' at $(x, y) = (\pi/6, \pi/3)$ if $y = 2x \sin(x + y)$.

3. Evaluate the limit $\lim_{x \rightarrow 0} \left(\frac{\ln(1+x)}{x} \right)^{1/x}$.

4. a. Show that $\int_0^\infty t^n e^{-t} dt = n \int_0^\infty t^{n-1} e^{-t} dt$ for all $n \geq 1$.

b. Let R be the region between the graph of $y = \sqrt{x} e^{-ax}$ and the x -axis where a is a positive constant. Let V be the volume of the solid generated by revolving R about the x -axis, and let W be the volume of the solid generated by revolving R about the y -axis. Find all values of a for which $V = W$.

[In this question you may use the fact that $\int_0^\infty \sqrt{x} e^{-x} dx = \frac{\sqrt{\pi}}{2}$.]

5. In each of the following if there exists a function f that satisfies the given condition, give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No explanation is required. No partial credit will be given.

- a. f is differentiable on $(-\infty, \infty)$, f has at least two distinct zeros and f' has no zeros.

$f(x) =$

4a. Assume that y is a differentiable function of x satisfying $y = xg(x^2 + y^2) - 13$, where g is a function such that $g(5) = 7$, $g'(5) = -2$, and $g''(5) = 12$. Find $\left. \frac{dy}{dx} \right|_{(x,y)=(2,1)}$.

4b. Does the equation $x^2 = f(x)$ have a solution for every function f that is continuous on $[-1, 2]$, and has values $f(-1) = 4$ and $f(2) = 1$? If YES, prove that it does; if NO, give an example of a function for which it does not.

5. In each of the following if there exists a function f whose domain is the entire real line and which satisfies the given condition, give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No further explanation is required. No partial credit will be given.

a. f is differentiable, f is not constant and $(f(x))^2 + (f'(x))^2$ is constant.

b. $|f(x)| \leq |x|$ for all $x \neq 0$ and $\lim_{x \rightarrow 0} f(x)$ does not exist.

c. f is continuous at 0 and $f'(0)$ does not exist.

d. f is differentiable and $f(x) > f'(x) > 0$ for all x .

e. $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ exist and are different.

Fall 2010 Midterm II

1. Evaluate the following integrals.

a. $\int \frac{x^2}{x^3 - 1} dx$

b. $\int_0^{2\pi} \sqrt{1 - \cos 2x} dx$

2a. Let f be a differentiable function satisfying

$$\int_2^x f(t) dt = \sin(f(x)) - \frac{1}{2}$$

for all x in an open interval containing 2. Find $f'(a)$ if $f(a) = \frac{\pi}{3}$.

2b. Evaluate the limit $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 + k^2}$.

3. Let R be the region lying between the curve $y = \frac{e^x}{x}$ and the line $y = x$ for $1 \leq x \leq 2$.

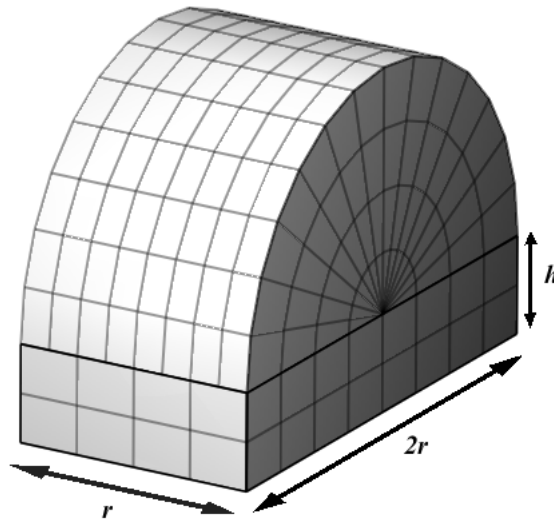
a. Show that $\frac{e^x}{x} \geq x$ for $1 \leq x \leq 2$.

b. Express the volume V of the solid generated by revolving the region R about the x -axis. (*Do not compute!*)

c. Express the volume W of the solid generated by revolving the region R about the y -axis. (*Do not compute!*)

d. Compute either $\square V$ or $\square W$. (Indicate the one you choose by putting a **X** in the square in front of it.)

4. We want to build a greenhouse that has a half cylinder roof of radius r and height r mounted horizontally on top of four rectangular walls of height h as shown in the figure. We have 200π m² of plastic sheet to be used in the construction of this structure. Find the value of r for the greenhouse with the largest possible volume we can build.



5. In each of the following if there exists a function f that is defined and differentiable on the entire real line and that satisfies the given condition, give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No further explanation is required. No partial credit will be given.

a. f has exactly one local maximum, exactly two zeros, and no local minimums.

$$f(x) = \boxed{\phantom{f(x) = \text{DOES NOT EXIST}}}$$

b. f has exactly one local maximum, no zeros, and no local minimums.

$$f(x) = \boxed{\phantom{f(x) = \text{DOES NOT EXIST}}}$$

c. f has no local maximums, no zeros, and no local minimums.

$$f(x) =$$

- d. f has exactly one local maximum, exactly three zeros, and exactly one local minimum.

$$f(x) =$$

Fall 2010 Final

- 1a. Find the equation of the tangent line to the curve $y = x^2 2^x$ at $x = 1$.

- 1b. Evaluate the limit $\lim_{x \rightarrow \pi/4} (\tan x)^{\tan 2x}$.

2. Evaluate the following integrals.

a. $\int \frac{dx}{x - \sqrt[3]{x}}$

b. $\int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx$

- 3a. Let $A = \int_0^\infty e^{-x^2} dx$. Express the value of the improper integral $\int_0^\infty x^2 e^{-x^2} dx$ in terms of A .

- 3b. Find all real numbers x that satisfy the equality $x + x^3 + x^5 + \cdots = -1$.

4. Consider the sequence satisfying the conditions $a_1 = 2$ and $a_n = a_{n-1} - \frac{1}{a_{n-1}}$ for $n \geq 2$.

- a. Compute a_2, a_3, a_4 .

- b. Exactly one of the following statements is true. Choose the true statement and mark the box in front of it with a ✓.

☐ The sequence $\{a_n\}$ converges.

☐ The sequence $\{a_n\}$ diverges.

- c. Prove the statement you chose in part (b).

5. In each of the following indicate whether the given series converges or diverges, and also indicate the *best* way of determining this by marking the corresponding boxes with a ✗ and filling in the corresponding blank.

a. $\sum_{n=2}^{\infty} \frac{n}{\sqrt{n^5 - 1}}$ ☐ converges ☐ diverges

☐ p -series with $p = \underline{\hspace{2cm}}$ ☐ Direct Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ geometric series with $r = \underline{\hspace{2cm}}$ ☐ Limit Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ n th Term Test ☐ Ratio Test

☐ Integral Test ☐ Root Test

b. $\sum_{n=1}^{\infty} \sin^2\left(\frac{1}{n}\right) \sin^2 n$ ☐ converges ☐ diverges

☐ p -series with $p = \underline{\hspace{2cm}}$ ☐ Direct Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ geometric series with $r = \underline{\hspace{2cm}}$ ☐ Limit Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ n th Term Test ☐ Ratio Test

☐ Integral Test ☐ Root Test

c. $\sum_{n=1}^{\infty} \left(\frac{n-2}{n}\right)^n$ ☐ converges ☐ diverges

☐ p -series with $p = \underline{\hspace{2cm}}$ ☐ Direct Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ geometric series with $r = \underline{\hspace{2cm}}$ ☐ Limit Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ n th Term Test ☐ Ratio Test

☐ Integral Test ☐ Root Test

d. $\sum_{n=1}^{\infty} \frac{n^3}{\left(2 - \frac{\sin n}{n}\right)^n}$ ☐ converges ☐ diverges

☐ p -series with $p = \underline{\hspace{2cm}}$ ☐ Direct Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ geometric series with $r = \underline{\hspace{2cm}}$ ☐ Limit Comparison Test with $\sum \underline{\hspace{2cm}}$

☐ n th Term Test ☐ Ratio Test

☐ Integral Test ☐ Root Test

Fall 2009 Midterm I

1. Consider the function

$$f(x) = \begin{cases} \frac{\sqrt{1+x}-1}{x} & \text{if } x \neq 0 \\ k & \text{if } x = 0 \end{cases}$$

where k is a constant.

a. Determine the value of the constant k for which f is continuous at 0.

b. Write the limit defining $f'(0)$. (*Do not compute.*)

2a. Evaluate the limit $\lim_{x \rightarrow 0} \frac{5 - 6 \cos x + \cos^2 x}{x \sin x}$. (*Do not use L'Hôpital's Rule.*)

2b. Evaluate the limit $\lim_{x \rightarrow 1} \frac{x^5 + x^3 - 2}{x^3 + x^2 - 2}$. (*Do not use L'Hôpital's Rule.*)

3. Consider a point P moving along the x -axis, and let Q be a point moving on the curve $y = x^2$ such that the line PQ is normal to the curve $y = x^2$ at the point Q at all times. Determine the rate of change of the x -coordinate of the point Q at the moment when P is at $(3, 0)$ and the x -coordinate of P is increasing at a rate of 1 cm/sec.

4a. Let f be a differentiable function and $g(x) = (f(f(x^3) + x))^2$. If $f(1) = 1$, $f(2) = 5$, $f'(1) = -3$, and $g'(1) = 2$, find $f'(2)$.

4b. If $y = \frac{\sin(x^2)}{x^3}$, find $\left. \frac{dy}{dx} \right|_{x=\sqrt{\pi/2}}$.

5a. Is there a continuous function f on $(0, \infty)$ such that $f(2x)f(x) < 0$ for all $x > 0$? If YES, give an example; if NO, prove that no such function exists.

5b. Is there a continuous function f on $(0, \infty)$ such that $\lim_{x \rightarrow \infty} f(x)$ does not exist, but $\lim_{x \rightarrow \infty} (f(x+1) - f(x))$ exists? If YES, give an example; if NO, prove that no such function exists.

Fall 2009 Midterm II

1. A function f that is continuous on the entire real line and differentiable for $x \neq 0$ satisfies the following conditions: *i.* $f(0) = 0$; *ii.* $f(1) = 1$; *iii.* $f'(x) < 0$ for $x < 0$, $f'(x) > 0$ for $0 < x < 1$, and $f'(x) < 0$ for $x > 1$; *iv.* $f''(x) < 0$ for $x \neq 0$.

a. Sketch the graph of $y = f(x)$.

b. Give an example of a function in the form $f(x) = ax^b + cx^d$, where a, b, c, d are constants, that satisfies the conditions (*i-iv*). (You do not need to explain how you found the example, but you have to verify that your example satisfies the conditions.)

2. Find the maximum possible total surface area of a cylinder inscribed in a hemisphere of radius 1.

3. Evaluate the limit $\lim_{x \rightarrow 0^+} \frac{\sin x + 2 \cos \sqrt{x} - 2}{x^2}$.

4a. If $f(x) = x^2 \int_0^x \sin\left(\frac{\pi}{t^2 + 1}\right) dt$, express $f'(1)$ in terms of $A = f(1)$.

4b. Find $y(1)$ if y is a differentiable function of x such that $\frac{dy}{dx} = xy^2$ and $y(0) = 1$.

5. Evaluate the following integrals.

a. $\int_1^2 \sqrt[3]{\frac{2-x}{x^7}} dx$

b. $\int \frac{\sin^2 \sqrt{x} \cos \sqrt{x}}{\sqrt{x}} dx$

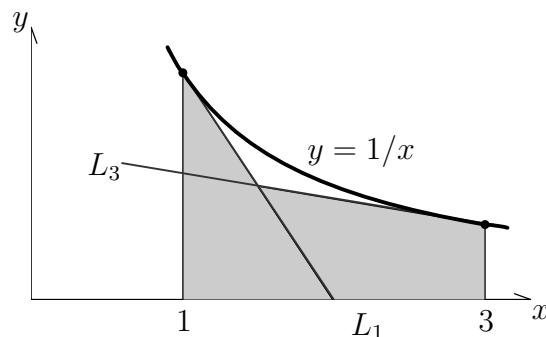
Fall 2009 Final

1. Consider the region bounded by the curve $y = \sin(kx)$ and the x -axis for $0 \leq x \leq \pi/k$, where k is a positive constant. Let V be the volume of the solid generated by revolving this region about the x -axis, and W be the volume of the solid generated by revolving this region about the y -axis. Find all values of k for which $V = W$.

2a. Let $a > 0$ be a constant. Show that $\frac{1}{x} \geq -\frac{1}{a^2}(x - a) + \frac{1}{a}$ for all $x > 0$.

2b. Express in one sentence what the inequality in part (a) means in geometric terms.

2c. In the figure, the line L_1 and the line L_3 are tangent to the graph of $y = 1/x$ at the points $(1, 1)$ and $(3, 1/3)$, respectively. Compute the area of the shaded region.



2d. Explain in one sentence why part (c) implies that $3 > e$.

3. Evaluate the following limits:

a. $\lim_{x \rightarrow 0^+} (e^x - 1)^{1/\ln x}$

b. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{(en + k)(\ln(en + k) - \ln n)}$

4a. Express the value of the integral $\int_0^1 \frac{\ln x}{x^2 + 1} dx$ in terms of $A = \int_1^\infty \frac{\ln x}{x^2 + 1} dx$.

4b. Express the value of the integral $\int_0^1 \frac{\ln x}{x^2 + 1} dx$ in terms of $B = \int_0^1 \frac{\arctan x}{x} dx$.

5. In each of the following if there exists a function f that satisfies the given conditions, give an example of such a function; otherwise just write DOES NOT EXIST. No further explanation is required. No partial credit will be given.

a. f is differentiable on $(-\infty, \infty)$, and $f'(x) > f(x) > 0$ for all x .

b. f is differentiable on $(0, \infty)$, $f(2x) = -f(x)$ for all $x > 0$, and $f(3) \neq 0$.

c. f'' is continuous on $[0, 1]$, $\int_0^1 (f'(x))^2 dx = 1$, and $\int_0^1 f(x)f''(x) dx = -1$.

d. $f'(0)$ exists, but f is not continuous at $x = 0$.

Fall 2008 Midterm I

1. Evaluate the following limits: (*Do not use L'Hôpital's Rule.*)

a. $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x^3 - 82x + 9}$

b. $\lim_{x \rightarrow 2} \frac{1 + \sin(3\pi/x)}{(x - 2)^2}$

2. Consider the limit $\lim_{x \rightarrow 2} \frac{x \ln(x) - 2 \ln(2)}{x - 2}$.

a. Interpret this limit as the derivative $f'(a)$ of a function $f(x)$ at a point $x = a$.

b. Compute the limit using the interpretation in part (a).

3. Suppose that $f(1) = 1$, $f'(1) = 2$, $f''(1) = -5$, and that y is a differentiable function satisfying the equation $f(y^2x) = 2yf(x) - 1$. Find $\left. \frac{d^2y}{dx^2} \right|_{(x,y)=(1,1)}$.

4a. A differentiable parametric curve $x = f(t)$, $y = g(t)$ satisfies $\frac{dy}{dx} = \sqrt{t^2 + 1} \cos(\pi t)$. If

$$\left. \frac{d^2 y}{dx^2} \right|_{t=3/4} = 2, \text{ find } \left. \frac{dy}{dt} \right|_{t=3/4}.$$

4b. Show that there is a real number c such that the tangent line to the graph of $u(x) = x^3 - x$ at the point $(c, u(c))$ and the tangent line to the graph of $v(x) = e^{-x^2}$ at the point $(c, v(c))$ are perpendicular to each other.

5. In each of the following if there exists a function f whose domain is the entire real line and that satisfies the given condition, give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No further explanation is required.

- a.** $\lim_{x \rightarrow 0} x f(x) = 0$ and $\lim_{x \rightarrow 0} f(x) = \infty$.
- b.** $\lim_{x \rightarrow \infty} f(x) = 0$, but $\lim_{x \rightarrow 0} f(1/x)$ does not exist.
- c.** $\lim_{h \rightarrow 0} \frac{f(h) - f(-h)}{h}$ exists, but $f'(0)$ does not exist.
- d.** $f'(0)$ exists, but f is not continuous at $x = 0$.
- e.** $f(x)f'(x) < 0$ for all x .

Fall 2008 Midterm II

1. Evaluate the following limits:

- a.** $\lim_{x \rightarrow 0} \frac{\cos^2(x) - \cos(\sqrt{2}x)}{x^4}$
- b.** $\lim_{x \rightarrow \infty} \left(\frac{2}{\pi} \arctan x \right)^x$

2a. One morning of a day when the sun will pass directly overhead, the shadow of an 40-m building is 30 m long. At the moment in question, the angle the sun makes with the ground is increasing at the rate of $0.27^\circ/\text{min}$. At what rate and how is the length of the shadow changing?

2b. Find $y(x)$ if $\frac{dy}{dx} = e^x - e^{-x}$ and $y(0) = 0$.

3. Sketch the graph of the function $y = e^{-x/2}(x^2 + 4x + 3)$ by computing y' and y'' , and determining their signs; finding the critical points, the inflections points, the intercepts, and the asymptotes; and clearly labeling them in the picture.

4. A hangar consisting of two quarter-spheres joined by a half-cylinder, and having a volume of $2000\pi/3 \text{ m}^3$ will be designed. If the construction cost for the unit area of the spherical parts is $3/4$ of the construction cost of the unit area of the cylindrical part, find the dimensions of the least expensive hangar that can be built.

5. Determine the number of solutions of the equation $e^x = kx$ for each value of the positive constant k .

Fall 2008 Final

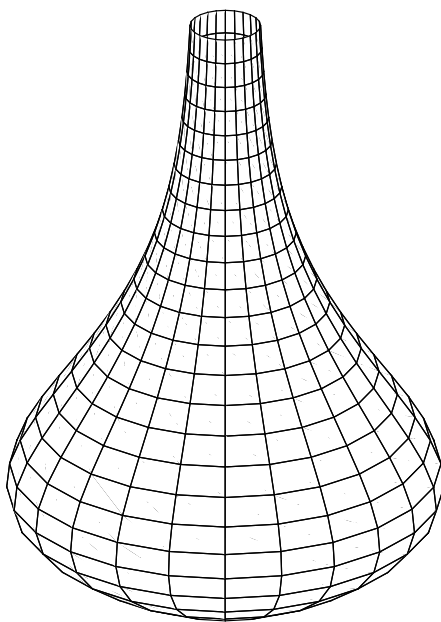
1. Show that $x \geq \ln(1+x) \geq \frac{x}{1+x}$ for all $x > -1$.

2. Evaluate the following limits.

a. $\lim_{x \rightarrow 0} \frac{x^3 - \sin^3 x}{\sin(x^5)}$

b. $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{2k-1}{(2k-1)^2 + n^2}$

3a. The radius of a vase at a height of h cm from its base is $r(h) = \frac{1}{2}(h+3)^2 e^{-h/5} + 2$ cm. If the water is running into the vase at a rate of $150 \text{ cm}^3/\text{sec}$, how fast is the level of water changing at the moment when it is 10 cm deep?



3b. Find the area of the surface generated by revolving the curve $y = \sqrt{x}$, $0 \leq x \leq 1$, about the x -axis.

4a. Evaluate the integral $\int_0^{\pi/3} \frac{\sin x \, dx}{2 \sin^2 x + 3 \cos x}$.

4b. Evaluate the integral $\int_0^1 x(f(x)f''(x) + (f'(x))^2) \, dx$ if $f(0) = 3$, $f'(0) = -5$, $f''(0) = 2$, $f(1) = 7$, $f'(1) = 4$, $f''(1) = -1$.

5a. Show that the improper integral $\int_1^\infty e^{-a(x+\frac{1}{x})} dx$ converges for all positive values of the constant a .

5b. Show that $\int_1^\infty e^{-a(x+\frac{1}{x})} dx = \frac{e^{-2a}}{a} + \int_0^1 e^{-a(x+\frac{1}{x})} dx$ for all $a > 0$.

Fall 2007 Midterm I

1. Define the function f as below

$$f(x) = \begin{cases} \frac{\cos x - 1}{x} & \text{for } x \neq 0, \\ 0 & \text{for } x = 0. \end{cases}$$

a. Is the function f continuous at $x = 0$? (*Don't use L'Hôpital's rule.*)

b. Find $f'(0)$ if it exists. (*Don't use L'Hôpital's rule.*)

2a. Define $g(x) = x^2 - \cos x$. Show that there exists a number c in $(0, \pi/2)$ such that $g(c) = -1/2$.

2b. Show that the function $p(x) = x^3 - 3x + 1$ has all its three roots in $[-2, 2]$.

3a. Let a , b and c be constants. Define

$$f(x) = \begin{cases} \cos x & \text{for } x \leq 0, \\ a + bx + cx^2 & \text{for } x > 0. \end{cases}$$

Find a , b , c so that $f'''(0)$ (the third derivative of f at $x = 0$) exists. Does $f^{(4)}(0)$ (the fourth derivative of f at $x = 0$) exist?

3b. Find the horizontal and vertical asymptotes of the curve

$$y = \frac{3x^2 - 1 + 2 \sin x}{x^2 - 1}.$$

4. Let f be a differentiable function on $(-5, 10)$. Assume

$$f\left(\frac{5}{2}\right) = 10, \quad f\left(\frac{3}{2}\right) = 0, \quad \text{and} \quad f'\left(\frac{3}{2}\right) = 3.$$

a. Find $\lim_{x \rightarrow 3} \left(3 + f\left(x - \frac{1}{2}\right)\right)$. (Explain!)

b. Find $\lim_{x \rightarrow 1} \frac{f\left(x + \frac{1}{2}\right)}{x - 1}$. (Explain!)

5a. Let a curve C be defined through the equation $x + \sin(xy) = y$. Find an equation of the tangent line and an equation of the normal line of C at the point $(0, 0)$.

5b. If one perpendicular side of a right triangle decreases at 1 cm/min and the other perpendicular side increases at 2 cm/min then how fast is the length of the hypotenuse changing at the time where the first perpendicular side is 8 cm and the other one is 4 cm long?

Fall 2007 Midterm II

1a. Evaluate $\lim_{x \rightarrow 0} \frac{1}{\sin x} \int_0^{\sin 2x} \cos 5t \, dt$.

1b. A hallway of width 4 ft meets a hallway of width $12\sqrt{3}$ ft at a right angle. Find the length of the longest ladder that can be carried horizontally around the corner from one hallway to the other hallway. (*i.* While solving this problem, to find the global extrema of a function that is not defined on a closed interval you have to use the first derivative test or the second derivative test.) (*ii.* Find the exact answer.)

2. Define a function f as follows

$$f(x) = \frac{x^2}{x^2 - 4}.$$

You have to explicitly write the answer to each part.

a. Find the domain of the function f .

b. Find all asymptotes, the x -intercept(s) and the y -intercept(s) of the graph of f .

c. Find $f'(x)$ and $f''(x)$.

d. Find the intervals on which the function f is increasing and decreasing. Identify the function's local extreme values, if any, saying where they are taken on.

e. Find where the graph of f is concave up and where it is concave down. Are there any inflection points on the graph of f ?

f. Sketch the graph of f .

3a. Calculate the following limit $\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{i=1}^n i^2 \sin \left(5 + \frac{\pi i^3}{n^3} \right)$.

3b. Calculate $\int \frac{\sin 2x}{3 + 2 \cos^2 x} \, dx$.

4. Let f be a continuous function on $[2, 5]$. Assume that

$$f(7 - x) = f(x) \neq 0$$

for all x in $[2, 5]$ and

$$\int_4^5 f(u) \, du = 26.$$

We define

$$F(x) = \int_4^x f(u) du \quad \text{and} \quad G(x) = \int_3^x f(u) du$$

for all x in $[2, 5]$.

- a. Show that G is an increasing function on $[2, 5]$.
- b. Show that there exists only one number c in the interval $[2, 5]$ such that $F(c) = 0$.
- c. Calculate $F(2) + G(4)$.

5a. Find the volume of the solid generated by revolving the region bounded by the curves $x = \ln y$, $y = e^{2x}$, and $x = \ln 2$ about the x -axis.

5b. Find the surface area of the solid generated by revolving the curve $y = \sqrt{2x - x^2}$ for $1 \leq x \leq 3/2$.

Fall 2007 Final

1a. Is there any real number that is one less than its cube? Explain and support your answer with well known theorems.

1b. Prove that $\left| xe^{-x^2} + \cos x - \sin x \right| < \frac{1 + 2\sqrt{2}}{2}$ for all x in $(-\infty, \infty)$.

2a. Show that $\int_0^{\pi/2} \sin^{2n} x dx = \frac{2n-1}{2n} \int_0^{\pi/2} \sin^{2n-2} x dx$ where n is a positive integer.

2b. Show that $\int_0^{\pi/2} \sin^6 x dx = \frac{1 \cdot 3 \cdot 5 \pi}{2 \cdot 4 \cdot 6 \cdot 2} = \frac{5\pi}{32}$.

2c. Show that $\int_0^{\pi/2} \sin^{2n} x dx = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1) \pi}{2 \cdot 4 \cdot 6 \cdots (2n)} \frac{\pi}{2}$ (for any integer $n > 0$).

3a. Find $\lim_{x \rightarrow \infty} \left(\frac{x+a}{x-a} \right)^x$ where a is a nonzero real number.

3b. Find $\lim_{n \rightarrow \infty} \frac{1}{n^{r+1}} \sum_{k=1}^n k^r$ where r is a real number greater than 1.

3c. Find $\lim_{n \rightarrow \infty} \frac{1}{n^{r+2}} \sum_{k=1}^n k^r$ where r is a real number greater than 1.

4. Evaluate the following integrals.

a. $\int_{\sqrt{e}}^e \frac{\sin^{-1}(\ln x)}{x} dx$

b. $\int \frac{x-1}{x(x+1)^2} dx$

c. $\int_{-\infty}^0 2^{3x} dx$

5a. Show that the linearization $L(x)$ of e^x about $x = 0$ is $L(x) = x + 1$.

5b. Let $L(x)$ be the linearization of a function f at $x = 2$. Assume that the function f'' is continuous on $(1, 4)$ and the graph of $y = f(x)$ is concave up on $(0, 5)$. Define a function h as follows: $h(x) = f(x) - L(x)$ for all x in $(1, 3)$.

i. Identify the intervals on which h is increasing and decreasing.

ii. Determine the concavity of graph $y = h(x)$.

iii. Find, if any, absolute extreme values of the function h .

Fall 2006 Midterm I

1. Evaluate the following limits. (*Do not use L'Hôpital's Rule.*)

a. $\lim_{x \rightarrow 1} \frac{x^2 - 4x + 3}{\sqrt{x} - 1}$

b. $\lim_{x \rightarrow \infty} \left((x+2) \sin\left(\frac{2}{x+1}\right) \right)$

2a. Find the equations of the tangent lines to the curve $y = 2x^3 - 3x^2 - 12x + 20$ which are perpendicular to the line $y = 1 - x/24$.

2b. Find $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{3}$ if $x = t - \sin t$ and $y = 1 - \cos t$.

3a. Evaluate $g'(0)$ and $g''(0)$ if $g(x) = \sec(x^2 + 5x)$.

3b. If $f(1) = -3$, $f'(1) = 4$ and if $y = f(xy^2 + f(x^2)) + 5$ defines y as a differentiable function of x , determine $\frac{dy}{dx}$ at the point $(x, y) = (1, 2)$.

4. In each of the following if there exists a function f whose domain is the entire real line and which satisfies the given condition, then give an example of such a function; otherwise just write DOES NOT EXIST inside the box. No further explanation is required.

a. $f(x)$ is not continuous at $x = 4$, but $(f(\sqrt{x}))^2$ is continuous at $x = 4$.

b. $f(x)$ is not continuous at $x = 1$, but $(f(x))^2$ is continuous at $x = 1$.

c. $\lim_{x \rightarrow 0} f(x)$ does not exist, but $\lim_{x \rightarrow 0} f(x^2)$ exists.

d. $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x}$ exists, but f is not differentiable at 0.

e. f is continuous and $f(x)f(x+1) < 0$ for all x .

5. For each of the following functions determine if there is a positive constant C such that for all $0 < \varepsilon < 1$ and for all x ,

$$|x| < C\varepsilon \implies |f(x)| < \varepsilon.$$

In particular, if such C exists, find one; if such C does not exist, explain why not.

a. $f(x) = x^2 + 5x$

b. $f(x) = x^{1/3}$

Fall 2006 Midterm II

1a. Evaluate the integral $\int_0^1 t \sin^2(\pi(t^2 + 1)) dt$.

1b. Find the area of the region enclosed by the curves $y = 2x^3 - x^2 - 5x$ and $y = x^2 - x$.

2a. Evaluate the limit $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n^2}{(n+k)^3}$.

2b. Show that the equation $x^5 + x^3 - 3x^2 + 3x + 7 = 0$ has exactly one real root.

3a. Let $f(x) = \frac{1}{x} \int_0^x \sin(t^2) dt$ for $x > 0$. If $A = f(\sqrt{\pi/2})$, express $f'(\sqrt{\pi/2})$ in terms of A .

3b. Show that if g is a twice differentiable function such that $g(0) = 1$, $g'(0) = -1$, $g(1) = 2$, $g'(1) = 5$, and $g''(x) \geq 0$ for all x , then $g(x) \geq 1/3$ for all $0 \leq x \leq 1$.

4. Find the smallest possible total surface area for a cone circumscribed about a sphere of radius R .

5. Sketch the graph of $y = \frac{x^3}{x^2 - 1}$ by finding y' , y'' ; determining their signs; finding and classifying the critical points, the inflection points and the intercepts; and finding the asymptotes.

Fall 2006 Final

1. Evaluate the following integrals.

a. $\int \frac{1 + e^x}{1 - e^x} dx$

b. $\int \frac{dx}{x^2 \sqrt{x^2 + 4}}$

2. Evaluate the following integrals.

a. $\int_{-1/2}^{1/2} \frac{dt}{t + \sqrt{1-t^2}}$

b. $\int_{-1/2}^{1/2} \sqrt{\frac{1-x}{1+x}} \arcsin x \, dx$

3. Consider the region bounded by the parabolas $y = 4x - x^2$ and $y = x^2 - 8x + 12$. Express the following volumes in terms of definite integrals, but do not evaluate.

a. The volume of the solid generated by revolving this region about the y -axis.

b. The volume of the solid generated by revolving this region about the x -axis.

4a. Express $\int_0^1 x^2 f''(x) \, dx$ in terms of $k = \int_0^1 f(x) \, dx$ if $f'(1) = 2f(1)$ and f'' is continuous.

4b. Prove that $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi}{4\sqrt{2}}$.

5a. Find constants A and B such that the limit $\lim_{x \rightarrow \infty} x^3 \left(A + \frac{B}{x} + \arctan x \right)$ exists.

5b. Let $f(x) = x^{1/x}$ for $x > 0$. Find the absolute maximum value of f .

Fall 2005 Midterm I

1. Evaluate the following limits:

a. $\lim_{x \rightarrow \infty} (\sqrt[3]{x^3 + x^2} - x)$

b. $\lim_{x \rightarrow 5/2} \frac{\cos \pi x}{8x^3 - 16x^2 - 25}$

2. Assume that $\lim_{x \rightarrow 0^+} f(x) = A$ and $\lim_{x \rightarrow 0^-} f(x) = B$ where A and B are real numbers. In each of the following, if possible, express the limit in terms of A and B ; otherwise, write CANNOT BE DONE in the box. No further explanation is required.

a. $\lim_{x \rightarrow 0^+} f(x^2 - x)$

b. $\lim_{x \rightarrow 0^-} f(x^2 - x)$

c. $\lim_{x \rightarrow 0^-} (f(x^2) - f(x))$

d. $\lim_{x \rightarrow \infty} f((x^2 - x)^{-1})$

e. $\lim_{x \rightarrow \infty} f(x^{-2} - x^{-1})$

f. $\lim_{x \rightarrow 1^-} f(x^{-2} - x^{-1})$

3a. Let f be a differentiable function such that $f(1) = 1$ and the slope of the tangent line to the curve $y = f(xf(xy))^2$ at the point $(1, 1)$ is 3. Find all possible values of $f'(1)$.

3b. Show that if u is differentiable at 0, then the limit

$$\lim_{t \rightarrow 0} \frac{u(3t) - u(-2t)}{t}$$

exists.

4a. The angle of elevation from a point 3 m away from the base of a flag pole to its top is $60^\circ \pm 1^\circ$. Estimate the error in the height of the pole calculated from this measurement.

4b. Suppose that a drop of mist is a perfect sphere and that, through condensation, the drop picks up moisture at a rate proportional to its surface area. Show that under these circumstances the drop's radius increases at a constant rate.

5. Consider the function $f(x) = x^{4/3} - x - x^{1/3}$.

a. Find the absolute maximum value and the absolute minimum value of f on the interval $[-1, 6]$.

b. Determine the number of solutions of the equation $f(x) = 0$.

Fall 2005 Midterm II

1. Find the closest point on the curve $y = x^2$ to the point $(0, 1)$

2. Evaluate the limit: $\lim_{x \rightarrow 0} \frac{x \int_0^x \sin(t^2) dt}{\sin(x^4)}$.

3. Let a be a positive constant. Consider the region bounded by the parabola $y = a(x - x^2)$ and the x -axis in the first quadrant. Let V be the volume of the solid generated by revolving this region about the x -axis. Let W be the volume of the solid generated by revolving this region about the y -axis. Find all values of a for which $V = W$.

4. Evaluate the following integrals:

a. $\int_0^1 x^5 \sqrt{x^3 + 1} dx$

b. $\int \tan^2 x dx$

5a. True or false? No explanation is required.

$$\int x^2 dx = \frac{x^3}{3} + C \quad \text{True} \square \quad \text{False} \square$$

$$\int (5x - 7)^2 dx = \frac{(5x - 7)^3}{3 \cdot 5} + C \quad \text{True} \square \quad \text{False} \square$$

$$\int \sin^2 x dx = \frac{\sin^3 x}{3 \cos x} + C \quad \text{True} \square \quad \text{False} \square$$

5b. Determine all differentiable functions f which satisfy

$$\int (f(x))^2 dx = \frac{(f(x))^3}{3f'(x)} + C .$$

Fall 2005 Final

1. Find y' . (*Do not simplify!*)

a. $y = (x^2 + 1)(x^3 + 1)$

b. $y = \frac{x^2 + 1}{x^3 + 1}$

2. Evaluate the following limits:

a. $\lim_{x \rightarrow \infty} x(2^{1/x} - 1)$

b. $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$

3. Evaluate the following integrals:

a. $\int_0^1 \arctan x dx$

b. $\int_0^{\pi/4} \tan^5 x \sec^4 x dx$

4. Sketch the graph of $y = \frac{\ln x}{x}$ by finding y' , y'' , determining their signs, finding and classifying the critical points, the inflection points and the intercepts, and finding the asymptotes.

5. Assume that as an ice cube melts it retains its cubical shape, and its volume decreases at a rate that is proportional to its surface area. If the cube loses $1/4$ of its volume during the first hour, how long will it take for the entire ice cube to melt?

Fall 2004 Midterm I

1. Evaluate the following limits: (*Do not use L'Hôpital's Rule.*)

a. $\lim_{x \rightarrow 5} \frac{x^3 - 24x - 5}{x^3 - 2x^2 - 75}$

b. $\lim_{x \rightarrow 8} \frac{1 - \cos(x^{1/3} - 2)}{(x - 8)^2}$

2. Find d^2y/dx^2 at the point $(x, y) = (1, -1)$ if y is a differentiable function of x satisfying

$$y^3 + x^2y + x = x^3 + 2y .$$

3a. Show that $\lim_{x \rightarrow 3} \frac{1}{x - 1} = \frac{1}{2}$ using the ε - δ definition of the limit.

3b. Suppose that for all $0 < \varepsilon < 1$,

$$|x - 1| < \frac{\varepsilon^2}{4} \implies |f(x) - 3| < \varepsilon$$

and

$$|x - 1| < \frac{\varepsilon}{35} \implies |g(x) - 4| < \varepsilon .$$

Find a real number $\delta > 0$ such that

$$|x - 1| < \delta \implies |f(x) + g(x) - 7| < \frac{1}{5} .$$

4a. Show that the equation $x^4 + 1 = 7x^3$ has at least 2 real roots.

4b. Find $\frac{dy}{dx}$ if $y = \sin^7(\cos(x^5))$.

5. In each of the following if there exists a function f whose domain is the entire real line and which satisfies the given condition, then give an example of such a function; otherwise just write DOES NOT EXIST. No further explanation is required.

a. $\lim_{x \rightarrow 1} f\left(\frac{x - 1}{x + 1}\right) \neq f(0)$

b. $y = f(x)$ is not differentiable at $x = -1$, but $y = (f(x))^2$ is differentiable at $x = -1$.

c. $y = f(x)$ is not continuous at $x = 5$, but it is differentiable at $x = 5$.

d. $\lim_{x \rightarrow 0} f(x) = 2$ and $f(0.000000000000001) = -1$.

e. $f(2) = 3$, $f(4) = -1$ and $f(x) \neq 0$ for all x .

Fall 2004 Midterm II

1. Sketch the graph of $y = x^3 - 6x^2 + 5$ by computing y' and y'' , determining their signs, finding the local extreme points, the inflection points and the intercepts.

2a. Let $x = \int_{1/2}^y \frac{dt}{\sqrt{t-t^3}}$. Find $\frac{dy}{dx}$ at a point where $\frac{d^2y}{dx^2} = 0$.

2b. A ball of radius 5 cm is dropped directly under a light bulb which is 3 m above the floor. If at the moment the ball is 35 cm away from the bulb it is moving at a speed of 2 m/sec, how fast is the radius of the shadow of the ball on the floor changing?

3. Evaluate the following integrals:

a. $\int_3^4 \sqrt{x^4 - 4x^3 + 4x^2 - 1} dx$

b. $\int_0^\pi \sin^3 x \cos^2 x dx$

c. $\int x(x+1)^{1/3} dx$

d. $\int \frac{\tan \theta}{\sqrt{\sec \theta}} d\theta$

4. A cylinder is drawn inside a sphere of radius 1. Find the maximum possible area of the cylinder.

5a. Find f if $f''(x) = 1$ for all x , $f(0) = 1$ and $f(2) = 2$.

5b. Show that if $g''(x) \leq 0$ for all x , $g(0) = 0$ and $g(2) = 0$, then $g(x) \geq 0$ for all $0 \leq x \leq 2$.

5c. Show that if $h''(x) \leq 1$ for all x , $h(0) = 1$ and $h(2) = 2$, then $h(x) \geq \frac{7}{8}$ for all $0 \leq x \leq 2$.

Fall 2004 Final

1. Evaluate the following integrals.

a. $\int_{-\ln 2}^0 \frac{dx}{e^x - 2e^{-x} - 1}$

b. $\int x \tan^2 x dx$

2. Evaluate the following integrals:

a. $\int_{-\pi/2}^{\pi/2} \frac{dx}{\sin x - 2 \cos x + 3}$

b. $\int \sqrt{1 + e^x} dx$

3a. Find the linearization of $f(x) = \arcsin x = \sin^{-1} x$ centered at $x = 1/2$.

3b. Evaluate the limit $\lim_{x \rightarrow 0} \left(\frac{(1+x)^{1/x}}{e} \right)^{1/x}$.

4. A solid cylinder whose axis is the x -axis intersects another solid cylinder whose axis is the y -axis. Both cylinders have radius 2 units. Find the volume of the part of the intersection that lies in the first octant.

5. Suppose that the function f is continuous, increasing and $f(x) \geq 0$ on $[0, \infty)$.

a. Write down an integral expressing the volume $V(a)$ of the solid generated by revolving the region between the graph $y = f(x)$ and the x -axis for $0 \leq x \leq a$ about the x -axis where a is a constant.

b. Write down an integral expressing the volume $W(a)$ of the solid generated by revolving the region between the graph $y = f(x)$ and the line $y = f(a)$ for $0 \leq x \leq a$ about the y -axis where a is a constant.

c. Find $f'(3)$ if $V(a) = W(a)$ for all $a \geq 0$ and $f(3) = 2$.

d. Find $f(x)$ for all $x \geq 0$ if $V(a) = W(a)$ for all $a \geq 0$ and $f(3) = 2$.

Fall 2003 Midterm I

1. For a real number t , let L_t be the line passing through the point $(-1, 0)$ with slope t , and let M_t be the line passing through the point $(1, 0)$ and perpendicular to L_t .

a. Find the intersection point P_t of the lines L_t and M_t .

b. Describe the curve traced by P_t for $-\infty < t < \infty$ using Cartesian coordinates.

2. Determine all values of the constant a for which the function $f(x) = e^x + ae^{-x}$ is one-to-one. For these values of a , find $f^{-1}(x)$ and the domain of f^{-1} .

3. Find the following limits:

a. $\lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h}$

b. $\lim_{x \rightarrow 9} \frac{x^3 - 729}{x - \sqrt{x} - 6}$

4a. Find the limit $\lim_{x \rightarrow 0^+} \frac{1 - \sqrt{\cos x}}{(1 - \cos \sqrt{x}) \sin x}$

- 4b. Prove that if $|f(x)| < 7$ for $0 < |x - 2| < 1$ and $\lim_{x \rightarrow 2} g(x) = 0$, then $\lim_{x \rightarrow 2} (f(x)g(x)) = 0$
5. Show that $\lim_{x \rightarrow 1} (x^2 + x) = 2$ using the ε - δ definition of the limit.

Fall 2003 Midterm II

1. Find the values of h , k and $r > 0$ that make the circle $(x - h)^2 + (y - k)^2 = r^2$ tangent to the parabola $y = x^2 + 1$ at the point $(1, 2)$ and that also make the second derivatives d^2y/dx^2 have the same value on both curves there.
2. A particle moves along the curve $2x^3 + 2y^3 = 9xy$. If the distance of the particle to the origin is increasing at a rate of 3 m/sec at the moment it passes through the point $(2, 1)$, how fast is its x -coordinate changing at this moment?
- 3a. Find y' if $y = \tan^3(x^2 + \cos(5x))$. (*Do not simplify!*)
- 3b. Assume that f is continuous on $[0, 2]$ and $f(0) = f(2) = 0$. Show that $f(c + 1) = f(c)$ for some c .
4. A cone is constructed from a disk of radius a by removing a sector AOC of arc length x and then connecting the edges OA and OC . Find the maximum possible volume of the cone.
5. Graph $y = \frac{x - 1}{\sqrt{2x^2 + 1}}$ by computing y' and y'' and determining their signs and the shape of the graph, finding the critical points, the inflection points and the intercepts, and finding the asymptotes.

Fall 2003 Final

1. Find the volume of the solid generated by revolving the region bounded by the curve $y = \ln x$, the x -axis and the line $x = e$ about
- a. the x -axis.
 - b. the y -axis.
2. Evaluate the following integrals:
- a. $\int_0^1 \frac{dx}{(x^2 + 1)^2}$
 - b. $\int \frac{dx}{e^x - e^{-x}}$

3. Evaluate the following limits:

a. $\lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$

b. $\lim_{x \rightarrow \infty} \left(\frac{2}{\pi} \arctan 3x \right)^x$

4. Find $\frac{dy}{dx}$ at $(x, y) = (1, 2)$ if $x^y + y^x = 3$.

5a. Show that $|\arctan a - \arctan b| \leq |a - b|$ for all a, b .

5b. Find $f''(1)$ if $f(x) = x \int_0^x \frac{t^2}{\sqrt{1+t^4}} dt$.

Fall 2002 Midterm I

1) a) Find $\lim_{x \rightarrow 0^+} \frac{\sqrt{\sin 2x} - \sqrt{\sin x}}{\sqrt{x}}$.

b. Use the ε - δ definition of the limit to show that $\lim_{x \rightarrow 0} (\sin(\sqrt[3]{x}) \sin(1/x)) = 0$.

2. Let f be a differentiable function such that $xf(x) + f(x^2) = 2$ for all $x > 0$.

a. Find $f'(1)$.

b. Show that if $f(x_0) = 0$ for some $x_0 > 1$, then there exists $x_1 > x_0$ such that $f(x_1) = 0$.

3. Find the derivatives of the following functions:

a. $y = \tan^5(\pi x^3)$

b. $y = \sqrt[3]{\frac{x}{x^2 + 1}}$

4. Find y'' at the point $(x, y) = (\pi/2, 0)$ if y is a differentiable function of x satisfying the equation $y^2 = x \cos(x + y)$.

5. Points A and B move along the x - and y -axes, respectively, in such a way that the distance r (meters) along the perpendicular from the origin to the line AB remains constant. How fast is OA changing, and is it increasing, or decreasing, when $OB = 2r$ and B is moving toward O at the rate of $0.3r$ m/sec?

Fall 2002 Midterm II

1. Graph $y = \frac{x^2 - 4}{x + 1}$ by computing y' and y'' and determining their signs and the shape of the graph, finding the critical points, the inflection points and the intercepts, and finding the asymptotes.

2. Consider the area of the region between the parabola $y = x^2$ and the line $y = c$ for $0 \leq x \leq 1$ as a function of c . Find the absolute maximum and the absolute minimum values of this function on the interval $0 \leq c \leq 1$.

3a. Evaluate the limit $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(n+k)^2}$.

3b. Evaluate $y' + y''$ at $x = \pi/4$ if f is a differentiable function with $f'(1/\sqrt{2}) = 4$ and $y = \int_0^{\sin x} f(t) dt$.

4. Evaluate the following integrals:

a. $\int \frac{x^3}{\sqrt{x^2 - 9}} dx$

b. $\int_{\pi^2/36}^{\pi^2/4} \frac{\cos(\sqrt{\theta})}{\sqrt{\theta} \sin^2(\sqrt{\theta})} d\theta$

5. Find the area of the region enclosed by the curves $y^4 = 2y^3 + x$ and $(y - 1)^2 = x + 1$.

Fall 2002 Final

1. Let c be a positive constant. Find the point on the curve $y = \sqrt{x}$ closest to the point $(c, 0)$.

2. Find the volume generated by revolving the region enclosed by the curves $y = \frac{1}{\sqrt{3+x^2}}$

and $y = \frac{x^2}{2}$ about

a. the x -axis

b. the y -axis

3a. Find $h''(1)$ if $h(x) = (g(x^2))^3$ and $g^{(n)}(1) = n + 1$ for $n \geq 0$.

3b. Find $f'(1)$ if $\int_1^x f(t) dt = x + \ln(f(x))$ for all x .

4. Evaluate the following limits:

a. $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{3x}\right)^x$

b. $\lim_{x \rightarrow 1} \frac{\int_1^x t^t dt}{x^{x+1} - 1}$

5. Evaluate the following integrals:

a. $\int \frac{(1 + e^x)^2}{1 + e^{2x}} dx$

b. $\int_0^1 \sqrt{\frac{1-x}{1+x}} dx$

Fall 2001 Midterm I

1. Find the following limits: (*Do not use L'Hôpital's Rule.*)

a. $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x^2 + x - 2}$

b. $\lim_{x \rightarrow 0} \frac{\sqrt{2 - \cos x} - 1}{\sin^2 \pi x}$

2. Find the derivatives of the following functions:

a. $y = \left(\frac{\sin x}{1 + \cos x} \right)^2$

b. $y = \sqrt{x} \sin(\sqrt{x})$

3. Find y'' at the point $(1, 1)$ if y is a differentiable function of x satisfying the equation $x^3 + y^3 = 6xy - 4$.

4. a. Let A and B the points of intersection of the tangent line to the curve $y = 1/x$ at a point $P(x_0, y_0)$ with the y - and x -axes, respectively. Find the length of the line segment AB as a function of x_0 .

b. Suppose that the point P is moving along the curve $y = 1/x$. If at the moment P passes through the point $(2, 1/2)$, the length of the line segment AB is increasing at a rate of 4 mm/sec, how fast is x_0 changing?

5. Graph $y = 4x^3 - x^4$ by computing y' and y'' , finding the critical points and the inflection points, determining the signs of the derivatives, determining the shape of the graph and finding the intercepts.

Fall 2001 Midterm I

1. An 8 feet high wall is 27 feet away from a building. Find the length of the shortest straight beam that will reach to the side of the building from the ground outside the wall.

2. Suppose that u is a continuous function and a is a real constant satisfying

$$\int_0^x u(t) e^{a(t^2-x^2)} dt = \sin x$$

for all x . Find a if $u'(\pi/2) = 5$.

3. Find the volumes of the solids generated by revolving each of the following regions about the x -axis.

a. The region lying between $y = \cos x$ and $y = \sin x$ for $0 \leq x \leq \pi/3$.

b. The region bounded on the left by $x = 3y^2 - 2$, on the right by $x = y^2$, and below by the x -axis.

4. Find the following integrals:

a. $\int \frac{x}{\sqrt{1+x^2} + \sqrt{(1+x^2)^3}} dx$

b. $\int_0^{\ln 3} \frac{1}{e^x + 2} dx$

5. Let $g(x) = x^{-5}$. Find a function f such that $f(1) = 1$ and

$$\frac{d}{dx} (f(x)g(x)) = \frac{d}{dx} (f(x)) \cdot \frac{d}{dx} (g(x))$$

for all $x > 0$.

Fall 2001 Final

1. Find the following limits.

a. $\lim_{x \rightarrow 0^+} (\sin(x^2))^{1/\ln x}$

b. $\lim_{x \rightarrow 0} \left(\frac{1}{x^3} \int_0^x \frac{e^{t^4} - e^{-t^4}}{t^2} dt \right)$

2. Evaluate the following integrals.

a. $\int x^2 (\ln x)^2 dx$

b. $\int \frac{1}{(x^2 + 4x + 5)^2} dx$

3. Evaluate the improper integral $\int_2^\infty \frac{x+3}{(x-1)(x^2+1)} dx$.

4. Graph the function $f(x) = xe^{-x}$ by finding the intervals on which it is increasing, decreasing, concave up, concave down; its critical points and inflections points, and its asymptotes.

5. Let $f(x) = x^{1/x}$ for $x > 0$.

a. Find the absolute maximum value of $f(x)$.

b. Show that $\sqrt{2} \leq \int_2^3 f(x) dx \leq \sqrt{3}$.

Fall 2000 Midterm I

1. Evaluate the following limits. (*Do not use L'Hôpital's Rule.*)

a. $\lim_{x \rightarrow 0} \frac{x}{\sqrt{1 + \tan 3x} - 1}$

b. $\lim_{x \rightarrow 0} \frac{\sin 4x - 2 \sin 2x}{x^3}$

2. Let

$$f(x) = \begin{cases} 1 + (x - 1) \sin \left(\frac{1}{x - 1} \right) & \text{if } x > 1, \\ |x|/x & \text{if } 0 \neq |x| \leq 1 \\ -1 + \sin \left(\frac{1}{x + 1} \right) & \text{if } x < -1. \end{cases}$$

Determine if each of the following limits exists. Explain your reasoning in detail.

a. $\lim_{x \rightarrow 1} f(x)$

b. $\lim_{x \rightarrow 0} f(x)$

c. $\lim_{x \rightarrow -1} f(x)$

3. Find the equation of the tangent line of the curve $y = \sqrt{x}$ which is perpendicular to the line $2x + 3y = 5$.

4. A particle moves along the curve $y = x^3 + 1$ in such a way that its distance from the origin increases at a rate of 7 units per second. Find the rate of change of the x -coordinate of the particle when it passes through the point $(1, 2)$.

5. Assume that the function f satisfies the equation $f'(x) = f(x/2)$ for all x , and $f(0) = 1$.

a. Show that if $f(x_0) = 0$ for some $x_0 > 0$, then there is x_1 such that $0 < x_1 < x_0$ and $f(x_1) = 0$.

b. Find $f'''(0)$.

6. Graph the function $y = x^{4/3} + 4x^{1/3}$ by computing y' and y'' , determining their signs, and finding the extreme points, the inflection points and x -intercepts.

Fall 2000 Midterm II

1. A wire of length L is bent to form the shape where the sides of the square have length a and the radius of the semicircle is r , and then revolved about the line containing the fourth side of the square to sweep out the surface of a sphere and a cylinder. Find r and a which give the maximum surface area.

(1. This question requires no integration: you can use the surface area formulas for these geometric objects. 2. Note that the bottom and the top of the cylinder as well as its side are included in the surface area.)

2. a. Evaluate the limit $\lim_{x \rightarrow 0^+} \int_0^x \frac{1}{(x-t)^4 + x^4} dt$.

b. If $f(x) = \int_2^{\sqrt{x}} \sqrt{9+t^4} dt$, find $f'(4)$.

3. Evaluate the following integrals:

a. $\int_0^{\pi/2} \frac{\sin 2x}{(a \sin^2 x + b \cos^2 x)^2} dx$ where $a, b > 0$.

b. $\int (1 + x^{-3/4})^{1/3} dx$.

4. a. If $y' = \frac{\sin^3(1/x) \cos(1/x)}{x^2}$ and $y(3/\pi) = 1$, find $y(4/\pi)$.

b. Evaluate the limit $\lim_{n \rightarrow \infty} \left(\frac{1}{2n+1} + \frac{1}{2n+3} + \cdots + \frac{1}{4n-3} + \frac{1}{4n-1} \right)$.

5. a. Find the volume of the solid obtained by revolving the region between the curve $y = x^2 - x^3$ and the x -axis about the y -axis.

b. Find the area of the smaller piece if a sphere with radius 3 is cut into two by a plane passing 2 units away from its center.

6. Show that $\int_0^\pi \frac{t \sin^2 t}{1+t^2} dt \leq \frac{\pi}{4}$.

Fall 2000 Final

1. Consider the rectangles bounded by the axes and the lines perpendicular to the axes through a point on the curve $y = (\ln x)^4/x^3$ for $x \geq 1$. Find the maximum value of the area of these rectangles if it exists.

2. Sketch the graph of the function $y = (x^2 - 2x)e^x$ by computing y', y'' , determining the shape of the graph, finding the critical points, the inflection points, and the x -intercepts, and the asymptotes.

3. a. Find the volume of the solid obtained by revolving the region between the graph of $f(x) = \begin{cases} \tan x/x & \text{if } 0 < x \leq \pi/4 \\ 1 & \text{if } x = 0 \end{cases}$ and the x -axis about the y -axis.

b. Evaluate the improper integral $\int_0^\infty \frac{dx}{a + be^x}$.

4. Evaluate the integrals:

a. $\int_{-\pi/2}^{\pi/2} \frac{\cos x}{(9 + 2 \sin x)(5 + \sin x)} dx$

b. $\int (\arcsin x)^2 dx$

5. Evaluate the following limits:

a. $\lim_{x \rightarrow \infty} x \int_0^1 e^{-t^2/x^2} \ln \left(1 + \frac{t}{x} \right) dt$

b. $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$

6. Let f be a continuous function and define

$$g(x) = \int_{-1}^1 f(t)|x - t| dt$$

for all x . Express $g''(x)$ in terms of $f(x)$ for x in $(-1, 1)$.

Fall 1999 Midterm I

1. Evaluate the following limits: (You are not allowed to use the L'Hôpital's Rule.)

a. $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x(\cos^3 2x - 1)}$

b. $\lim_{x \rightarrow 2} \frac{\sin(x - 2)}{\sqrt{x + 3} - \sqrt{2x + 1}}$

2. Find $f'(8)$ if the function $f(x)$ satisfies the following properties:

- $f(ab) = f(a) + f(b)$ for all $a, b > 0$.

- $\lim_{x \rightarrow 1} \frac{f(x)}{x-1} = 2$.

3. Sketch the graph of $y = x^{-1/3} - x^{-2/3}$ by computing y' and y'' , determining their signs, finding the critical and the inflection points, indicating the intervals on which the function is increasing, decreasing, concave up, concave down, and finding the asymptotes.

4. Find y''' at $(x, y) = (1, 0)$ if $\sin(xy) + x = 1$.

5. a. The length of the hypotenuse of a right triangle is constant at 5 cm, and the length of one of its sides is increasing at a rate of 2 cm/sec. Find the rate of change of the area of the triangle, when this side is 4 cm long.

b. Show that for any twice differentiable function $f(x)$ on the entire real line such that $f(0) = 0, f(1) = 1, f(2) = 4$, there exists a point c in $(0, 2)$ such that $f''(c) = 2$.

Fall 1999 Midterm II

1. A 36π m³ container with cylindrical side and hemispherical ends is going to be constructed. The construction of the cylindrical side surface costs 3×10^9 TL/m² and the construction of the hemispheres costs 10×10^9 TL/m². Find the minimum cost of constructing this container.

2. Show that the function $f(x) = \int_{e^{-x}}^{e^x} \frac{\sin t}{t} dt$ has a critical point at $x = a$ where a is the positive real number satisfying $e^a + e^{-a} = 2\pi$ and determine whether this is a local maximum or minimum.

3. a. Find the volume of the solid generated by revolving the region between the curve $y = x^3$ and the x -axis for $0 \leq x \leq 1$ about the y -axis.

b. Find the area of the surface generated by revolving the curve $y = x^3$ for $0 \leq x \leq 1$ about the x -axis.

4. a. Evaluate the integral $\int_1^8 \frac{1}{x + 5\sqrt[3]{x}} dx$.

b. Show that $\frac{5\pi}{24} \leq \int_{\pi/3}^{\pi/2} \sqrt{1 + \sin^4 x} dx \leq \frac{\pi}{3\sqrt{2}}$.

5. Evaluate the limit

$$\lim_{x \rightarrow \infty} \left(1 + \ln(\cos(a/x)) \right)^{x^2}$$

where a is a constant.

Fall 1999 Final

1. a. Let

$$f(x) = \begin{cases} \frac{1}{x^2} \int_0^{|x|} \sin t^2 dt & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Does $f'(0)$ exist? Explain.

b. Suppose $g(x)$ and $h(x)$ are differentiable functions on $[0, 1]$ and $g(0) = 4, g(1) = 1, h(0) = 3, h(1) = 0$. Show that there exists a point c in $(0, 1)$ such that $g'(c) = h'(c)$.

2. a. Find constants a and b such that $\lim_{x \rightarrow \infty} (xe^{1/x} - (ax + b)) = 0$.

b. Sketch the graph of $y = xe^{1/x}$.

3. Find $\int_{\pi/2}^{3\pi/2} f(x) dx$ if $f'(x) = \frac{\cos x}{x}$, $f(\pi/2) = a$ and $f(3\pi/2) = b$.

4. Evaluate the following integrals:

a. $\int \frac{1}{2x^2 + 5x + 2} dx$

b. $\int x \tan^2 x dx$

5. Evaluate the integral $\int_{-\infty}^{\infty} \frac{1}{e^x + ae^{-x}} dx$ where a is a positive constant.

Fall 1997 Midterm I

1. Assume the functions f and g have the following properties: for every ε and x ,

$$\begin{aligned} &\text{if } 0 < |x - 2| < \frac{3\varepsilon}{2} \text{ then } |f(x) - 3| < \varepsilon, \\ &\text{and if } 0 < |x - 2| < \varepsilon + \sqrt{\varepsilon} \text{ then } |g(x) - 4| < \varepsilon. \end{aligned}$$

Find a number $\delta > 0$ such that for all x ,

$$\text{if } 0 < |x - 2| < \delta \text{ then } |f(x) + g(x) - 7| < \frac{1}{50}.$$

2a. Evaluate the limit $\lim_{x \rightarrow 0} \frac{\cos^3 3x - \cos^3 2x}{x^2}$.

- 2b.** Find $\frac{d^2y}{dx^2}$ by implicit differentiation at the point $(1, -1)$: $x \cos^2(\pi y) + y^3 = 0$.
- 3.** Graph the function $f(x) = \frac{x^3 - x^2 + 3}{x^2 - 3}$ by computing f' and f'' , determining their signs, finding the critical points and inflection points, indicating the intervals on which the function is increasing, decreasing, concave up, down, and finding the asymptotes.
- 4.** A sphere of radius r is inscribed into a cone. Find the radius and the height of the cone (in terms of r) if its volume is to be minimum.
- 5.** A point A is moving in the counterclockwise direction along the ellipse $\frac{x^2}{9} + \frac{y^2}{16} = 1$ so that the ray OA has constant angular speed of 6 radians per second (where O is the origin). Find the rate of change of the area of the triangle formed by OA , the x -axis and the perpendicular from A to the x -axis at a moment when A is $(-12/5, 12/5)$.

Fall 1997 Midterm II

- 1.** Let the function f be defined as follows:

$$f(x) = \begin{cases} \int_0^x t \sin(1/t) dt & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Find $f'(0)$.

- 2. a.** Evaluate the limit: $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x + x^2/2}{x^3}$.

b. Find $\frac{dy}{dx}$ where $y = (\ln x)^{\ln x}$.

- 3.** Let C_1 be the graph of $y = x^m$, C be the graph of $y = kx^m$, and C_2 be the graph of $y = f(x)$ where $m > 0$ and $k > 1$ are constants, and f is an increasing continuous function with $f(0) = 0$. Find f if C bisects C_1 and C_2 in the following sense: for every (a, b) on C , the area of the region between C and C_1 for $0 \leq x \leq a$ is equal to the area of the region between C and C_2 for $0 \leq y \leq b$.

- 4.** The top and the bottom of a truncated rectangular pyramid of height h have sides of length a and b , and A and B , respectively. Find the volume.

- 5.** The plane region between the parabolas $y = 4 - x^2$ and $y = 8 - 2x^2$ has density $\delta(x, y) = 3y$. Find the coordinates of the center of mass.

Fall 1997 Final

1. Evaluate the following limits:

a. $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}.$

b. $\lim_{n \rightarrow \infty} \int_0^n \frac{(1 + \pi/x)^x}{n} dx.$

2. Evaluate the following integrals. You may not use any reduction formulas or formulas from the integration tables.

a. $\int \frac{x dx}{\sqrt{x^2 + 2x}}.$

b. $\int_0^\infty \frac{x^2 + x}{(x^2 + 1)^2} dx.$

3. Let R be the region in the xy -plane bounded by the curve $y = 1/x^{3/2}$, the x -axis, the lines $x = u$ and $x = u^2$ for $u \geq 1$. Assume that R is revolved about the x -axis and a solid is generated. Let $V(u)$ denote the volume of this solid. Find the maximum value of $V(u)$ for $u \geq 1$.

4. An irregularly shaped water container is 5 units deep. It has a flat bottom and cross-sectional areas $A(z) = (z^3 + 2z + 15)^{1/3}$ units³ where z is the depth measured from the bottom and the cross-sections are taken by planes parallel to the bottom. Initially the container is full of water. Then the water is pumped out of the container at the constant rate of 0.3 units³/sec. How fast is the depth of the water decreasing when water is 2 units deep?

5. a. Determine whether the following improper integral converges or diverges:

$$\int_0^\infty \sqrt{x^2 + 1} e^{-x} dx$$

b. Let $p > q > 0$ be constants. Find all possible values of p and q for which the following improper integral is convergent:

$$\int_0^\infty \frac{dx}{x^p + x^q}$$

Fall 1996 Midterm I

1. Show that $(1 + x)^\alpha < 1 + \alpha x$ for all $x > 0$ and $0 < \alpha < 1$.

2. Find $(f \circ g \circ h)'(0)$ if $f(x) = \cos x$, $g(y) = \frac{\pi}{6} y^2 + y - 1$ and $h(z) = \frac{1}{\sqrt{1 + z + z^2}}.$

3. Consider the curve $y = \frac{3x^2 + 1}{2x^2 + 5}$. Find the points where the slope of this curve assumes its absolute maximum and absolute minimum.

4. A silo (base not included) is to be constructed in the form of a cylinder surmounted by a hemisphere. The cost of construction per square unit surface area is twice as great for the hemisphere as it is for the cylindrical sidewall. Determine the dimensions to be use if the volume is fixed and the cost of construction is to be kept to a minimum.

5. Let

$$f(x) = \begin{cases} 2x + x^2 \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Find $f'(x)$ for all x . Is f' continuous at $x = 0$? Explain.

Fall 1996 Midterm II

1. Find $y'(1)$ if $x^y + y^x = 3$.

2. Find the area of the surface of revolution obtained by rotating the curve $y = \frac{1}{2} \left(\frac{1}{1-\alpha} x^{1-\alpha} - \frac{1}{1+\alpha} x^{1+\alpha} \right)$, $0 \leq x \leq 1$, about the x -axis where $0 < \alpha < 1$.

3. Evaluate the integral $\int_0^1 \frac{x^2 + 1}{x^2 - x - 6} dx$.

4. Evaluate the limit $\lim_{n \rightarrow \infty} \left(\ln n - \frac{1}{n} \sum_{k=n+1}^{2n} \ln k \right)$.

5. Evaluate the limit $\lim_{x \rightarrow \infty} \left(x \int_0^1 \frac{e^{-xt}}{1+t^2} dt \right)$.

Fall 1996 Final

1. If $f(x) = x \int_e^x \frac{dt}{\ln t}$ for $x > 1$, find $f''(e)$.

2. For $0 \leq t \leq 1$, let $A(t)$ denote the area of the triangle bounded by the x -axis, the y -axis and the tangent line to the curve $y = \ln x$ at $(t, \ln t)$. Find the maximum value of $A(t)$.

3. Sketch the graph of $y = x(\ln x)^2$.

4. [missing]

5. Evaluate the improper integral $\int_1^\infty \left(\frac{1}{x} + \tan^{-1} x - \frac{\pi}{2} \right) dx$.

Fall 1995 Midterm I

1. Let $A(0, t^2/\sqrt{3})$ and $B(4, 0)$ where the time $0 \leq t < \infty$ is measured in seconds. Let θ be the measure of the angle OBA . At what times is θ changing the fastest and the slowest?

2. Find y'' at $x = 0$ if $\sin xy + x^2 + y = \pi$.

3. Find the constant k for which the limit

$$\lim_{x \rightarrow 0} \frac{\sin(x + kx^3) - x}{x^5}$$

exists, and find the value of the limit then for this k .

4. You are drawing up plans for the piping that will connect a drilling rig 30 kilometers offshore to a refinery on shore 35 kilometers down the coast. How far away from the refinery should the pipeline reach the coast to give the least expensive connection if underwater pipe costs 5 BTL per kilometer and land based pipe costs 4 BTL per kilometer. (1 BTL=1 billion Turkish Liras.)

5. Graph the function $y = \frac{3x^3 - 2x}{x^2 - 1}$ by finding its asymptotes, local minimums and maximums, inflection points and intercepts.

Fall 1995 Midterm II

1. Let D_n denote the region that is under the curve $y = \frac{1}{x^2\sqrt{x^2-1}}$, above the x -axis and between the lines $x = 1 + \frac{1}{n}$ and $x = n$, where n is a positive integer. Revolve the region D_n around the y -axis and denote the volume of the resulting solid by V_n . Find $\lim_{n \rightarrow \infty} V_n$.

2. For which value of α , $\int_{\pi/6}^{\pi/3} (\log_{\alpha} 2)(\log_2 \sin x)' \tan x \, dx = \frac{\pi}{6}$?

3. Evaluate the limit $\lim_{x \rightarrow \infty} x \int_0^x e^{t^2 - x^2} \, dt$.

4. Consider the region inside the circle $x^2 + (y - 2)^2 = 1$ and between the lines $x = -1/2$ and $x = 1/2$. Find the surface area of the solid obtained by revolving this region around the x -axis.

5. Graph the function $y = e^{1/x}$ by finding its

a. asymptotes and $\lim_{x \rightarrow 0^-} y'$,

b. local minimums and maximums,

- c. inflection points,
- d. making a table of the relevant values.

Fall 1995 Final

1a. Explain whether $\int_{-\infty}^{\infty} \frac{x^2}{e^x + e^{-x}} dx$ converges or diverges.

1b. Evaluate $\int_{10}^{\infty} \frac{dx}{x(\ln x)(\ln \ln x)}$.

2a. Find $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$.

2b. Given that $t \rightarrow \infty$, simplify $(t - t^{1/2} + O(1)) \cdot (t^{1/2} + 1 + O(t^{-1/2}))$.

2c. Explain briefly whether true or false: As $n \rightarrow \infty$,

$$\frac{1}{n+1} - \frac{1}{n+2} = o(n^{-3/2}).$$

3. According to Newton's law of cooling, the rate at which the temperature of an object is changing at any given time is proportional to the difference between its temperature and the temperature of its environment (which is assumed to be constant). A pan of water at 25°C was put in a refrigerator. Ten minutes later, the temperature of the water was 13° ; 20 minutes after that, it was 4° . Use Newton's law of cooling to calculate how cold the refrigerator was.

4. Evaluate $\int_{1/2}^2 \frac{dx}{x^3 + 6x + 7}$.

5. The region bounded by the curve $y = (\arctan x)^2$, the x -axis and the line $x = 1$ is revolved about the y -axis to generate a solid. Find the volume of this solid.

Fall 1994 Midterm I

1. The graph of f is as shown. Draw the graph of the function $g(x) = \frac{f(x)}{1 + f(x)}$. Indicate the asymptotes, extrema and the inflection points.

[The picture shows a graph $y = f(x)$ such that $f(-4) = 0$, $f(-3) = -1$, $f(-2) = -2$, $f(-1) = -1$, $f(0) = 0$, $f(2) = 2$, $f(3) = 0$, $f(7/2) = -1$; f is increasing on $[-2, 2]$ and decreasing on $(-\infty, -2]$ and $[2, \infty)$. It is not clear from the picture if f is piecewise-linear.]

2. Find all positive integer values of n for which $\cos(x^{2/n})$ is differentiable at $x = 0$.

- 3.** Estimate $\sin 29^\circ$ by using the quadratic approximation. Give an upper bound for the error.
- 4.** How many real solutions to $x^3 + 3x - 5 = 0$ are there? Apply Newton's Method two steps to locate any root.
- 5.** A straight line passing through the point $(1/2, 1)$ divides the rectangular region with corners at $(0, 0)$, $(2, 0)$, $(2, 4)$, $(0, 4)$ into two subregions. Among all triangular subregions with the origin as one vertex that are formed by such a division, find the ones which have maximum and minimum areas.

Fall 1994 Midterm II

- 1a.** Given that $g(x) = \int_1^{f^{-1}(x)} \sqrt{1+t^3} dt$ and $f(3) = 2$, $f'(3) = 5$, find $g'(2)$.
- 1b.** Find the volume of the solid described as follows: The base of the solid is the region between the curve $y = 3\sqrt{\sin x}$ and the interval $[0, \pi]$ on the x -axis. The cross-sections perpendicular to the x -axis are squares.
- 2a.** Find $\lim_{n \rightarrow \infty} \left(\frac{1}{5n^2 + 1} + \frac{1}{5n^2 + 2} + \cdots + \frac{1}{6n^2} \right)$.
- 2b.** Find $\lim_{x \rightarrow \infty} \frac{\sqrt{\ln x}}{x} \int_2^x \frac{dy}{\sqrt{\ln y^2}}$.
- 3.** Find the centroid of the quarter ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $x, y \geq 0$.
- 4.** Solve the initial value problem:
- $$\frac{1}{y+2} \frac{dy}{dx} = x \cos^2 x^2, \quad y = e^2 - 2 \text{ when } x = 0.$$
- 5a.** Estimate $\ln 0.6$ by quadratic approximation.
- 5b.** Estimate $\ln 0.6$ by using Simpson's rule with $n = 4$.

Fall 1994 Final

- 1a.** Evaluate $\int_0^{\ln \sqrt{3}} e^{-x} \arctan(e^x) dx$.
- 1b.** Evaluate $\int \frac{dx}{(x^2 + 2x + 2)^{3/2}}$.

2. Evaluate $\int x^5 e^{x^3} dx$.

3a. Determine whether $\int_0^\infty \frac{dx}{\sqrt{x^3+1}}$ converges or diverges. Give reasons.

3b. For a certain number A the integral $\int_2^\infty \left(\frac{Ax}{x^2+1} - \frac{1}{2x+1} \right) dx$ converges. Find that value of A and the integral.

4. A boat is going in a straight line at 80 m/min when the engine of the boat is shut off. The boat is subject to a deceleration (i.e. negative acceleration) proportional to its speed. One minute later the speed of the boat is reduced to 40 m/min. How far has the boat drifted in that one minute?

5. Consider $y = \operatorname{sech} x$.

a. Plot the graph of the function by considering the asymptotes and the signs of y' and y'' .

b. Find the quadratic approximation of $y = \operatorname{sech} x$ about $x = 0$.

c. Find the volume of the solid obtained by revolving the region between $y = \operatorname{sech} x$ and the x -axis about the x -axis.

Fall 1993 Midterm I

1a. Let $f(x) = x + \frac{\sin x}{2x - \frac{12}{x-1}}$.

i. Find all the points at which f is discontinuous.

ii. Find all discontinuities at which f has a continuous extension.

1b. Let f be a continuous function and assume that for each $0 \leq x \leq 1$ we have $0 \leq f(x) \leq 1$. Show that the equation $f(x) = x$ has a solution in the interval $[0, 1]$.

2a. Let $y = t + \sin^2 u$ where at $t = 0$, $u = \sqrt{\frac{\pi}{3}}$ and $\frac{du}{dt} = 1$. Find y and $\frac{dy}{dt}$ at $t = 0$.

2b. Consider the equation $\cot \pi x = x$.

i. Show that this equation has exactly one solution r_k in $I_k = (k, k+1)$ for every $k \in \mathbb{Z}$.

ii. Apply Newton's method once to estimate r_k .

3a. Let $f(x) = \sqrt{x}$.

i. Find the quadratic approximation of $f(x)$ for $x \approx 1$.

ii. Determine the accuracy of this approximation when $\frac{3}{4} \leq x \leq \frac{5}{4}$.

- 3b.** Show that the equation $x^8 + x - 1 = 0$ has exactly two real roots.
- 4a.** Evaluate $\lim_{x \rightarrow \infty} (\sqrt{3x^2 + 2x - 1} - \sqrt{3x^2 - x + 5})$.
- 4b.** Let $f(x) = x^{-3} \sin 3x + Ax^{-2} + B$ where A and B are constants. Determine the values of A and B for which $\lim_{x \rightarrow 0} f(x) = 2$.
- 5.** Graph the function $f(x) = \frac{x^2}{4 + x|x|}$ by computing f' and f'' ; determining their signs; finding the critical points and the inflection points; indicating the intervals on which the graph is rising, falling, concave up, down; and finding the asymptotes.
- 6.** If the sum of the volumes of a cube and a sphere is held constant, what ratio of an edge of the cube to the radius of the sphere will make the sum of the surface areas
- a.** as small as possible?
 - b.** as large as possible?

Fall 1993 Midterm II

- 1a.** Find $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{3n} \right)$.
- 1b.** Given that $f(1) = 3$, $f(2) = 5$, $f(4) = 7$ and $f(14) = 23$, evaluate $\int_1^2 (x^2 + 1)f'(x^3 + 3x) dx$.
- 2a.** Find $\lim_{y \rightarrow 0} \frac{1}{y} \int_0^y (\cos 2t)^{1/t^2} dt$.
- 2b.** If $x^2 y'' = 1$ for all $x > 0$ and $y' = -1$, $y = 1$ when $x = 1$, find y for all $x > 0$.
- 3a.** Estimate $\ln(1+2x)$ about $x = 0$ by using the quadratic approximation and calculate $\ln(1.4)$ approximately.
- 3b.** Estimate $\ln(1.4)$ using Simpson's method by choosing 4 subintervals.
- 4a.** The curve $x = y^{1/2} - y^{3/2}/3$ where $0 \leq y \leq 1$ is rotated about the y -axis. Find the area of the surface of revolution.
- 4b.** Find the x -coordinate of the centroid of the curve in part **(a)**.
- 5a.** Find the area of the region bounded by the curves $y = \tan x$, $y = 0$, $y = 1$ and $x = \pi/2$.
- 5b.** Find the volume of the solid generated by revolution of the region in part **(a)** about the x -axis.

6a. Plot the graph of the function $y = x^2 e^x$.

6b. Find $\int \frac{dx}{1 + 2e^{-3x}}$.

Fall 1993 Final

1. a. Draw the graph of $y = \operatorname{sech} x$ by computing y' , y'' , and finding the critical points, the inflection points, and the asymptotes.

b. Find the area between the curves $y = \operatorname{sech} x$ and $y = \frac{3}{4} \cosh x$.

2. Find the volume generated by revolving the region bounded by $y^2 = x^5 - x^8$ about the y -axis.

3. a. Find the quadratic approximation of the function $f(x) = x^x - \frac{1}{2-x}$ around $x = 1$.

b. Estimate the value of x^x for $x = 10/11$.

4. Let $f(x) = \int_1^e t^x (\ln t)^2 dt$.

a. Compute $f(x)$.

b. Prove that f is continuous at $x = -1$.

5. Evaluate $\int_0^\infty \frac{x dx}{(x+1)(x^2+2)}$.

6. a. Find y as a function of x if $x \frac{dy}{dx} = \tan y$ and $y = \frac{\pi}{6}$ when $x = \frac{1}{2}$.

b. Prove that

$$\int_0^x \left(\int_0^u f(t) dt \right) du = \int_0^x f(u)(x-u) du.$$

Fall 1992 Midterm I

1. Let $f(x) = g(x + g(x))g(x)$ where g is such that $g(0) = 2$, $g(1) = 1$, $g(2) = -5$, $g'(0) = -3$, $g'(1) = 7$, $g'(2) = 4$. Find $f'(0)$.

2. Find the normal to the curve $y = 4x^2 + \sin(\pi \tan(x^2))$ at the point where $x = \frac{\sqrt{\pi}}{2}$.

3a. Consider $f(x) = \sqrt{x^2 + x} - \sqrt{x^2 - x}$.

i. Find the domain of the function f .

ii. Evaluate $\lim_{x \rightarrow -\infty} f(x)$.

3b. Evaluate $\lim_{x \rightarrow 0} \frac{\sin \alpha x - \tan \alpha x}{x^3}$ where α is a constant.

4. a. Find the values of α , β , γ for which the function

$$f(x) = \begin{cases} 2 & \text{if } x = 0, \\ -x^2 + 4x + \alpha & \text{if } 0 < x < 1, \\ \beta x + \gamma & \text{if } 1 \leq x \leq 3. \end{cases}$$

is differentiable on $[0, 3]$.

b. Find the local maxima and minima of f for these values of α , β , γ .

5. A 3 m long wire is used for making a circle and an equilateral triangle. How should the wire be distributed between the two shapes to maximize the sum of the enclosed areas?

6. Graph the function $f(x) = \frac{x}{x^3 + 1}$ by computing f' and f'' ; determining their signs; finding the critical points and the inflection points; indicating the intervals on which the graph is rising, falling, concave up, down; and finding the asymptotes.

Fall 1992 Midterm II

1. a. Find the quadratic approximation at $x = 0$ of

$$f(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt.$$

b. Find the error in this approximation for $|x| \leq 1$.

2. a. A solid is generated by revolving the region bounded by the graph of $y = f(x)$, the x -axis, the lines $x = 1$ and $x = a$ about the x -axis. Its volume, for all $a \geq 1$, is $a^2 - a$. Find $f(x)$.

b. A surface is generated by revolving the curve $y = \frac{\sqrt{x}}{3}(3 - x)$, $0 \leq x \leq 3$, about the x -axis. Find its area.

3. a. Using the Mean Value Theorem for Definite Integrals, find upper and lower bounds for the value of $\int_0^1 \frac{dx}{x^3 + 1}$.

b. Obtain better bounds by considering

$$\int_0^1 \frac{dx}{x^3 + 1} = \int_0^{1/2} \frac{dx}{x^3 + 1} + \int_{1/2}^1 \frac{dx}{x^3 + 1}$$

and using the same theorem for the integrals on the right.

4. Evaluate:

a. $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{1/x^2}$

b. $\lim_{x \rightarrow 0} \frac{x}{5^x - 3^x}$

5. Evaluate:

a. $\int_0^{\pi/2} \frac{\sin x \, dx}{3 + \cos^2 x}$

b. $\int_e^{e^2} \frac{dx}{x \ln x}$

6. a. Show that the difference of the functions $f(x) = \sin^{-1} \left(\frac{x^2 - 1}{x^2 + 1} \right)$ and $g(x) = 2 \tan^{-1} x$ is a constant for $x \geq 0$.

b. Find this constant.

Fall 1992 Final

1. Let f and g be continuous on $[a, b]$ and that $f(a) \leq g(a)$ and $f(b) \geq g(b)$. Show that there exists a point c in $[a, b]$ such that $f(c) = g(c)$.

2. a. Given that

$$\int_0^\pi (f(x) + f''(x)) \sin x \, dx = A \quad \text{and} \quad f(\pi) = B,$$

find $f(0)$ in terms of A and B .

b. Let

$$f(x) = \int_0^x \frac{dt}{\sqrt{1+t^3}} \quad \text{for } x \geq 0,$$

and let $g = f^{-1}$. Show that g'' is proportional to g^2 .

3. a. Show that

$$\Gamma(x) = \int_0^\infty e^{-x} t^{x-1} \, dt$$

converges for all $x > 0$.

b. Show that $\Gamma(x+1) = x\Gamma(x)$ for $x > 0$.

4. a. Given

$$f(x) = \begin{cases} \frac{g(x)}{x} & \text{for } x \neq 0, \\ 0 & \text{for } x = 0, \end{cases}$$

and $g(0) = 0$, $g'(0) = 0$, $g''(0) = 17$, find $f'(0)$.

b. Evaluate the limit $\lim_{x \rightarrow 0^+} x^{\arcsin x}$.

5. Evaluate the integral $\int \ln(\sqrt{x} + \sqrt{1+x}) dx$.

6. A garden is to be designed in the shape of a circular sector with radius R and the central angle θ . The garden is to have fixed area A . For what values of R and θ will the length of the fencing around the perimeter be minimized?

Fall 1991 Midterm I

1. Use the ε - δ definition to show that $\lim_{x \rightarrow 2} (x^2 + 1) = 5$.

2. Let

$$f(x) = \begin{cases} g(x) \cos(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0 \end{cases}$$

where g is a differentiable function with $g(0) = g'(0) = 0$. Show that f is differentiable at $x = 0$ and $f'(0) = 0$.

3. Find the local minimum, local maximum and the inflection points of the function

$$f(x) = \begin{cases} x^2 & \text{if } x \leq 1, \\ (2-x)^3 & \text{if } x > 1. \end{cases}$$

4. Sketch the graph of $y = \frac{2+x-x^2}{(x-1)^2}$.

5. Let $y = \sin x + \cos x$ and

$$z = \sin \left(\frac{y^3}{\sin \left(\frac{y^3}{\sin y} \right)} \right).$$

Find the derivative of z with respect to x at the point $\frac{\pi}{4}$.

6. An isosceles triangle is inscribed into a circle of radius r . If the top angle of the triangle is required to lie between 0 and $\frac{\pi}{2}$, find the largest and the smallest values possible of the perimeter of the triangle.

Fall 1991 Midterm II

1. Evaluate the following limits:

a. $\lim_{x \rightarrow 0} \frac{\int_0^x \sin t^3 dt}{\sin x^4}$

b. $\lim_{x \rightarrow 0} (e^x + x)^{1/x}$

2. Evaluate the following integrals:

a. $\int \frac{x^{49}}{1 + x^{100}} dx$

b. $\lim_{x \rightarrow 0} \frac{dx}{\sqrt{e^{2x} - 1}}$

3. Let $0 < a < b$. Show that

$$\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}.$$

4. Find the area of the surface generated by revolving the curve $x = t^2$, $y = t^3/3 - t$, $-\sqrt{3} \leq t \leq \sqrt{3}$, about the y -axis.

5. The region between the graphs of $y = x^2$ and $y = \frac{1}{2} - x^2$ is revolved about the y -axis to form a lens. Compute the volume of the lens.

6. A point $P(t)$ is moving along the x -axis in the positive direction at a speed of 5 cm/sec, and is at the origin at time $t = 0$. Find the rate of change, at time $t = 0$, of the area under the curve $y = (\tanh^2 x + \cos^3 x)^4$ and above the interval $[P(t), 100]$.

Fall 1991 Final

1. Discuss the continuity of the following functions in their domain of definition:

a. $f(x) = \begin{cases} \frac{1}{1 + e^{1/x}} & \text{if } x \neq 0, \\ 1 & \text{if } x = 0. \end{cases}$

b. $f(x) = \begin{cases} \frac{x - |x|}{x} & \text{if } x < 0, \\ 2 & \text{if } x = 0. \end{cases}$

3. Evaluate

$$\lim_{n \rightarrow \infty} \left(\ln \sqrt[n]{1 + \frac{1}{n}} + \ln \sqrt[n]{1 + \frac{2}{n}} + \cdots \ln \sqrt[n]{1 + \frac{n}{n}} \right)$$

by relating it to a definite integral.

4. Evaluate:

a. $\int x^3 e^{x^2} dx$

b. $\int \frac{\sqrt{x}}{\sqrt{1-x^3}} dx$

5. Evaluate $\int \frac{x}{\sqrt{x^2 + 2x + 5}} dx$.

6. Evaluate $\int \frac{x dx}{(x+1)^2(x^2+1)}$.

7. Find the volume of the solid generated by revolving the region R bounded by $y = x^3 - 8x$ and $y = -x^2 + 4x$ for $0 \leq x \leq 3$ about the line $x = -1$.

8. The continuously differentiable curve $y = f(x)$ passes through the point $(0, 0)$, and for every $x > 0$ the length of the curve from $(0, 0)$ to $(x, f(x))$ is $e^x + f(x)$. Find $f(x)$.

9. a. Evaluate $\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x} + \frac{1}{x\sqrt{x}}\right)^{2x}$.

b. Differentiate $\log_x(x^2 + 5)$ with respect to x .

Fall 1990 Midterm I

1. Discuss the existence of the limit where for any real number a , denotes the greatest integer in a .

2. A function f satisfies the following condition at a point $x = x_0$: There are numbers $M > 0$, $\beta > 0$ and α such that

$$0 < |x - x_0| < \beta \implies |f(x) - f(x_0)| < M|x - x_0|^\alpha.$$

Then show that

a. if $\alpha > 0$, then f is continuous at $x = x_0$.

b. if $\alpha > 1$, then f is differentiable at $x = x_0$.

3. Find the equation of the line through the point $(3, 0)$ that is normal to the parabola $y = x^2$.

4. Let $f(x) = \sqrt{x}$. Find a formula for the n th derivative $f^{(n)}$ and prove your formula using induction.

5. Let $g(x) = 1 + \sqrt{x}$ and $(f \circ g)(x) = 3 + 2\sqrt{x} + x$. Find $f'(2)$ by

a. by first solving for f and then differentiating,

b. by the chain rule.

6. Consider the equation $2 \sin x = x$.

a. Show that the equation above has a solution $c \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$.

b. Can we find this solution by using Picard's method? If your answer is YES, find an interval I such that every $x_0 \in I$ is a valid initial value for Picard's method.

Fall 1990 Midterm II

1. A rectangle of fixed perimeter P is rotated about one of its sides to generate a cylinder. Of all such possible rectangles, find the dimensions of the one which generates the cylinder of greatest volume.

2. For $a > 1$ show that $\frac{a-1}{a} \leq \ln a \leq a-1$.

3. Evaluate:

a. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x - \sin x}$

b. $\lim_{x \rightarrow 0^+} \frac{\ln(\tan 2x)}{\ln(\tan 3x)}$

4. Find the volume of the solid generated by revolving the region bounded by the graphs of $y = 4 - x^2$ and $y = 0$ about the line $x = 3$.

5. Show that the length of the sine curve $y = \sin x$ for $0 \leq x \leq \pi$ is equal to half of the circumference of the ellipse $2x^2 + y^2 = 2$.

6. Evaluate:

a. $\int e^{(x+e^x)} dx$

b. $\int_0^4 \frac{2dx}{\sqrt{x}(4 + \sqrt{x})}$

Fall 1990 Final

1. When y is given by the following relation find y'' in terms of x and y . Simplify your answer.

$$\ln(x - y) = x + y$$

2. Calculate the following limit: $\lim_{x \rightarrow 0} \left(\frac{3}{x^4} + \frac{1}{x^2} - \frac{3 \tan x}{x^5} \right)$.

3. Find the following antiderivative: $\int \sqrt{4 + 3^x} dx$.

4. Find the following antiderivative: $\int \ln(x + \sqrt{1 + x^2}) dx$.

5. Find the area of the surface obtained by revolving $y = \sin x$ around the x -axis for $0 \leq x \leq \pi$.

6. A rectangle with sides of length a and b is inscribed in a isosceles triangle in such a way that one of the sides with length b is contained in the base of the triangle. Find the minimum possible area of triangle.

7. A point P at the origin of the xy -plane is attached by a string of length a to a mass M located at the point $(a, 0)$. The point P then moves up the y -axis, dragging the mass M by the string. As it moves, the mass traces a differentiable curve $y = f(x)$. Find the equation of the curve $y = f(x)$.

8. At points of the curve $y = 2\sqrt{x}$, lines of length $h = y$ are drawn perpendicular to the coordinate plane. Find the area of the surface formed by these lines from $(0, 0)$ to $(3, 2\sqrt{3})$.

Fall 1989 Midterm

1. Let $f(x)$ be a differentiable function on $(0, \infty)$ and assume that $\lim_{x \rightarrow 0} \frac{f(x)}{Ax^4 + Bx^3} = C$, where A, B, C are constants and $A \neq 0$.

a. Find $\lim_{x \rightarrow 0} \frac{f(x)}{x^2}$.

b. Let

$$H(x) = \begin{cases} \frac{f(x)}{x} & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -\frac{f(-x)}{x} & \text{if } x < 0. \end{cases}$$

Show that $H'(0)$ exists, and find its value.

2. Let $f_1(x)$, $f_2(x)$ and $f_3(x)$ be differentiable functions on $\mathbb{R}$ such that $f_i(m) = 4 - i + m$ and $f'_i(m) = i + m$ where $i = 1, 2, 3$ and m is an integer. Let $F(x) = (f_1 \circ f_2 \circ f_3)(x)$. Find $F'(1)$.

Remark: The data above is enough to solve this problem. However if you want to know if such functions do exist, here are three functions satisfying these conditions: $f_i(x) = 4 - i + x + \frac{i-1}{2\pi} \sin 2\pi x + \frac{1}{4\pi} \sin 2\pi x^2$, $i = 1, 2, 3$. Note that you should solve the problem without using this example.

3. Sketch the graph of $y = \frac{x^3 - x^2 + 4}{x - 1}$.

4. Let $P(x)$ be a polynomial of degree n , and assume that $P(x) \geq 0$ for all $x \in \mathbb{R}$. Show that for all $x \in \mathbb{R}$,

$$P(x) + P'(x) + P''(x) + \cdots + P^{(n)}(x) \geq 0.$$

5. Assume that in an analogue clock the minute arm is twice as long as the hour arm. Find the first time of the day when tips of the minute arm and the hour arm are moving away from each other the fastest.

Remark: You may assume that the arms of the clock lie in the same plane.